

CS1231S Midterm Exercise Set

Exercises from Susanna S. Epp, Discrete Mathematics with Applications, 5th ed. · only those within the scope of Lectures 1-6.

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962 of 1006 exercises kept. Blue numbers have solutions in Epp's Appendix B; H marks a hint; * marks a harder exercise.

What was dropped, and why:

- 1.2 #15-16 · strings (Epp 1.2 'Strings'); not in Lecture 1 or 5
- 1.3 #9-20 (12 exercises) · functions and strings · the midterm covers only the relation half of Epp 1.3
- 6.3 #25 · mathematical induction (Epp Ch. 5)
- 6.4 #1-23, 28 (24 exercises) · Boolean-algebra axioms (the lecture only names Boolean algebra, via Table 6.4.1), liar sentences, and the halting problem
- 8.1 #12 · one-to-one / onto functions (Epp Ch. 7)
- 8.3 #35 · digital logic circuits (Epp 2.4)
- 8.5 #11, 39, 51 · lexicographic order, mathematical induction, and PERT/CPM timing

EXERCISE SET 1.1

Variables

13 of 13 exercises in scope

In each of 1–6, fill in the blanks using a variable or variables to rewrite the given statement.

1. Is there a real number whose square is -1 ?
 - a. Is there a real number x such that _____?
 - b. Does there exist _____ such that $x^2 = -1$?
2. Is there an integer that has a remainder of 2 when it is divided by 5 and a remainder of 3 when it is divided by 6?
 - a. Is there an integer n such that n has _____?
 - b. Does there exist _____ such that if n is divided by 5 the remainder is 2 and if _____?

Note: There are integers with this property. Can you think of one?

3. Given any two distinct real numbers, there is a real number in between them.
 - a. Given any two distinct real numbers a and b , there is a real number c such that c is _____.
 - b. For any two _____, _____ such that c is between a and b .
4. Given any real number, there is a real number that is greater.
 - a. Given any real number r , there is _____ s such that s is _____.
 - b. For any _____, _____ such that $s > r$.
5. The reciprocal of any positive real number is positive.
 - a. Given any positive real number r , the reciprocal of _____.
 - b. For any real number r , if r is _____, then _____.
 - c. If a real number r _____, then _____.
6. The cube root of any negative real number is negative.
 - a. Given any negative real number s , the cube root of _____.
 - b. For any real number s , if s is _____, then _____.
 - c. If a real number s _____, then _____.

7. Rewrite the following statements less formally, without using variables. Determine, as best as you can, whether the statements are true or false.
 - a. There are real numbers u and v with the property that $u + v < u - v$.
 - b. There is a real number x such that $x^2 < x$.
 - c. For every positive integer n , $n^2 \geq n$.
 - d. For all real numbers a and b , $|a + b| \leq |a| + |b|$.

In each of 8–13, fill in the blanks to rewrite the given statement.

8. For every object J , if J is a square then J has four sides.
 - a. All squares _____.
 - b. Every square _____.
 - c. If an object is a square, then it _____.
 - d. If J _____, then J _____.
 - e. For every square J , _____.
9. For every equation E , if E is quadratic then E has at most two real solutions.
 - a. All quadratic equations _____.
 - b. Every quadratic equation _____.
 - c. If an equation is quadratic, then it _____.
 - d. If E _____, then E _____.
 - e. For every quadratic equation E , _____.
10. Every nonzero real number has a reciprocal.
 - a. All nonzero real numbers _____.
 - b. For every nonzero real number r , there is _____ for r .
 - c. For every nonzero real number r , there is a real number s such that _____.
11. Every positive number has a positive square root.
 - a. All positive numbers _____.
 - b. For every positive number e , there is _____ for e .
 - c. For every positive number e , there is a positive number r such that _____.
12. There is a real number whose product with every number leaves the number unchanged.
 - a. Some _____ has the property that its _____.
 - b. There is a real number r such that the product of r _____.
 - c. There is a real number r with the property that for every real number s , _____.

13. There is a real number whose product with every real number equals zero.
- Some _____ has the property that its _____.
 - There is a real number a such that the product of a _____.
 - There is a real number a with the property that for every real number b , _____.

EXERCISE SET 1.2

The Language of Sets

14 of 16 exercises in scope · omitted: strings (Epp 1.2 'Strings'); not in Lecture 1 or 5

1. Which of the following sets are equal?

$$\begin{aligned} A &= \{a, b, c, d\} & B &= \{d, e, a, c\} \\ C &= \{d, b, a, c\} & D &= \{a, a, d, e, c, e\} \end{aligned}$$

2. Write in words how to read each of the following out loud.

- $\{x \in \mathbf{R}^+ \mid 0 < x < 1\}$
- $\{x \in \mathbf{R} \mid x \leq 0 \text{ or } x \geq 1\}$
- $\{n \in \mathbf{Z} \mid n \text{ is a factor of } 6\}$
- $\{n \in \mathbf{Z}^+ \mid n \text{ is a factor of } 6\}$

3. a. Is $4 = \{4\}$?
 b. How many elements are in the set $\{3, 4, 3, 5\}$?
 c. How many elements are in the set $\{1, \{1\}, \{1, \{1\}\}\}$?
4. a. Is $2 \in \{2\}$?
 b. How many elements are in the set $\{2, 2, 2, 2\}$?
 c. How many elements are in the set $\{0, \{0\}\}$?
 d. Is $\{0\} \in \{\{0\}, \{1\}\}$?
 e. Is $0 \in \{\{0\}, \{1\}\}$?

- H5. Which of the following sets are equal?

$$\begin{aligned} A &= \{0, 1, 2\} \\ B &= \{x \in \mathbf{R} \mid -1 \leq x < 3\} \\ C &= \{x \in \mathbf{R} \mid -1 < x < 3\} \\ D &= \{x \in \mathbf{Z} \mid -1 < x < 3\} \\ E &= \{x \in \mathbf{Z}^+ \mid -1 < x < 3\} \end{aligned}$$

- H6. For each integer n , let $T_n = \{n, n^2\}$. How many elements are in each of T_2 , T_{-3} , T_1 , and T_0 ? Justify your answers.

7. Use the set-roster notation to indicate the elements in each of the following sets.
- $S = \{n \in \mathbf{Z} \mid n = (-1)^k, \text{ for some integer } k\}$.
 - $T = \{m \in \mathbf{Z} \mid m = 1 + (-1)^i, \text{ for some integer } i\}$.

- $U = \{r \in \mathbf{Z} \mid 2 \leq r \leq -2\}$
- $V = \{s \in \mathbf{Z} \mid s > 2 \text{ or } s < 3\}$
- $W = \{t \in \mathbf{Z} \mid 1 < t < -3\}$
- $X = \{u \in \mathbf{Z} \mid u \leq 4 \text{ or } u \geq 1\}$

8. Let $A = \{c, d, f, g\}$, $B = \{f, j\}$, and $C = \{d, g\}$. Answer each of the following questions. Give reasons for your answers.

- Is $B \subseteq A$?
- Is $C \subseteq A$?
- Is $C \subseteq C$?
- Is C a proper subset of A ?

9. a. Is $3 \in \{1, 2, 3\}$?
 b. Is $1 \subseteq \{1\}$?
 c. Is $\{2\} \in \{1, 2\}$?
 d. Is $\{3\} \in \{1, \{2\}, \{3\}\}$?
 e. Is $1 \in \{1\}$?
 f. Is $\{2\} \subseteq \{1, \{2\}, \{3\}\}$?
 g. Is $\{1\} \subseteq \{1, 2\}$?
 h. Is $1 \in \{\{1\}, 2\}$?
 i. Is $\{1\} \subseteq \{1, \{2\}\}$?
 j. Is $\{1\} \subseteq \{1\}$?

10. a. Is $((-2)^2, -2^2) = (-2^2, (-2)^2)$?
 b. Is $(5, -5) = (-5, 5)$?
 c. Is $(8 - 9, \sqrt[3]{-1}) = (-1, -1)$?
 d. Is $(\frac{-2}{4}, (-2)^3) = (\frac{3}{6}, -8)$?

11. Let $A = \{w, x, y, z\}$ and $B = \{a, b\}$. Use the set-roster notation to write each of the following sets, and indicate the number of elements that are in each set.

- $A \times B$
- $B \times A$
- $A \times A$
- $B \times B$

12. Let $S = \{2, 4, 6\}$ and $T = \{1, 3, 5\}$. Use the set-roster notation to write each of the following sets, and indicate the number of elements that are in each set.

- $S \times T$
- $T \times S$
- $S \times S$
- $T \times T$

13. Let $A = \{1, 2, 3\}$, $B = \{u\}$, and $C = \{m, n\}$. Find each of the following sets.

- $A \times (B \times C)$
- $(A \times B) \times C$
- $A \times B \times C$

14. Let $R = \{a\}$, $S = \{x, y\}$, and $T = \{p, q, r\}$. Find each of the following sets.
- $R \times (S \times T)$
 - $(R \times S) \times T$
 - $R \times S \times T$

EXERCISE SET 1.3

The Language of Relations and Functions

8 of 20 exercises in scope · omitted: functions and strings · the midterm covers only the relation half of Epp 1.3

- Let $A = \{2, 3, 4\}$ and $B = \{6, 8, 10\}$ and define a relation R from A to B as follows: For every $(x, y) \in A \times B$,
 $(x, y) \in R$ means that $\frac{y}{x}$ is an integer.
 - Is $4 R 6$? Is $4 R 8$? Is $(3, 8) \in R$? Is $(2, 10) \in R$?
 - Write R as a set of ordered pairs.
 - Write the domain and co-domain of R .
 - Draw an arrow diagram for R .
- Let $C = D = \{-3, -2, -1, 1, 2, 3\}$ and define a relation S from C to D as follows: For every $(x, y) \in C \times D$,
 $(x, y) \in S$ means that $\frac{1}{x} - \frac{1}{y}$ is an integer.
 - Is $2 S 2$? Is $-1 S -1$? Is $(3, 3) \in S$?
Is $(3, -3) \in S$?
 - Write S as a set of ordered pairs.
 - Write the domain and co-domain of S .
 - Draw an arrow diagram for S .
- Let $E = \{1, 2, 3\}$ and $F = \{-2, -1, 0\}$ and define a relation T from E to F as follows: For every $(x, y) \in E \times F$,
 $(x, y) \in T$ means that $\frac{x-y}{3}$ is an integer.
 - Is $3 T 0$? Is $1 T (-1)$? Is $(2, -1) \in T$?
Is $(3, -2) \in T$?
 - Write T as a set of ordered pairs.
 - Write the domain and co-domain of T .
 - Draw an arrow diagram for T .

- Let $G = \{-2, 0, 2\}$ and $H = \{4, 6, 8\}$ and define a relation V from G to H as follows: For every $(x, y) \in G \times H$,
 $(x, y) \in V$ means that $\frac{x-y}{4}$ is an integer.
 - Is $2 V 6$? Is $(-2) V (8)$? Is $(0, 6) \in V$?
Is $(2, 4) \in V$?
 - Write V as a set of ordered pairs.
 - Write the domain and co-domain of V .
 - Draw an arrow diagram for V .
- Define a relation S from $\mathbf{R}$ to $\mathbf{R}$ as follows:
 For every $(x, y) \in \mathbf{R} \times \mathbf{R}$,
 $(x, y) \in S$ means that $x \geq y$.
 - Is $(2, 1) \in S$? Is $(2, 2) \in S$? Is $2 S 3$?
Is $(-1) S (-2)$?
 - Draw the graph of S in the Cartesian plane.
- Define a relation R from $\mathbf{R}$ to $\mathbf{R}$ as follows:
 For every $(x, y) \in \mathbf{R} \times \mathbf{R}$,
 $(x, y) \in R$ means that $y = x^2$.
 - Is $(2, 4) \in R$? Is $(4, 2) \in R$? Is $(-3) R 9$?
Is $9 R (-3)$?
 - Draw the graph of R in the Cartesian plane.
- Let $A = \{4, 5, 6\}$ and $B = \{5, 6, 7\}$ and define relations R , S , and T from A to B as follows:
 For every $(x, y) \in A \times B$:
 $(x, y) \in R$ means that $x \geq y$.
 $(x, y) \in S$ means that $\frac{x-y}{2}$ is an integer.
 $T = \{(4, 7), (6, 5), (6, 7)\}$.
 - Draw arrow diagrams for R , S , and T .
 - Indicate whether any of the relations R , S , and T are functions.
- Let $A = \{2, 4\}$ and $B = \{1, 3, 5\}$ and define relations U , V , and W from A to B as follows:
 For every $(x, y) \in A \times B$:
 $(x, y) \in U$ means that $y - x > 2$.
 $(x, y) \in V$ means that $y - 1 = \frac{x}{2}$.
 $W = \{(2, 5), (4, 1), (2, 3)\}$.
 - Draw arrow diagrams for U , V , and W .
 - Indicate whether any of the relations U , V , and W are functions.

EXERCISE SET 2.1

Logical Form and Logical Equivalence

54 of 54 exercises in scope

In each of 1–4 represent the common form of each argument using letters to stand for component sentences, and fill in the blanks so that the argument in part (b) has the same logical form as the argument in part (a).

1. a. If all integers are rational, then the number 1 is rational.
All integers are rational.
Therefore, the number 1 is rational.
b. If all algebraic expressions can be written in prefix notation, then _____.
_____.
Therefore, $(a + 2b)(a^2 - b)$ can be written in prefix notation.
2. a. If all computer programs contain errors, then this program contains an error.
This program does not contain an error.
Therefore, it is not the case that all computer programs contain errors.
b. If _____, then _____.
2 is not odd.
Therefore, it is not the case that all prime numbers are odd.
3. a. This number is even or this number is odd.
This number is not even.
Therefore, this number is odd.
b. _____ or logic is confusing.
My mind is not shot.
Therefore, _____.
4. a. If the program syntax is faulty, then the computer will generate an error message.
If the computer generates an error message, then the program will not run.
Therefore, if the program syntax is faulty, then the program will not run.
b. If this simple graph _____, then it is complete.
If this graph _____, then any two of its vertices can be joined by a path.
Therefore, if this simple graph has 4 vertices and 6 edges, then _____.

5. Indicate which of the following sentences are statements.
 - a. 1,024 is the smallest four-digit number that is a perfect square.
 - b. She is a mathematics major.
 - c. $128 = 2^6$
 - d. $x = 2^6$

Write the statements in 6–9 in symbolic form using the symbols $\sim$, $\vee$, and $\wedge$ and the indicated letters to represent component statements.

6. Let s = “stocks are increasing” and i = “interest rates are steady.”
 - a. Stocks are increasing but interest rates are steady.
 - b. Neither are stocks increasing nor are interest rates steady.
7. Juan is a math major but not a computer science major. (m = “Juan is a math major,” c = “Juan is a computer science major”)
8. Let h = “John is healthy,” w = “John is wealthy,” and s = “John is wise.”
 - a. John is healthy and wealthy but not wise.
 - b. John is not wealthy but he is healthy and wise.
 - c. John is neither healthy, wealthy, nor wise.
 - d. John is neither wealthy nor wise, but he is healthy.
 - e. John is wealthy, but he is not both healthy and wise.
9. Let p = “ $x > 5$,” q = “ $x = 5$,” and r = “ $10 > x$.”
 - a. $x \geq 5$
 - b. $10 > x > 5$
 - c. $10 > x \geq 5$
10. Let p be the statement “DATAENDFLAG is off,” q the statement “ERROR equals 0,” and r the statement “SUM is less than 1,000.” Express the following sentences in symbolic notation.
 - a. DATAENDFLAG is off, ERROR equals 0, and SUM is less than 1,000.
 - b. DATAENDFLAG is off but ERROR is not equal to 0.
 - c. DATAENDFLAG is off; however, ERROR is not 0 or SUM is greater than or equal to 1,000.
 - d. DATAENDFLAG is on and ERROR equals 0 but SUM is greater than or equal to 1,000.
 - e. Either DATAENDFLAG is on or it is the case that both ERROR equals 0 and SUM is less than 1,000.

11. In the following sentence, is the word *or* used in its inclusive or exclusive sense? A team wins the playoffs if it wins two games in a row or a total of three games.

Write truth tables for the statement forms in 12–15.

12. $\sim p \wedge q$
13. $\sim(p \wedge q) \vee (p \vee q)$
14. $p \wedge (q \wedge r)$
15. $p \wedge (\sim q \vee r)$

Determine whether the statement forms in 16–24 are logically equivalent. In each case, construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

16. $p \vee (p \wedge q)$ and p
17. $\sim(p \wedge q)$ and $\sim p \wedge \sim q$
18. $p \vee \mathbf{t}$ and $\mathbf{t}$
19. $p \wedge \mathbf{t}$ and p
20. $p \wedge \mathbf{c}$ and $p \vee \mathbf{c}$
21. $(p \wedge q) \wedge r$ and $p \wedge (q \wedge r)$
22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$
23. $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$
24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$

Use De Morgan's laws to write negations for the statements in 25–30.

25. Hal is a math major and Hal's sister is a computer science major.
26. Sam is an orange belt and Kate is a red belt.
27. The connector is loose or the machine is unplugged.

28. The train is late or my watch is fast.
29. This computer program has a logical error in the first ten lines or it is being run with an incomplete data set.
30. The dollar is at an all-time high and the stock market is at a record low.
31. Let s be a string of length 2 with characters from $\{0, 1, 2\}$, and define statements a , b , c , and d as follows:
- a = "the first character of s is 0"
- b = "the first character of s is 1"
- c = "the second character of s is 1"
- d = "the second character of s is 2".

Describe the set of all strings for which each of the following is true.

- a. $(a \vee b) \wedge (c \vee d)$
- b. $(\sim(a \vee b)) \wedge (c \vee d)$
- c. $((\sim a) \vee b) \wedge (c \vee (\sim d))$

Assume x is a particular real number and use De Morgan's laws to write negations for the statements in 32–37.

32. $-2 < x < 7$
33. $-10 < x < 2$
34. $x < 2$ or $x > 5$
35. $x \leq -1$ or $x > 1$
36. $1 > x \geq -3$
37. $0 > x \geq -7$

In 38 and 39, imagine that num_orders and num_instock are particular values, such as might occur during execution of a computer program. Write negations for the following statements.

38. $(\text{num_orders} > 100 \text{ and } \text{num_instock} \leq 500)$ or $\text{num_instock} < 200$
39. $(\text{num_orders} < 50 \text{ and } \text{num_instock} > 300)$ or $(50 \leq \text{num_orders} < 75 \text{ and } \text{num_instock} > 500)$

Use truth tables to establish which of the statement forms in 40–43 are tautologies and which are contradictions.

40. $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$

41. $(p \wedge \sim q) \wedge (\sim p \vee q)$

42. $((\sim p \wedge q) \wedge (q \wedge r)) \wedge \sim q$

43. $(\sim p \vee q) \vee (p \wedge \sim q)$

44. Recall that $a < x < b$ means that $a < x$ and $x < b$. Also $a \leq b$ means that $a < b$ or $a = b$. Find all real numbers that satisfy the following inequalities.

a. $2 < x \leq 0$

b. $1 \leq x < -1$

45. Determine whether the statements in (a) and (b) are logically equivalent.

- a. Bob is both a math and computer science major and Ann is a math major, but Ann is not both a math and computer science major.
b. It is not the case that both Bob and Ann are both math and computer science majors, but it is the case that Ann is a math major and Bob is both a math and computer science major.

- *46. Let the symbol $\oplus$ denote *exclusive or*; so $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$. Hence the truth table for $p \oplus q$ is as follows:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

- a. Find simpler statement forms that are logically equivalent to $p \oplus p$ and $(p \oplus p) \oplus p$.
b. Is $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$? Justify your answer.
c. Is $(p \oplus q) \wedge r \equiv (p \wedge r) \oplus (q \wedge r)$? Justify your answer.

- *47. In logic and in standard English, a double negative is equivalent to a positive. There is one fairly common English usage in which a “double positive” is equivalent to a negative. What is it? Can you think of others?

In 48 and 49 below, a logical equivalence is derived from Theorem 2.1.1. Supply a reason for each step.

48. $(p \wedge \sim q) \vee (p \wedge q) \equiv p \wedge (\sim q \vee q)$ by (a)
 $\equiv p \wedge (q \vee \sim q)$ by (b)
 $\equiv p \wedge \mathbf{t}$ by (c)
 $\equiv p$ by (d)

Therefore, $(p \wedge \sim q) \vee (p \wedge q) \equiv p$.

49. $(p \vee \sim q) \wedge (\sim p \vee \sim q)$
 $\equiv (\sim q \vee p) \wedge (\sim q \vee \sim p)$ by (a)
 $\equiv \sim q \vee (p \wedge \sim p)$ by (b)
 $\equiv \sim q \vee \mathbf{c}$ by (c)
 $\equiv \sim q$ by (d)

Therefore, $(p \vee \sim q) \wedge (\sim p \vee \sim q) \equiv \sim q$.

Use Theorem 2.1.1 to verify the logical equivalences in 50–54. Supply a reason for each step.

50. $(p \wedge \sim q) \vee p \equiv p$

51. $p \wedge (\sim q \vee p) \equiv p$

52. $\sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$

53. $\sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$

54. $(p \wedge (\sim(\sim p \vee q))) \vee (p \wedge q) \equiv p$

EXERCISE SET 2.2

Conditional Statements
51 of 51 exercises in scope

Rewrite the statements in 1–4 in if-then form.

- This loop will repeat exactly N times if it does not contain a **stop** or a **go to**.
- I am on time for work if I catch the 8:05 bus.
- Freeze or I'll shoot.
- Fix my ceiling or I won't pay my rent.

Construct truth tables for the statement forms in 5–11.

5. $\sim p \vee q \rightarrow \sim q$

6. $(p \vee q) \vee (\sim p \wedge q) \rightarrow q$

7. $p \wedge \sim q \rightarrow r$

8. $\sim p \vee q \rightarrow r$

9. $p \wedge \sim r \leftrightarrow q \vee r$

10. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

11. $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$

12. Use the logical equivalence established in Example 2.2.3, $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$, to rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

13. Use truth tables to verify the following logical equivalences. Include a few words of explanation with your answers.
- a. $p \rightarrow q \equiv \sim p \vee q$
- b. $\sim(p \rightarrow q) \equiv p \wedge \sim q$.

- H 14. a. Show that the following statement forms are all logically equivalent:

$p \rightarrow q \vee r$, $p \wedge \sim q \rightarrow r$, and $p \wedge \sim r \rightarrow q$

- b. Use the logical equivalences established in part (a) to rewrite the following sentence in two different ways. (Assume that n represents a fixed integer.)

If n is prime, then n is odd or n is 2.

15. Determine whether the following statement forms are logically equivalent:

$p \rightarrow (q \rightarrow r)$ and $(p \rightarrow q) \rightarrow r$

In 16 and 17, write each of the two statements in symbolic form and determine whether they are logically equivalent. Include a truth table and a few words of explanation to show that you understand what it means for statements to be logically equivalent.

16. If you paid full price, you didn't buy it at Crown Books. You didn't buy it at Crown Books or you paid full price.
17. If 2 is a factor of n and 3 is a factor of n , then 6 is a factor of n . 2 is not a factor of n or 3 is not a factor of n or 6 is a factor of n .

18. Write each of the following three statements in symbolic form and determine which pairs are logically equivalent. Include truth tables and a few words of explanation.

If it walks like a duck and it talks like a duck, then it is a duck.

Either it does not walk like a duck or it does not talk like a duck, or it is a duck.

If it does not walk like a duck and it does not talk like a duck, then it is not a duck.

19. True or false? The negation of "If Sue is Luiz's mother, then Ali is his cousin" is "If Sue is Luiz's mother, then Ali is not his cousin."

20. Write negations for each of the following statements. (Assume that all variables represent fixed quantities or entities, as appropriate.)

- a. If P is a square, then P is a rectangle.
- b. If today is New Year's Eve, then tomorrow is January.
- c. If the decimal expansion of r is terminating, then r is rational.
- d. If n is prime, then n is odd or n is 2.
- e. If x is nonnegative, then x is positive or x is 0.
- f. If Tom is Ann's father, then Jim is her uncle and Sue is her aunt.
- g. If n is divisible by 6, then n is divisible by 2 and n is divisible by 3.

21. Suppose that p and q are statements so that $p \rightarrow q$ is false. Find the truth values of each of the following:

a. $\sim p \rightarrow q$ b. $p \vee q$ c. $q \rightarrow p$

- H 22. Write contrapositives for the statements of exercise 20.

- H 23. Write the converse and inverse for each statement of exercise 20.

Use truth tables to establish the truth of each statement in 24–27.

24. A conditional statement is not logically equivalent to its converse.
25. A conditional statement is not logically equivalent to its inverse.

26. A conditional statement and its contrapositive are logically equivalent to each other.

27. The converse and inverse of a conditional statement are logically equivalent to each other.

H 28. “Do you mean that you think you can find out the answer to it?” said the March Hare.

“Exactly so,” said Alice.

“Then you should say what you mean,” the March Hare went on.

“I do,” Alice hastily replied; “at least—at least I mean what I say—that’s the same thing, you know.”

“Not the same thing a bit!” said the Hatter.

“Why, you might just as well say that ‘I see what I eat’ is the same thing as ‘I eat what I see!’”

—from “A Mad Tea-Party” in *Alice in Wonderland*, by Lewis Carroll

The Hatter is right. “I say what I mean” is not the same thing as “I mean what I say.” Rewrite each of these two sentences in if-then form and explain the logical relation between them. (This exercise is referred to in the introduction to Chapter 4.)

If statement forms P and Q are logically equivalent, then $P \leftrightarrow Q$ is a tautology. Conversely, if $P \leftrightarrow Q$ is a tautology, then P and Q are logically equivalent. Use $\leftrightarrow$ to convert each of the logical equivalences in 29–31 to a tautology. Then use a truth table to verify each tautology.

29. $p \rightarrow (q \vee r) \equiv (p \wedge \sim q) \rightarrow r$

30. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

31. $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Rewrite each of the statements in 32 and 33 as a conjunction of two if-then statements.

32. This quadratic equation has two distinct real roots if, and only if, its discriminant is greater than zero.

33. This integer is even if, and only if, it equals twice some integer.

Rewrite the statements in 34 and 35 in if-then form in two ways, one of which is the contrapositive of the other. Use the formal definition of “only if.”

34. The Cubs will win the pennant only if they win tomorrow’s game.

35. Sam will be allowed on Signe’s racing boat only if he is an expert sailor.

36. Taking the long view on your education, you go to the Prestige Corporation and ask what you should do in college to be hired when you graduate. The personnel director replies that you will be hired *only if* you major in mathematics or computer science, get a B average or better, and take accounting. You do, in fact, become a math major, get a B+ average, and take accounting. You return to Prestige Corporation, make a formal application, and are turned down. Did the personnel director lie to you?

In 37–39, rewrite the statements in if-then form.

37. Payment will be made on fifth unless a new hearing is granted.

38. Ann will go unless it rains.

39. This door will not open unless a security code is entered.

Rewrite the statements in 40 and 41 in if-then form.

40. Catching the 8:05 bus is a sufficient condition for my being on time for work.

41. Having two 45° angles is a sufficient condition for this triangle to be a right triangle.

Use the contrapositive to rewrite the statements in 42 and 43 in if-then form in two ways.

42. Being divisible by 3 is a necessary condition for this number to be divisible by 9.

43. Doing homework regularly is a necessary condition for Jim to pass the course.

Note that “a sufficient condition for s is r ” means r is a sufficient condition for s and that “a necessary condition for s is r ” means r is a necessary condition for s . Rewrite the statements in 44 and 45 in if-then form.

44. A sufficient condition for Jon’s team to win the championship is that it win the rest of its games.

45. A necessary condition for this computer program to be correct is that it not produce error messages during translation.
46. "If compound X is boiling, then its temperature must be at least 150°C ." Assuming that this statement is true, which of the following must also be true?
- If the temperature of compound X is at least 150°C , then compound X is boiling.
 - If the temperature of compound X is less than 150°C , then compound X is not boiling.
 - Compound X will boil only if its temperature is at least 150°C .
 - If compound X is not boiling, then its temperature is less than 150°C .
 - A necessary condition for compound X to boil is that its temperature be at least 150°C .
 - A sufficient condition for compound X to boil is that its temperature be at least 150°C .

In 47–50 (a) use the logical equivalences $p \rightarrow q \equiv \sim p \vee q$ and $p \leftrightarrow q \equiv (\sim p \vee q) \wedge (\sim q \vee p)$ to rewrite the given statement forms without using the symbol $\rightarrow$ or $\leftrightarrow$, and (b) use the logical equivalence $p \vee q \equiv \sim(\sim p \wedge \sim q)$ to rewrite each statement form using only $\wedge$ and $\sim$.

47. $p \wedge \sim q \rightarrow r$

48. $p \vee \sim q \rightarrow r \vee q$

49. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

50. $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$

51. Given any statement form, is it possible to find a logically equivalent form that uses only $\sim$ and $\wedge$? Justify your answer.

EXERCISE SET 2.3

Valid and Invalid Arguments
44 of 44 exercises in scope

Use modus ponens or modus tollens to fill in the blanks in the arguments of 1–5 so as to produce valid inferences.

1. If $\sqrt{2}$ is rational, then $\sqrt{2} = a/b$ for some integers a and b .
It is not true that $\sqrt{2} = a/b$ for some integers a and b .
 $\therefore$ _____.

2. If $1 - 0.99999 \dots$ is less than every positive real number, then it equals zero.

_____.
 $\therefore$ The number $1 - 0.99999 \dots$ equals zero.

3. If logic is easy, then I am a monkey's uncle.
I am not a monkey's uncle.

_____.

4. If this graph can be colored with three colors, then it can colored with four colors.
This graph cannot be colored with four colors.

_____.

5. If they were unsure of the address, then they would have telephoned.

_____.
 $\therefore$ They were sure of the address.

Use truth tables to determine whether the argument forms in 6–11 are valid. Indicate which columns represent the premises and which represent the conclusion, and include a sentence explaining how the truth table supports your answer. Your explanation should show that you understand what it means for a form of argument to be valid or invalid.

6. $p \rightarrow q$
 $q \rightarrow p$
 $\therefore p \vee q$

7. p
 $p \rightarrow q$
 $\sim q \vee r$
 $\therefore r$

8. $p \vee q$
 $p \rightarrow \sim q$
 $p \rightarrow r$
 $\therefore r$

9. $p \wedge q \rightarrow \sim r$
 $p \vee \sim q$
 $\sim q \rightarrow p$
 $\therefore \sim r$

10. $p \vee q \rightarrow r$
 $\therefore \sim r \rightarrow \sim p \wedge \sim q$
(This is the form of argument shown on pages 37 and 38.)

$$\begin{array}{l}
 11. \quad p \rightarrow q \vee r \\
 \quad \quad \sim q \vee \sim r \\
 \quad \therefore \sim p \vee \sim r
 \end{array}$$

12. Use truth tables to show that the following forms of argument are invalid.

$$\begin{array}{ll}
 \text{a.} & p \rightarrow q \\
 & q \\
 & \therefore p \\
 & \text{(converse error)} \\
 \text{b.} & p \rightarrow q \\
 & \sim p \\
 & \therefore \sim q \\
 & \text{(inverse error)}
 \end{array}$$

Use truth tables to show that the argument forms referred to in 13–21 are valid. Indicate which columns represent the premises and which represent the conclusion, and include a sentence explaining how the truth table supports your answer. Your explanation should show that you understand what it means for a form of argument to be valid.

13. Modus tollens:

$$\begin{array}{l}
 p \rightarrow q \\
 \sim q \\
 \therefore \sim p
 \end{array}$$

14. Example 2.3.3(a)

15. Example 2.3.3(b)

16. Example 2.3.4(a)

17. Example 2.3.4(b)

18. Example 2.3.5(a)

19. Example 2.3.5(b)

20. Example 2.3.6

21. Example 2.3.7

Use symbols to write the logical form of each argument in 22 and 23, and then use a truth table to test the argument for validity. Indicate which columns represent the premises and which represent the conclusion, and include a few words of explanation showing that you understand the meaning of validity.

22. If Tom is not on team A, then Hua is on team B.
If Hua is not on team B, then Tom is on team A.
 $\therefore$ Tom is not on team A or Hua is not on team B.

23. Oleg is a math major or Oleg is an economics major.
If Oleg is a math major, then Oleg is required to take Math 362.
 $\therefore$ Oleg is an economics major or Oleg is not required to take Math 362.

Some of the arguments in 24–32 are valid, whereas others exhibit the converse or the inverse error. Use symbols to write the logical form of each argument. If the argument is valid, identify the rule of inference that guarantees its validity. Otherwise, state whether the converse or the inverse error is made.

24. If Jules solved this problem correctly, then Jules obtained the answer 2.
Jules obtained the answer 2.
 $\therefore$ Jules solved this problem correctly.
25. This real number is rational or it is irrational.
This real number is not rational.
 $\therefore$ This real number is irrational.
26. If I go to the movies, I won't finish my homework.
If I don't finish my homework, I won't do well on the exam tomorrow.
 $\therefore$ If I go to the movies, I won't do well on the exam tomorrow.
27. If this number is larger than 2, then its square is larger than 4.
This number is not larger than 2.
 $\therefore$ The square of this number is not larger than 4.
28. If there are as many rational numbers as there are irrational numbers, then the set of all irrational numbers is infinite.
The set of all irrational numbers is infinite.
 $\therefore$ There are as many rational numbers as there are irrational numbers.
29. If at least one of these two numbers is divisible by 6, then the product of these two numbers is divisible by 6.
Neither of these two numbers is divisible by 6.
 $\therefore$ The product of these two numbers is not divisible by 6.

30. If this computer program is correct, then it produces the correct output when run with the test data my teacher gave me.
This computer program produces the correct output when run with the test data my teacher gave me.
∴ This computer program is correct.
31. Sandra knows Java and Sandra knows C++.
∴ Sandra knows C++.
32. If I get a Christmas bonus, I'll buy a stereo.
If I sell my motorcycle, I'll buy a stereo.
∴ If I get a Christmas bonus or I sell my motorcycle, then I'll buy a stereo.
33. Give an example (other than Example 2.3.11) of a valid argument with a false conclusion.
34. Give an example (other than Example 2.3.12) of an invalid argument with a true conclusion.
35. Explain in your own words what distinguishes a valid form of argument from an invalid one.
36. Given the following information about a computer program, find the mistake in the program.
- There is an undeclared variable or there is a syntax error in the first five lines.
 - If there is a syntax error in the first five lines, then there is a missing semicolon or a variable name is misspelled.
 - There is not a missing semicolon.
 - There is not a misspelled variable name.
37. In the back of an old cupboard you discover a note signed by a pirate famous for his bizarre sense of humor and love of logical puzzles. In the note he wrote that he had hidden treasure somewhere on the property. He listed five true statements (a–e below) and challenged the reader to use them to figure out the location of the treasure.
- If this house is next to a lake, then the treasure is not in the kitchen.
 - If the tree in the front yard is an elm, then the treasure is in the kitchen.
 - This house is next to a lake.
 - The tree in the front yard is an elm or the treasure is buried under the flagpole.
 - If the tree in the back yard is an oak, then the treasure is in the garage.

Where is the treasure hidden?

38. You are visiting the island described in Example 2.3.14 and have the following encounters with natives.
- Two natives *A* and *B* address you as follows:
A says: Both of us are knights.
B says: *A* is a knave.
What are *A* and *B*?
 - Another two natives *C* and *D* approach you but only *C* speaks.
C says: Both of us are knaves.
What are *C* and *D*?
 - You then encounter natives *E* and *F*.
E says: *F* is a knave.
F says: *E* is a knave.
How many knaves are there?
- H d.** Finally, you meet a group of six natives, *U*, *V*, *W*, *X*, *Y*, and *Z*, who speak to you as follows:
U says: None of us is a knight.
V says: At least three of us are knights.
W says: At most three of us are knights.
X says: Exactly five of us are knights.
Y says: Exactly two of us are knights.
Z says: Exactly one of us is a knight.
Which are knights and which are knaves?
39. The famous detective Percule Hoirot was called in to solve a baffling murder mystery. He determined the following facts:
- Lord Hazelton, the murdered man, was killed by a blow on the head with a brass candlestick.
 - Either Lady Hazelton or a maid, Sara, was in the dining room at the time of the murder.
 - If the cook was in the kitchen at the time of the murder, then the butler killed Lord Hazelton with a fatal dose of strychnine.
 - If Lady Hazelton was in the dining room at the time of the murder, then the chauffeur killed Lord Hazelton.
 - If the cook was not in the kitchen at the time of the murder, then Sara was not in the dining room when the murder was committed.
 - If Sara was in the dining room at the time the murder was committed, then the wine steward killed Lord Hazelton.
- Is it possible for the detective to deduce the identity of the murderer from these facts? If so, who did murder Lord Hazelton? (Assume there was only one cause of death.)

40. Sharky, a leader of the underworld, was killed by one of his own band of four henchmen. Detective Sharp interviewed the men and determined that all were lying except for one. He deduced who killed Sharky on the basis of the following statements:
- Socko: Lefty killed Sharky.
 - Fats: Muscles didn't kill Sharky.
 - Lefty: Muscles was shooting craps with Socko when Sharky was knocked off.
 - Muscles: Lefty didn't kill Sharky.
- Who did kill Sharky?

In 41–44 a set of premises and a conclusion are given. Use the valid argument forms listed in Table 2.3.1 to deduce the conclusion from the premises, giving a reason for each step as in Example 2.3.8. Assume all variables are statement variables.

41. a. $\sim p \vee q \rightarrow r$
 b. $s \vee \sim q$
 c. $\sim t$
 d. $p \rightarrow t$
 e. $\sim p \wedge r \rightarrow \sim s$
 f. $\therefore \sim q$
42. a. $p \vee q$
 b. $q \rightarrow r$
 c. $p \wedge s \rightarrow t$
 d. $\sim r$
 e. $\sim q \rightarrow u \wedge s$
 f. $\therefore t$
43. a. $\sim p \rightarrow r \wedge \sim s$
 b. $t \rightarrow s$
 c. $u \rightarrow \sim p$
 d. $\sim w$
 e. $u \vee w$
 f. $\therefore \sim t$
44. a. $p \rightarrow q$
 b. $r \vee s$
 c. $\sim s \rightarrow \sim t$
 d. $\sim q \vee s$
 e. $\sim s$
 f. $\sim p \wedge r \rightarrow u$
 g. $w \vee t$
 h. $\therefore u \wedge w$

EXERCISE SET 3.1

Predicates and Quantified Statements I

33 of 33 exercises in scope

- A menagerie consists of seven brown dogs, two black dogs, six gray cats, ten black cats, five blue birds, six yellow birds, and one black bird. Determine which of the following statements are true and which are false.
 - There is an animal in the menagerie that is red.
 - Every animal in the menagerie is a bird or a mammal.
 - Every animal in the menagerie is brown or gray or black.
 - There is an animal in the menagerie that is neither a cat nor a dog.
 - No animal in the menagerie is blue.
 - There are in the menagerie a dog, a cat, and a bird that all have the same color.
- Indicate which of the following statements are true and which are false. Justify your answers as best as you can.
 - Every integer is a real number.
 - 0 is a positive real number.
 - For every real number r , $-r$ is a negative real number.
 - Every real number is an integer.
- Let $R(m, n)$ be the predicate "If m is a factor of n^2 then m is a factor of n ," with domain for both m and n being $\mathbf{Z}$ the set of integers.
 - Explain why $R(m, n)$ is false if $m = 25$ and $n = 10$.
 - Give values different from those in part (a) for which $R(m, n)$ is false.
 - Explain why $R(m, n)$ is true if $m = 5$ and $n = 10$.
 - Give values different from those in part (c) for which $R(m, n)$ is true.

4. Let $Q(x, y)$ be the predicate “If $x < y$ then $x^2 < y^2$ ” with domain for both x and y being $\mathbf{R}$ the set of real numbers.
- Explain why $Q(x, y)$ is false if $x = -2$ and $y = 1$.
 - Give values different from those in part (a) for which $Q(x, y)$ is false.
 - Explain why $Q(x, y)$ is true if $x = 3$ and $y = 8$.
 - Give values different from those in part (c) for which $Q(x, y)$ is true.
5. Find the truth set of each predicate.
- Predicate: $6/d$ is an integer, domain: $\mathbf{Z}$
 - Predicate: $6/d$ is an integer, domain: $\mathbf{Z}^+$
 - Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{R}$
 - Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{Z}$
6. Let $B(x)$ be “ $-10 < x < 10$.” Find the truth set of $B(x)$ for each of the following domains.
- $\mathbf{Z}$
 - $\mathbf{Z}^+$
 - The set of all even integers
7. Let S be the set of all strings of length 3 consisting of a ’s, b ’s, and c ’s. List all the strings in S that satisfy the following conditions:
- Every string in S begins with b .
 - No string in S has more than one c .
8. Let T be the set of all strings of length 3 consisting of 0’s and 1’s. List all the strings in T that satisfy the following conditions:
- For every string s in T , the second character of s is 1 or the first two characters of s are the same.
 - No string in T has all three characters the same.

Find counterexamples to show that the statements in 9–12 are false.

- $\forall x \in \mathbf{R}, x \geq 1/x$.
- $\forall a \in \mathbf{Z}, (a - 1)/a$ is not an integer.
- $\forall$ positive integers m and $n, m \cdot n \geq m + n$.
- $\forall$ real numbers x and $y, \sqrt{x + y} = \sqrt{x} + \sqrt{y}$.

13. Consider the following statement:

$\forall$ basketball player x, x is tall.

Which of the following are equivalent ways of expressing this statement?

- Every basketball player is tall.
- Among all the basketball players, some are tall.
- Some of all the tall people are basketball players.
- Anyone who is tall is a basketball player.
- All people who are basketball players are tall.
- Anyone who is a basketball player is a tall person.

14. Consider the following statement:

$\exists x \in \mathbf{R}$ such that $x^2 = 2$.

Which of the following are equivalent ways of expressing this statement?

- The square of each real number is 2.
- Some real numbers have square 2.
- The number x has square 2, for some real number x .
- If x is a real number, then $x^2 = 2$.
- Some real number has square 2.
- There is at least one real number whose square is 2.

- H 15. Rewrite the following statements informally in at least two different ways without using variables or quantifiers.

- $\forall$ rectangle x, x is a quadrilateral.
- $\exists$ a set A such that A has 16 subsets.

16. Rewrite each of the following statements in the form “ $\forall$ _____ x , _____.”

- All dinosaurs are extinct.
- Every real number is positive, negative, or zero.
- No irrational numbers are integers.
- No logicians are lazy.
- The number 2,147,581,953 is not equal to the square of any integer.
- The number -1 is not equal to the square of any real number.

17. Rewrite each of the following in the form “ $\exists$ _____ x such that _____.”

- Some exercises have answers.
- Some real numbers are rational.

18. Let D be the set of all students at your school, and let $M(s)$ be “ s is a math major,” let $C(s)$ be “ s is a computer science student,” and let $E(s)$ be “ s is an engineering student.” Express each of the following statements using quantifiers, variables, and the predicates $M(s)$, $C(s)$, and $E(s)$.

- There is an engineering student who is a math major.
- Every computer science student is an engineering student.
- No computer science students are engineering students.
- Some computer science students are also math majors.
- Some computer science students are engineering students and some are not.

19. Consider the following statement:

$\forall$ integer n , if n^2 is even then n is even.

Which of the following are equivalent ways of expressing this statement?

- All integers have even squares and are even.
- Given any integer whose square is even, that integer is itself even.
- For all integers, there are some whose square is even.
- Any integer with an even square is even.
- If the square of an integer is even, then that integer is even.
- All even integers have even squares.

- H 20. Rewrite the following statement informally in at least two different ways without using variables or the symbol $\forall$ or the words “for all.”

$\forall$ real numbers x , if x is positive then the square root of x is positive.

21. Rewrite the following statements so that the quantifier trails the rest of the sentence.
- For any graph G , the total degree of G is even.
 - For any isosceles triangle T , the base angles of T are equal.
 - There exists a prime number p such that p is even.
 - There exists a continuous function f such that f is not differentiable.

22. Rewrite each of the following statements in the form “ $\forall$ _____ x , if _____ then _____.”

- All Java programs have at least 5 lines.
- Any valid argument with true premises has a true conclusion.

23. Rewrite each of the following statements in the two forms “ $\forall x$, if _____ then _____” and “ $\forall x$, _____” (without an if-then).

- All equilateral triangles are isosceles.
- Every computer science student needs to take data structures.

24. Rewrite the following statements in the two forms “ $\exists$ _____ x such that _____” and “ $\exists x$ such that _____ and _____.”

- Some hatters are mad.
- Some questions are easy.

25. The statement “The square of any rational number is rational” can be rewritten formally as “For all rational numbers x , x^2 is rational” or as “For all x ,

if x is rational then x^2 is rational.” Rewrite each of the following statements in the two forms “ $\forall$ _____ x , _____” and “ $\forall x$, if _____, then _____” or in the two forms “ $\forall$ _____ x and y , _____” and “ $\forall x$ and y , if _____, then _____.”

- The reciprocal of any nonzero fraction is a fraction.
- The derivative of any polynomial function is a polynomial function.
- The sum of the angles of any triangle is 180° .
- The negative of any irrational number is irrational.
- The sum of any two even integers is even.
- The product of any two fractions is a fraction.

26. Consider the statement “All integers are rational numbers but some rational numbers are not integers.”

- Write this statement in the form “ $\forall x$, if _____ then _____, but $\exists$ _____ x such that _____.”
- Let $\text{Ratl}(x)$ be “ x is a rational number” and $\text{Int}(x)$ be “ x is an integer.” Write the given statement formally using only the symbols $\text{Ratl}(x)$, $\text{Int}(x)$, $\forall$, $\exists$, $\wedge$, $\vee$, $\sim$, and $\rightarrow$.

27. Refer to the picture of Tarski's world given in Example 3.1.13. Let $\text{Above}(x, y)$ mean that x is above y (but possibly in a different column). Determine the truth or falsity of each of the following statements. Give reasons for your answers.
- $\forall u, \text{Circle}(u) \rightarrow \text{Gray}(u)$.
 - $\forall u, \text{Gray}(u) \rightarrow \text{Circle}(u)$.
 - $\exists y$ such that $\text{Square}(y) \wedge \text{Above}(y, d)$.
 - $\exists z$ such that $\text{Triangle}(z) \wedge \text{Above}(f, z)$.

In 28–30, rewrite each statement without using quantifiers or variables. Indicate which are true and which are false, and justify your answers as best as you can.

28. Let the domain of x be the set D of objects discussed in mathematics courses, and let $\text{Real}(x)$ be “ x is a real number,” $\text{Pos}(x)$ be “ x is a positive real number,” $\text{Neg}(x)$ be “ x is a negative real number,” and $\text{Int}(x)$ be “ x is an integer.”
- $\text{Pos}(0)$
 - $\forall x, \text{Real}(x) \wedge \text{Neg}(x) \rightarrow \text{Pos}(-x)$
 - $\forall x, \text{Int}(x) \rightarrow \text{Real}(x)$
 - $\exists x$ such that $\text{Real}(x) \wedge \sim \text{Int}(x)$
29. Let the domain of x be the set of geometric figures in the plane, and let $\text{Square}(x)$ be “ x is a square” and $\text{Rect}(x)$ be “ x is a rectangle.”
- $\exists x$ such that $\text{Rect}(x) \wedge \text{Square}(x)$
 - $\exists x$ such that $\text{Rect}(x) \wedge \sim \text{Square}(x)$
 - $\forall x, \text{Square}(x) \rightarrow \text{Rect}(x)$
30. Let the domain of x be $\mathbf{Z}$, the set of integers, and let $\text{Odd}(x)$ be “ x is odd,” $\text{Prime}(x)$ be “ x is prime,” and $\text{Square}(x)$ be “ x is a perfect square.” (An integer n is said to be a **perfect square** if, and only if, it equals the square of some integer. For example, 25 is a perfect square because $25 = 5^2$.)
- $\exists x$ such that $\text{Prime}(x) \wedge \sim \text{Odd}(x)$
 - $\forall x, \text{Prime}(x) \rightarrow \sim \text{Square}(x)$
 - $\exists x$ such that $\text{Odd}(x) \wedge \text{Square}(x)$

H 31. In any mathematics or computer science text other than this book, find an example of a statement that is universal but is implicitly quantified. Copy the statement as it appears and rewrite it making the quantification explicit. Give a complete citation for your example, including title, author, publisher, year, and page number.

32. Let $\mathbf{R}$ be the domain of the predicate variable x . Which of the following are true and which are false? Give counter examples for the statements that are false.
- $x > 2 \Rightarrow x > 1$
 - $x > 2 \Rightarrow x^2 > 4$
 - $x^2 > 4 \Rightarrow x > 2$
 - $x^2 > 4 \Leftrightarrow |x| > 2$
33. Let $\mathbf{R}$ be the domain of the predicate variables a , b , c , and d . Which of the following are true and which are false? Give counterexamples for the statements that are false.
- $a > 0$ and $b > 0 \Rightarrow ab > 0$
 - $a < 0$ and $b < 0 \Rightarrow ab < 0$
 - $ab = 0 \Rightarrow a = 0$ or $b = 0$
 - $a < b$ and $c < d \Rightarrow ac < bd$

EXERCISE SET 3.2

Predicates and Quantified Statements II

50 of 50 exercises in scope

- Which of the following is a negation for “All discrete mathematics students are athletic”? More than one answer may be correct.
 - There is a discrete mathematics student who is nonathletic.
 - All discrete mathematics students are nonathletic.
 - There is an athletic person who is not a discrete mathematics student.
 - No discrete mathematics students are athletic.
 - Some discrete mathematics students are nonathletic.
 - No athletic people are discrete mathematics students.
- Which of the following is a negation for “All dogs are loyal”? More than one answer may be correct.
 - All dogs are disloyal.
 - No dogs are loyal.
 - Some dogs are disloyal.
 - Some dogs are loyal.
 - There is a disloyal animal that is not a dog.
 - There is a dog that is disloyal.
 - No animals that are not dogs are loyal.
 - Some animals that are not dogs are loyal.

3. Write a formal negation for each of the following statements.
 - a. $\forall$ string s , s has at least one character.
 - b. $\forall$ computer c , c has a CPU.
 - c. $\exists$ a movie m such that m is over 6 hours long.
 - d. $\exists$ a band b such that b has won at least 10 Grammy awards.
4. Write an informal negation for each of the following statements. Be careful to avoid negations that are ambiguous.
 - a. All dogs are friendly.
 - b. All graphs are connected.
 - c. Some suspicions were substantiated.
 - d. Some estimates are accurate.
5. Write a negation for each of the following statements.
 - a. Every valid argument has a true conclusion.
 - b. All real numbers are positive, negative, or zero.

Write a negation for each statement in 6 and 7.

6.
 - a. Sets A and B do not have any points in common.
 - b. Towns P and Q are not connected by any road on the map.
7.
 - a. This vertex is not connected to any other vertex in the graph.
 - b. This number is not related to any even number.
8. Consider the statement “There are no simple solutions to life’s problems.” Write an informal negation for the statement, and then write the statement formally using quantifiers and variables.

Write a negation for each statement in 9 and 10.

9. $\forall$ real number x , if $x > 3$ then $x^2 > 9$.
10. $\forall$ computer program P , if P compiles without error messages, then P is correct.

In each of 11–14 determine whether the proposed negation is correct. If it is not, write a correct negation.

11.

Statement: The sum of any two irrational numbers is irrational.

Proposed negation: The sum of any two irrational numbers is rational.

12.

Statement: The product of any irrational number and any rational number is irrational.

Proposed negation: The product of any irrational number and any rational number is rational.
13.

Statement: For every integer n , if n^2 is even then n is even.

Proposed negation: For every integer n , if n^2 is even then n is not even.
14.

Statement: For all real numbers x_1 and x_2 , if $x_1^2 = x_2^2$ then $x_1 = x_2$.

Proposed negation: For all real numbers x_1 and x_2 , if $x_1^2 = x_2^2$ then $x_1 \neq x_2$.

15. Let $D = \{-48, -14, -8, 0, 1, 3, 16, 23, 26, 32, 36\}$. Determine which of the following statements are true and which are false. Provide counterexamples for the statements that are false.
 - a. $\forall x \in D$, if x is odd then $x > 0$.
 - b. $\forall x \in D$, if x is less than 0 then x is even.
 - c. $\forall x \in D$, if x is even then $x \leq 0$.
 - d. $\forall x \in D$, if the ones digit of x is 2, then the tens digit is 3 or 4.
 - e. $\forall x \in D$, if the ones digit of x is 6, then the tens digit is 1 or 2.

In 16–23, write a negation for each statement.

16. $\forall$ real number x , if $x^2 \geq 1$ then $x > 0$.
17. $\forall$ integer d , if $6/d$ is an integer then $d = 3$.
18. $\forall x \in \mathbf{R}$, if $x(x+1) > 0$ then $x > 0$ or $x < -1$.
19. $\forall n \in \mathbf{Z}$, if n is prime then n is odd or $n = 2$.
20. $\forall$ integers a , b , and c , if $a - b$ is even and $b - c$ is even, then $a - c$ is even.
21. $\forall$ integer n , if n is divisible by 6, then n is divisible by 2 and n is divisible by 3.
22. If the square of an integer is odd, then the integer is odd.
23. If a function is differentiable then it is continuous.

- 24.** Rewrite the statements in each pair in if-then form and indicate the logical relationship between them.
- All the children in Tom's family are female.
All the females in Tom's family are children.
 - All the integers that are greater than 5 and end in 1, 3, 7, or 9 are prime.
All the integers that are greater than 5 and are prime end in 1, 3, 7, or 9.
- 25.** Each of the following statements is true. In each case write the converse of the statement, and give a counterexample showing that the converse is false.
- If n is any prime number that is greater than 2, then $n + 1$ is even.
 - If m is any odd integer, then $2m$ is even.
 - If two circles intersect in exactly two points, then they do not have a common center.

In 26–33, for each statement in the referenced exercise write the contrapositive, converse, and inverse. Indicate as best as you can which of these statements are true and which are false. Give a counterexample for each that is false.

- 26.** Exercise 16
- 27.** Exercise 17
- 28.** Exercise 18
- 29.** Exercise 19
- 30.** Exercise 20
- 31.** Exercise 21
- 32.** Exercise 22
- 33.** Exercise 23
- 34.** Write the contrapositive for each of the following statements.
- If n is prime, then n is not divisible by any prime number from 2 through $\sqrt{n}$. (Assume that n is a fixed integer.)
 - If A and B do not have any elements in common, then they are disjoint. (Assume that A and B are fixed sets.)
- 35.** Give an example to show that a universal conditional statement is not logically equivalent to its inverse.

- *36.** If $P(x)$ is a predicate and the domain of x is the set of all real numbers, let R be " $\forall x \in \mathbf{Z}, P(x)$," let S be " $\forall x \in \mathbf{Q}, P(x)$," and let T be " $\forall x \in \mathbf{R}, P(x)$."
- Find a definition for $P(x)$ (but do not use " $x \in \mathbf{Z}$ ") so that R is true and both S and T are false.
 - Find a definition for $P(x)$ (but do not use " $x \in \mathbf{Q}$ ") so that both R and S are true and T is false.
- 37.** Consider the following sequence of digits: 0204. A person claims that all the 1's in the sequence are to the left of all the 0's in the sequence. Is this true? Justify your answer. (*Hint:* Write the claim formally and write a formal negation for it. Is the negation true or false?)
- 38.** True or false? All occurrences of the letter u in *Discrete Mathematics* are lowercase. Justify your answer.

Rewrite each statement of 39–44 in if-then form.

- 39.** Earning a grade of C– in this course is a sufficient condition for it to count toward graduation.
- 40.** Being divisible by 8 is a sufficient condition for being divisible by 4.
- 41.** Being on time each day is a necessary condition for keeping this job.
- 42.** Passing a comprehensive exam is a necessary condition for obtaining a master's degree.
- 43.** A number is prime only if it is greater than 1.
- 44.** A polygon is square only if it has four sides.

Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite each of the statements 45–48 without using the word *necessary* or *sufficient*.

- 45.** Being divisible by 8 is not a necessary condition for being divisible by 4.
- 46.** Having a large income is not a necessary condition for a person to be happy.
- 47.** Having a large income is not a sufficient condition for a person to be happy.

48. Being a polynomial is not a sufficient condition for a function to have a real root.

49. The computer scientists Richard Conway and David Gries once wrote:

The absence of error messages during translation of a computer program is only a necessary and not a sufficient condition for reasonable [program] correctness.

Rewrite this statement without using the words *necessary* or *sufficient*.

50. A frequent-flyer club brochure states, “You may select among carriers only if they offer the same lowest fare.” Assuming that “only if” has its formal, logical meaning, does this statement guarantee that if two carriers offer the same lowest fare, the customer will be free to choose between them? Explain.

EXERCISE SET 3.3

Statements with Multiple Quantifiers

61 of 61 exercises in scope

1. Let C be the set of cities in the world, let N be the set of nations in the world, and let $P(c, n)$ be “ c is the capital city of n .” Determine the truth values of the following statements.

- a. $P(\text{Tokyo, Japan})$
- b. $P(\text{Athens, Egypt})$
- c. $P(\text{Paris, France})$
- d. $P(\text{Miami, Brazil})$

2. Let $G(x, y)$ be “ $x^2 > y$.” Indicate which of the following statements are true and which are false.

- a. $G(2, 3)$
- b. $G(1, 1)$
- c. $G(\frac{1}{2}, \frac{1}{2})$
- d. $G(-2, 2)$

3. The following statement is true: “ $\forall$ nonzero number x , $\exists$ a real number y such that $xy = 1$.” For each x given below, find a y to make the predicate “ $xy = 1$ ” true.

- a. $x = 2$
- b. $x = -1$
- c. $x = 3/4$

4. The following statement is true: “ $\forall$ real number x , $\exists$ an integer n such that $n > x$.”* For each x given below, find an n to make the predicate “ $n > x$ ” true.

- a. $x = 15.83$
- b. $x = 10^8$
- c. $x = 10^{10}$

The statements in exercises 5–8 refer to the Tarski world given in Figure 3.3.1. Explain why each is true.

5. For every circle x there is a square y such that x and y have the same color.

6. For every square x there is a circle y such that x and y have different colors and y is above x .

7. There is a triangle x such that for every square y , x is above y .

8. There is a triangle x such that for every circle y , y is above x .

9. Let $D = E = \{-2, -1, 0, 1, 2\}$. Explain why the following statements are true.

- a. $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 0.$
- b. $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = y.$

10. This exercise refers to Example 3.3.3. Determine whether each of the following statements is true or false.

- a. $\forall$ student S , $\exists$ a dessert D such that S chose D .
- b. $\forall$ student S , $\exists$ a salad T such that S chose T .
- c. $\exists$ a dessert D such that $\forall$ student S , S chose D .
- d. $\exists$ a beverage B such that $\forall$ student D , D chose B .

*This is called the Archimedean principle because it was first formulated by Syracuse, who lived from about 287 to 212 B.C.E.

- e. $\exists$ an item I such that $\forall$ student S , S did not choose I .
- f. $\exists$ a station Z such that $\forall$ student S , $\exists$ an item I such that S chose I from Z .

11. Let S be the set of students at your school, let M be the set of movies that have ever been released, and let $V(s, m)$ be “student s has seen movie m .” Rewrite each of the following statements without using the symbol $\forall$, the symbol $\exists$, or variables.

- a. $\exists s \in S$ such that $V(s, \text{Casablanca})$.
- b. $\forall s \in S, V(s, \text{Star Wars})$.
- c. $\forall s \in S, \exists m \in M$ such that $V(s, m)$.
- d. $\exists m \in M$ such that $\forall s \in S, V(s, m)$.
- e. $\exists s \in S, \exists t \in S$, and $\exists m \in M$ such that $s \neq t$ and $V(s, m) \wedge V(t, m)$.
- f. $\exists s \in S$ and $\exists t \in S$ such that $s \neq t$ and $\forall m \in M, V(s, m) \rightarrow V(t, m)$.

12. Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.
- $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 1.$
 - $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = -y.$
 - $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } xy \geq y.$
 - $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x \leq y.$

In each of 13–19, (a) rewrite the statement in English without using the symbol $\forall$ or $\exists$ or variables and expressing your answer as simply as possible, and (b) write a negation for the statement.

- $\forall \text{ color } C, \exists \text{ an animal } A \text{ such that } A \text{ is colored } C.$
- $\exists \text{ a book } b \text{ such that } \forall \text{ person } p, p \text{ has read } b.$
- $\forall \text{ odd integer } n, \exists \text{ an integer } k \text{ such that } n = 2k + 1.$
- $\exists \text{ a real number } u \text{ such that } \forall \text{ real number } v, uv = v.$
- $\forall r \in \mathbf{Q}, \exists \text{ integers } a \text{ and } b \text{ such that } r = a/b.$
- $\forall x \in \mathbf{R}, \exists \text{ a real number } y \text{ such that } x + y = 0.$
- $\exists x \in \mathbf{R} \text{ such that for every real number } y, x + y = 0.$

20. Recall that reversing the order of the quantifiers in a statement with two different quantifiers may change the truth value of the statement—but it does not necessarily do so. All the statements in the pairs below refer to the Tarski world of Figure 3.3.1. In each pair, the order of the quantifiers is reversed but everything else is the same. For each pair, determine whether the statements have the same or opposite truth values. Justify your answers.

- (1) For every square y there is a triangle x such that x and y have different colors.
 - (2) There is a triangle x such that for every square y , x and y have different colors.
- (1) For every circle y there is a square x such that x and y have the same color.
 - (2) There is a square x such that for every circle y , x and y have the same color.

21. For each of the following equations, determine which of the following statements are true:
- (1) For every real number x , there exists a real number y such that the equation is true.
 - (2) There exists a real number x , such that for every real number y , the equation is true.

Note that it is possible for both statements to be true or for both to be false.

- $2x + y = 7$
- $y + x = x + y$
- $x^2 - 2xy + y^2 = 0$
- $(x - 5)(y - 1) = 0$
- $x^2 + y^2 = -1$

In 22 and 23, rewrite each statement without using variables or the symbol $\forall$ or $\exists$. Indicate whether the statement is true or false.

- $\forall \text{ real number } x, \exists \text{ a real number } y \text{ such that } x + y = 0.$
 - $\exists \text{ a real number } y \text{ such that } \forall \text{ real number } x, x + y = 0.$
 - $\forall \text{ nonzero real number } r, \exists \text{ a real number } s \text{ such that } rs = 1.$
 - $\exists \text{ a real number } r \text{ such that } \forall \text{ nonzero real number } s, rs = 1.$
 - Use the laws for negating universal and existential statements to derive the following rules:
 - $$\sim(\forall x \in D(\forall y \in E(P(x, y))))$$

$$\equiv \exists x \in D(\exists y \in E(\sim P(x, y)))$$
 - $$\sim(\exists x \in D(\exists y \in E(P(x, y))))$$

$$\equiv \forall x \in D(\forall y \in E(\sim P(x, y)))$$
- Each statement in 25–28 refers to the Tarski world of Figure 3.3.1. For each, (a) determine whether the statement is true or false and justify your answer, and (b) write a negation for the statement (referring, if you wish, to the result in exercise 24).
- $\forall \text{ circle } x \text{ and } \forall \text{ square } y, x \text{ is above } y.$
 - $\forall \text{ circle } x \text{ and } \forall \text{ triangle } y, x \text{ is above } y.$
 - $\exists \text{ a circle } x \text{ and } \exists \text{ a square } y \text{ such that } x \text{ is above } y \text{ and } x \text{ and } y \text{ have different colors.}$
 - $\exists \text{ a triangle } x \text{ and } \exists \text{ a square } y \text{ such that } x \text{ is above } y \text{ and } x \text{ and } y \text{ have the same color.}$

For each of the statements in 29 and 30, (a) write a new statement by interchanging the symbols $\forall$ and $\exists$, and (b) state which is true: the given statement, the version with interchanged quantifiers, neither, or both.

29. $\forall x \in \mathbf{R}, \exists y \in \mathbf{R}$ such that $x < y$.
30. $\exists x \in \mathbf{R}$ such that $\forall y \in \mathbf{R}^-$ (the set of negative real numbers), $x > y$.
31. Consider the statement “Everybody is older than somebody.” Rewrite this statement in the form “ $\forall$ people x , $\exists$ _____.”
32. Consider the statement “Somebody is older than everybody.” Rewrite this statement in the form “ $\exists$ a person x such that $\forall$ _____.”

In 33–39, (a) rewrite the statement formally using quantifiers and variables, and (b) write a negation for the statement.

33. Everybody loves somebody.
34. Somebody loves everybody.
35. Everybody trusts somebody.
36. Somebody trusts everybody.
37. Any even integer equals twice some integer.
38. Every action has an equal and opposite reaction.
39. There is a program that gives the correct answer to every question that is posed to it.
40. In informal speech most sentences of the form “There is _____ every _____” are intended to be understood as meaning “ $\forall$ _____ $\exists$ _____,” even though the existential quantifier *there is* comes before the universal quantifier *every*. Note that this interpretation applies to the following well-known sentences. Rewrite them using quantifiers and variables.
- There is a sucker born every minute.
 - There is a time for every purpose under heaven.

41. Indicate which of the following statements are true and which are false. Justify your answers as best you can.

- $\forall x \in \mathbf{Z}^+, \exists y \in \mathbf{Z}^+$ such that $x = y + 1$.
- $\forall x \in \mathbf{Z}, \exists y \in \mathbf{Z}$ such that $x = y + 1$.
- $\exists x \in \mathbf{R}$ such that $\forall y \in \mathbf{R}, x = y + 1$.
- $\forall x \in \mathbf{R}^+, \exists y \in \mathbf{R}^+$ such that $xy = 1$.
- $\forall x \in \mathbf{R}, \exists y \in \mathbf{R}$ such that $xy = 1$.
- $\exists x \in \mathbf{R}$ such that $\forall y \in \mathbf{R}, x + y = y$.
- $\forall x \in \mathbf{R}^+, \exists y \in \mathbf{R}^+$ such that $y < x$.
- $\exists x \in \mathbf{R}^+$ such that $\forall y \in \mathbf{R}^+, x \leq y$.

42. Write the negation of the definition of limit of a sequence given in Example 3.3.7.

43. The following is the definition for $\lim_{x \rightarrow a} f(x) = L$:

For every real number $\varepsilon > 0$, there exists a real number $\delta > 0$ such that for every real number x , if $a - \delta < x < a + \delta$ and $x \neq a$ then

$$L - \varepsilon < f(x) < L + \varepsilon.$$

Write what it means for $\lim_{x \rightarrow a} f(x) \neq L$. In other words, write the negation of the definition.

44. The notation $\exists!$ stands for the words “there exists a unique.” Thus, for instance, “ $\exists! x$ such that x is prime and x is even” means that there is one and only one even prime number. Which of the following statements are true and which are false? Explain.

- $\exists!$ real number x such that $\forall$ real number y , $xy = y$.
- $\exists!$ integer x such that $1/x$ is an integer.
- $\forall$ real number x , $\exists!$ real number y such that $x + y = 0$.

*45. Suppose that $P(x)$ is a predicate and D is the domain of x . Rewrite the statement “ $\exists! x \in D$ such that $P(x)$ ” without using the symbol $\exists!$. (See exercise 44 for the meaning of $\exists!$.)

In 46–54, refer to the Tarski world given in Figure 3.1.1, which is shown again here for reference. The domains of all variables consist of all the objects in the Tarski world. For each statement, (a) indicate whether the statement

is true or false and justify your answer, (b) write the given statement using the formal logical notation illustrated in Example 3.3.10, and (c) write a negation for the given statement using the formal logical notation of Example 3.3.10.

 <i>a</i>	 <i>b</i>			
		 <i>c</i>	 <i>d</i>	
	 <i>e</i>	 <i>f</i>		
	 <i>g</i>	 <i>h</i>	 <i>i</i>	
			 <i>j</i>	 <i>k</i>

46. There is a triangle x such that for every square y , x is above y .
47. There is a triangle x such that for every circle y , x is above y .
48. For every circle x , there is a square y such that y is to the right of x .
49. For every object x , if x is a circle then there is a square y such that y has the same color as x .
50. For every object x , if x is a triangle then there is a square y such that y is below x .
51. There is a square x such that for every triangle y , if y is above x then y has the same color as x .
52. For every circle x and for every triangle y , x is to the right of y .
53. There is a circle x and there is a square y such that x and y have the same color.
54. There is a circle x and there is a triangle y such that x has the same color as y .

Let $P(x)$ and $Q(x)$ be predicates and suppose D is the domain of x . In 55–58, for the statement forms in each pair, determine whether (a) they have the same truth value for every choice of $P(x)$, $Q(x)$, and D , or (b) there is a choice of $P(x)$, $Q(x)$, and D for which they have opposite truth values.

55. $\forall x \in D, (P(x) \wedge Q(x))$, and
 $(\forall x \in D, P(x)) \wedge (\forall x \in D, Q(x))$
56. $\exists x \in D, (P(x) \wedge Q(x))$, and
 $(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$
57. $\forall x \in D, (P(x) \vee Q(x))$, and
 $(\forall x \in D, P(x)) \vee (\forall x \in D, Q(x))$
58. $\exists x \in D, (P(x) \vee Q(x))$, and
 $(\exists x \in D, P(x)) \vee (\exists x \in D, Q(x))$

In 59–61, find the answers Prolog would give if the following questions were added to the program given in Example 3.3.11.

59. a. `?isabove( $b_1$ ,  $w_1$ )`
b. `?color( $X$ , white)`
c. `?isabove( $X$ ,  $b_3$ )`
60. a. `?isabove( $w_1$ ,  $g$ )`
b. `?color( $w_2$ , blue)`
c. `?isabove( $X$ ,  $b_1$ )`
61. a. `?isabove( $w_2$ ,  $b_3$ )`
b. `?color( $X$ , gray)`
c. `?isabove( $g$ ,  $X$ )`

EXERCISE SET 3.4

Arguments with Quantified Statements
36 of 36 exercises in scope

1. Let the following law of algebra be the first statement of an argument: For all real numbers a and b ,

$$(a + b)^2 = a^2 + 2ab + b^2.$$
Suppose each of the following statements is, in turn, the second statement of the argument. Use universal instantiation or universal modus ponens to write the conclusion that follows in each case.
 - a. $a = x$ and $b = y$ are particular real numbers.
 - b. $a = f_i$ and $b = f_j$ are particular real numbers.
 - c. $a = 3u$ and $b = 5v$ are particular real numbers.
 - d. $a = g(r)$ and $b = g(s)$ are particular real numbers.
 - e. $a = \log(t_1)$ and $b = \log(t_2)$ are particular real numbers.

Use universal instantiation or universal modus ponens to fill in valid conclusions for the arguments in 2–4.

2. If an integer n equals $2 \cdot k$ and k is an integer, then n is even.
 0 equals $2 \cdot 0$ and 0 is an integer.
 $\therefore$ _____

3. For all real numbers a , b , c , and d , if $b \neq 0$ and $d \neq 0$ then $a/b + c/d = (ad + bc)/bd$.
 $a = 2$, $b = 3$, $c = 4$, and $d = 5$ are particular real numbers such that $b \neq 0$ and $d \neq 0$.
 $\therefore$ _____

4. $\forall$ real numbers r , a , and b , if r is positive, then $(r^a)^b = r^{ab}$.
 $r = 3$, $a = 1/2$, and $b = 6$ are particular real numbers such that r is positive.
 $\therefore$ _____

Use universal modus tollens to fill in valid conclusions for the arguments in 5 and 6.

5. All irrational numbers are real numbers.
 $\frac{1}{0}$ is not a real number.
 $\therefore$ _____

6. If a computer program is correct, then compilation of the program does not produce error messages.
 Compilation of this program produces error messages.
 $\therefore$ _____

Some of the arguments in 7–18 are valid by universal modus ponens or universal modus tollens; others are invalid and exhibit the converse or the inverse error. State which are valid and which are invalid. Justify your answers.

7. All healthy people eat an apple a day.
 Keisha eats an apple a day.
 $\therefore$ Keisha is a healthy person.

8. All freshmen must take a writing course.
 Caroline is a freshman.
 $\therefore$ Caroline must take a writing course.

9. If a graph has no edges, then it has a vertex of degree zero.
 This graph has at least one edge.
 $\therefore$ This graph does not have a vertex of degree zero.

10. If a product of two numbers is 0, then at least one of the numbers is 0.
 For a particular number x , neither $(2x + 1)$ nor $(x - 7)$ equals 0.
 $\therefore$ The product $(2x + 1)(x - 7)$ is not 0.

11. All cheaters sit in the back row.
 Monty sits in the back row.
 $\therefore$ Monty is a cheater.

12. If an 8-bit two's complement represents a positive integer, then the 8-bit two's complement starts with a 0.
 The 8-bit two's complement for this integer does not start with a 0.
 $\therefore$ This integer is not positive.

13. For every student x , if x studies discrete mathematics, then x is good at logic.
 Tarik studies discrete mathematics.
 $\therefore$ Tarik is good at logic.

14. If compilation of a computer program produces error messages, then the program is not correct.
 Compilation of this program does not produce error messages.
 $\therefore$ This program is correct.

15. Any sum of two rational numbers is rational.
 The sum $r + s$ is rational.
 $\therefore$ The numbers r and s are both rational.

16. If a number is even, then twice that number is even.
 The number $2n$ is even, for a particular number n .
 $\therefore$ The particular number n is even.

17. If an infinite series converges, then the terms go to 0.
 The terms of the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ go to 0.
 $\therefore$ The infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ converges.

18. If an infinite series converges, then its terms go to 0.
 The terms of the infinite series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ do not go to 0.
 $\therefore$ The infinite series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ does not converge.

19. Rewrite the statement “No good cars are cheap” in the form “ $\forall x$, if $P(x)$ then $\sim Q(x)$.” Indicate whether each of the following arguments is valid or invalid, and justify your answers.

- a. No good car is cheap.
A Rimbaud is a good car.
 $\therefore$ A Rimbaud is not cheap.
- b. No good car is cheap.
A Simbaru is not cheap.
 $\therefore$ A Simbaru is a good car.
- c. No good car is cheap.
A VX Roadster is cheap.
 $\therefore$ A VX Roadster is not good.
- d. No good car is cheap.
An Omnex is not a good car.
 $\therefore$ An Omnex is cheap.

20. a. Use a diagram to show that the following argument can have true premises and a false conclusion.

All dogs are carnivorous.
Aaron is not a dog.
 $\therefore$ Aaron is not carnivorous.

b. What can you conclude about the validity or invalidity of the following argument form? Explain how the result from part (a) leads to this conclusion.

$\forall x$, if $P(x)$ then $Q(x)$.
 $\sim P(a)$ for a particular a .
 $\therefore \sim Q(a)$.

Indicate whether the arguments in 21–27 are valid or invalid. Support your answers by drawing diagrams.

- 21. All people are mice.
All mice are mortal.
 $\therefore$ All people are mortal.
- 22. All discrete mathematics students can tell a valid argument from an invalid one.
All thoughtful people can tell a valid argument from an invalid one.
 $\therefore$ All discrete mathematics students are thoughtful.
- 23. All teachers occasionally make mistakes.
No gods ever make mistakes.
 $\therefore$ No teachers are gods.

- 24. No vegetarians eat meat.
All vegans are vegetarian.
 $\therefore$ No vegans eat meat.
- 25. No college cafeteria food is good.
No good food is wasted.
 $\therefore$ No college cafeteria food is wasted.
- 26. All polynomial functions are differentiable.
All differentiable functions are continuous.
 $\therefore$ All polynomial functions are continuous.
- 27. [Adapted from Lewis Carroll.]
Nothing intelligible ever puzzles *me*.
Logic puzzles *me*.
 $\therefore$ Logic is unintelligible.

In exercises 28–32, reorder the premises in each of the arguments to show that the conclusion follows as a valid consequence from the premises. It may be helpful to rewrite the statements in if-then form and replace some of them by their contrapositives. Exercises 28–30 refer to the kinds of Tarski worlds discussed in Examples 3.1.13

and 3.3.1. Exercises 31 and 32 are adapted from *Symbolic Logic* by Lewis Carroll.*

- 28. 1. Every object that is to the right of all the blue objects is above all the triangles.
2. If an object is a circle, then it is to the right of all the blue objects.
3. If an object is not a circle, then it is not gray.
 $\therefore$ All the gray objects are above all the triangles.
- 29. 1. All the objects that are to the right of all the triangles are above all the circles.
2. If an object is not above all the black objects, then it is not a square.
3. All the objects that are above all the black objects are to the right of all the triangles.
 $\therefore$ All the squares are above all the circles.
- 30. 1. If an object is above all the triangles, then it is above all the blue objects.
2. If an object is not above all the gray objects, then it is not a square.
3. Every black object is a square.
4. Every object that is above all the gray objects is above all the triangles.
 $\therefore$ If an object is black, then it is above all the blue objects.

31. 1. I trust every animal that belongs to me.
 2. Dogs gnaw bones.
 3. I admit no animals into my study unless they will beg when told to do so.
 4. All the animals in the yard are mine.
 5. I admit every animal that I trust into my study.
 6. The only animals that are really willing to beg when told to do so are dogs.
 $\therefore$ All the animals in the yard gnaw bones.
32. 1. When I work a logic example without grumbling, you may be sure it is one I understand.
 2. The arguments in these examples are not arranged in regular order like the ones I am used to.
 3. No easy examples make my head ache.
 4. I can't understand examples if the arguments are not arranged in regular order like the ones I am used to.
 5. I never grumble at an example unless it gives me a headache.
 $\therefore$ These examples are not easy.

*Lewis Carroll, *Symbolic Logic* (New York: Dover, 1958), pp. 118, 120, 123.

In 33 and 34 a single conclusion follows when all the given premises are taken into consideration, but it is difficult to see because the premises are jumbled up. Reorder the premises to make it clear that a conclusion follows logically, and state the valid conclusion that can be drawn. (It may be helpful to rewrite some of the statements in if-then form and to replace some statements by their contrapositives.)

33. 1. No birds except ostriches are at least 9 feet tall.
 2. There are no birds in this aviary that belong to anyone but me.
 3. No ostrich lives on mince pies.
 4. I have no birds less than 9 feet high.
34. 1. All writers who understand human nature are clever.
 2. No one is a true poet unless he can stir the human heart.
 3. Shakespeare wrote *Hamlet*.
 4. No writer who does not understand human nature can stir the human heart.
 5. None but a true poet could have written *Hamlet*.

- *35. Derive the validity of universal modus tollens from the validity of universal instantiation and modus tollens.
- *36. Derive the validity of universal form of part (a) of the elimination rule from the validity of universal instantiation and the valid argument called elimination in Section 2.3.

EXERCISE SET 4.1

Direct Proof and Counterexample I: Introduction

31 of 31 exercises in scope

In 1–4 justify your answers by using the definitions of even, odd, prime, and composite numbers.

- Assume that k is a particular integer.
 - Is -17 an odd integer?
 - Is 0 neither even nor odd?
 - Is $2k - 1$ odd?
- Assume that c is a particular integer.
 - Is $-6c$ an even integer?
 - Is $8c + 5$ an odd integer?
 - Is $(c^2 + 1) - (c^2 - 1) - 2$ an even integer?
- Assume that m and n are particular integers.
 - Is $6m + 8n$ even?
 - Is $10mn + 7$ odd?
 - If $m > n > 0$, is $m^2 - n^2$ composite?
- Assume that r and s are particular integers.
 - Is $4rs$ even?
 - Is $6r + 4s^2 + 3$ odd?
 - If r and s are both positive, is $r^2 + 2rs + s^2$ composite?

*For exercises with blue numbers, solutions are given in Appendix B. ' symbol * signals that an exercise is more challenging than usual.

Prove the statements in 5–11.

- There are integers m and n such that $m > 1$ and $n > 1$ and $\frac{1}{m} + \frac{1}{n}$ is an integer.
- There are distinct integers m and n such that $\frac{1}{m} + \frac{1}{n}$ is an integer.
- There are real numbers a and b such that

$$\sqrt{a+b} = \sqrt{a} + \sqrt{b}.$$

8. There is an integer $n > 5$ such that $2^n - 1$ is prime.
9. There is a real number x such that $x > 1$ and $2^x > x^{10}$.

Definition: An integer n is called a **perfect square** if, and only if, $n = k^2$ for some integer k .

10. There is a perfect square that can be written as a sum of two other perfect squares.
11. There is an integer n such that $2n^2 - 5n + 2$ is prime.

In 12–13, (a) write a negation for the given statement, and (b) use a counterexample to disprove the given statement. Explain how the counterexample actually shows that the given statement is false.

12. For all real numbers a and b , if $a < b$ then $a^2 < b^2$.
13. For every integer n , if n is odd then $\frac{n-1}{2}$ is odd.

Disprove each of the statements in 14–16 by giving a counterexample. In each case explain how the counterexample actually disproves the statement.

14. For all integers m and n , if $2m + n$ is odd then m and n are both odd.
15. For every integer p , if p is prime then $p^2 - 1$ is even.
16. For every integer n , if n is even then $n^2 + 1$ is prime.

In 17–20, determine whether the property is true for all integers, true for no integers, or true for some integers and false for other integers. Justify your answers.

17. $(a + b)^2 = a^2 + b^2$
18. $\frac{\frac{a}{b} + \frac{c}{d}}{\frac{v}{u}} = \frac{a + c}{\frac{b \cdot d}{v \cdot u}}$
- H 19. $-a^n = (-a)^n$

20. The average of any two odd integers is odd.

Prove the statement in 21 and 22 by the method of exhaustion.

21. Every positive even integer less than 26 can be expressed as a sum of three or fewer perfect squares. (For instance, $10 = 1^2 + 3^2$ and $16 = 4^2$.)
22. For each integer n with $1 \leq n \leq 10$, $n^2 - n + 11$ is a prime number.

Each of the statements in 23–26 is true. For each, (a) rewrite the statement with the quantification implicit as If _____, then _____, and (b) write the first sentence

of a proof (the “starting point”) and the last sentence of a proof (the “conclusion to be shown”). (Note that you do not need to understand the statements in order to be able to do these exercises.)

23. For every integer m , if $m > 1$ then $0 < \frac{1}{m} < 1$.
24. For every real number x , if $x > 1$ then $x^2 > x$.
25. For all integers m and n , if $mn = 1$ then $m = n = 1$ or $m = n = -1$.
26. For every real number x , if $0 < x < 1$ then $x^2 < x$.

H 27. Fill in the blanks in the following proof.

Theorem: For every odd integer n , n^2 is odd.

Proof: Suppose n is any (a). By definition of odd, $n = 2k + 1$ for some integer k . Then

$$\begin{aligned} n^2 &= \left(\frac{(b)}{} \right)^2 && \text{by substitution} \\ &= 4k^2 + 4k + 1 && \text{by multiplying out} \\ &= 2(2k^2 + 2k) + 1 && \text{by factoring out a 2} \end{aligned}$$

Now $2k^2 + 2k$ is an integer because it is a sum of products of integers. Therefore, n^2 equals $2 \cdot$ (an integer) $+ 1$, and so (c) is odd by definition of odd.

Because we have not assumed anything about n except that it is an odd integer, it follows from the principle of (d) that for every odd integer n , n^2 is odd.

and as If _____, then _____ (without using an explicit universal quantifier).

b. Fill in the blanks in the proof of the theorem.

28. Theorem: The sum of any two odd integers is even.

Proof: Suppose m and n are any [particular but arbitrarily chosen] odd integers.

[We must show that $m + n$ is even.]

By (a), $m = 2r + 1$ and $n = 2s + 1$ for some integers r and s .

Then

$$\begin{aligned} m + n &= (2r + 1) + (2s + 1) && \text{by (b)} \\ &= 2r + 2s + 2 \\ &= 2(r + s + 1) && \text{by algebra.} \end{aligned}$$

Let $u = r + s + 1$. Then u is an integer because r , s , and 1 are integers and because (c).

Hence $m + n = 2u$, where u is an integer, and so, by (d), $m + n$ is even [as was to be shown].

29. Theorem: The negative of any even integer is even.

Proof: Suppose n is any [particular but arbitrarily chosen] even integer.

[We must show that $-n$ is even.]

By (a), $n = 2k$ for some integer k .

Then

$$\begin{aligned} -n &= -(2k) && \text{by (b)} \\ &= 2(-k) && \text{by algebra.} \end{aligned}$$

Let $r = -k$. Then r is an integer because (-1) and k are integers and (c).

Hence $-n = 2r$, where r is an integer, and so $-n$ is even by (d) [as was to be shown].

30. Theorem 4.1.2: The sum of any even integer and any odd integer is odd.

Proof: Suppose m is any even integer and n is (a). By definition of even, $m = 2r$ for some (b), and by definition of odd, $n = 2s + 1$ for

some integer s . By substitution and algebra,

$$m + n = \text{(c)} = 2(r + s) + 1.$$

Since r and s are both integers, so is their sum $r + s$. Hence $m + n$ has the form twice some integer plus one, and so (d) by definition of odd.

31. Theorem: Whenever n is an odd integer, $5n^2 + 7$ is even.

Proof: Suppose n is any [particular but arbitrarily chosen] odd integer.

[We must show that $5n^2 + 7$ is even.]

By definition of odd, $n = \text{(a)}$ for some integer k .

Then

$$\begin{aligned} 5n^2 + 7 &= \text{(b)} && \text{by substitution} \\ &= 5(4k^2 + 4k + 1) + 7 \\ &= 20k^2 + 20k + 12 \\ &= 2(10k^2 + 10k + 6) && \text{by algebra.} \end{aligned}$$

Let $t = \text{(c)}$. Then t is an integer because products and sums of integers are integers.

Hence $5n^2 + 7 = 2t$, where t is an integer, and thus (d) by definition of even [as was to be shown].

EXERCISE SET 4.2

Direct Proof and Counterexample II: Writing Advice
41 of 41 exercises in scope

Prove the statements in 1–11. In each case use only the definitions of the terms and the Assumptions listed on page 161, not any previously established properties of odd and even integers. Follow the directions given in this section for writing proofs of universal statements.

1. For every integer n , if n is odd then $3n + 5$ is even.

2. For every integer m , if m is even then $3m + 5$ is odd.

H 3. For every integer n , $2n - 1$ is odd.

4. **Theorem 4.2.2:** The difference of any even integer minus any odd integer is odd.

5. If a and b are any odd integers, then $a^2 + b^2$ is even.

6. If k is any odd integer and m is any even integer, then $k^2 + m^2$ is odd.

H 7. The difference between the squares of any two consecutive integers is odd.

8. For any integers m and n , if m is even and n is odd then $5m + 3n$ is odd.

9. If an integer greater than 4 is a perfect square, then the immediately preceding integer is not prime.

10. If n is any even integer, then $(-1)^n = 1$.

11. If n is any odd integer, then $(-1)^n = -1$.

Prove that the statements in 12–14 are false.

12. There exists an integer $m \geq 3$ such that $m^2 - 1$ is prime.

13. There exists an integer n such that $6n^2 + 27$ is prime.

14. There exists an integer $k \geq 4$ such that $2k^2 - 5k + 2$ is prime.

Find the mistakes in the “proofs” shown in 15–19.

15. **Theorem:** For every integer k , if $k > 0$ then $k^2 + 2k + 1$ is composite.

“**Proof:** For $k = 2$, $k > 0$ and $k^2 + 2k + 1 = 2^2 + 2 \cdot 2 + 1 = 9$. And since $9 = 3 \cdot 3$, then 9 is composite. Hence the theorem is true.”

16. **Theorem:** The difference between any odd integer and any even integer is odd.

“**Proof:** Suppose n is any odd integer, and m is any even integer. By definition of odd, $n = 2k + 1$ where k is an integer, and by definition of even, $m = 2k$ where k is an integer. Then

$$n - m = (2k + 1) - 2k = 1,$$

and 1 is odd. Therefore, the difference between any odd integer and any even integer is odd.”

17. **Theorem:** For every integer k , if $k > 0$ then $k^2 + 2k + 1$ is composite.

“**Proof:** Suppose k is any integer such that $k > 0$. If $k^2 + 2k + 1$ is composite, then $k^2 + 2k + 1 = rs$ for some integers r and s such that

$$1 < r < k^2 + 2k + 1$$

and $1 < s < k^2 + 2k + 1$.

Since $k^2 + 2k + 1 = rs$

and both r and s are strictly between 1 and $k^2 + 2k + 1$, then $k^2 + 2k + 1$ is not prime. Hence $k^2 + 2k + 1$ is composite as was to be shown.”

18. **Theorem:** The product of any even integer and any odd integer is even.

“**Proof:** Suppose m is any even integer and n is any odd integer. If $m \cdot n$ is even, then by definition of even there exists an integer r such that $m \cdot n = 2r$. Also since m is even, there exists an integer p such that $m = 2p$, and since n is odd there exists an integer q such that $n = 2q + 1$. Thus

$$mn = (2p)(2q + 1) = 2r,$$

where r is an integer. By definition of even, then, $m \cdot n$ is even, as was to be shown.”

19. **Theorem:** The sum of any two even integers equals $4k$ for some integer k .

“**Proof:** Suppose m and n are any two even integers. By definition of even, $m = 2k$ for some integer k and $n = 2k$ for some integer k . By substitution,

$$m + n = 2k + 2k = 4k.$$

This is what was to be shown.”

In 20–38 determine whether the statement is true or false. Justify your answer with a proof or a counterexample, as appropriate. In each case use only the definitions of the terms and the Assumptions listed on page 161, not any previously established properties.

20. The product of any two odd integers is odd.

H 21. The negative of any odd integer is odd.

22. For all integers a and b , $4a + 5b + 3$ is even.

23. The product of any even integer and any integer is even.

24. If a sum of two integers is even, then one of the summands is even. (In the expression $a + b$, a and b are called **summands**.)

25. The difference of any two even integers is even.

26. For all integers a , b , and c , if a , b , and c are consecutive, then $a + b + c$ is even.

27. The difference of any two odd integers is even.

H 28. For all integers n and m , if $n - m$ is even then $n^3 - m^3$ is even.

29. For every integer n , if n is prime then $(-1)^n = -1$.
30. For every integer m , if $m > 2$ then $m^2 - 4$ is composite.
31. For every integer n , $n^2 - n + 11$ is a prime number.
32. For every integer n , $4(n^2 + n + 1) - 3n^2$ is a perfect square.
33. Every positive integer can be expressed as a sum of three or fewer perfect squares.
- I*** 34. (Two integers are **consecutive** if, and only if, one is one more than the other.) Any product of four consecutive integers is one less than a perfect square.
35. If m and n are any positive integers and mn is a perfect square, then m and n are perfect squares.
36. The difference of the squares of any two consecutive integers is odd.
- H** 37. For all nonnegative real numbers a and b , $\sqrt{ab} = \sqrt{a}\sqrt{b}$. (Note that if x is a nonnegative real number, then there is a unique nonnegative real number y , denoted $\sqrt{x}$, such that $y^2 = x$.)
38. For all nonnegative real numbers a and b ,
- $$\sqrt{a+b} = \sqrt{a} + \sqrt{b}.$$
39. Suppose that integers m and n are perfect squares. Then $m + n + 2\sqrt{mn}$ is also a perfect square. Why?
- I*** 40. If p is a prime number, must $2^p - 1$ also be prime? Prove or give a counterexample.
- *** 41. If n is a nonnegative integer, must $2^{2n} + 1$ be prime? Prove or give a counterexample.
2. 4.6037
3. $\frac{4}{5} + \frac{2}{9}$
4. 0.37373737...
5. 0.56565656...
6. 320.5492492492...
7. 52.4672167216721...
8. The zero product property, says that if a product of two real numbers is 0, then one of the numbers must be 0.
- a. Write this property formally using quantifiers and variables.
- b. Write the contrapositive of your answer to part (a).
- c. Write an informal version (without quantifier symbols or variables) for your answer to part (b).
9. Assume that a and b are both integers and that $a \neq 0$ and $b \neq 0$. Explain why $(b-a)/(ab^2)$ must be a rational number.
10. Assume that m and n are both integers and that $n \neq 0$. Explain why $(5m-12n)/(4n)$ must be a rational number.
11. Prove that every integer is a rational number.
12. Let S be the statement "The square of any rational number is rational." A formal version of S is "For every rational number r , r^2 is rational." Fill in the blanks in the proof for S .
- Proof:** Suppose that r is $\frac{(a)}{(b)}$. By definition of rational, $r = a/b$ for some $\frac{(b)}{(c)}$ with $b \neq 0$. By substitution,
- $$r^2 = \frac{(c)}{(d)} = a^2/b^2.$$

Since a and b are both integers, so are the products a^2 and $\frac{(d)}{(e)}$. Also $b^2 \neq 0$ by the $\frac{(e)}{(f)}$. Hence r^2 is a ratio of two integers with a non-zero denominator, and so $\frac{(f)}{(g)}$ by definition of rational.

EXERCISE SET 4.3

Direct Proof and Counterexample III: Rational Numbers
39 of 39 exercises in scope

The numbers in 1–7 are all rational. Write each number as a ratio of two integers.

1. $-\frac{35}{6}$

13. Consider the following statement: The negative of any rational number is rational.
- Write the statement formally using a quantifier and a variable.
 - Determine whether the statement is true or false and justify your answer.
14. Consider the statement: The cube of any rational number is a rational number.
- Write the statement formally using a quantifier and a variable.
 - Determine whether the statement is true or false and justify your answer.

Determine which of the statements in 15–19 are true and which are false. Prove each true statement directly from the definitions, and give a counterexample for each false statement. For a statement that is false, determine whether a small change would make it true. If so, make the change and prove the new statement. Follow the directions for writing proofs on page 173.

15. The product of any two rational numbers is a rational number.
- H 16. The quotient of any two rational numbers is a rational number.
- H 17. The difference of any two rational numbers is a rational number.
- H 18. If r and s are any two rational numbers, then $\frac{r+s}{2}$ is rational.
- H 19. For all real numbers a and b , if $a < b$ then $a < \frac{a+b}{2} < b$.
- (You may use the properties of inequalities in T17–T27 of Appendix A.)
20. Use the results of exercises 18 and 19 to prove that given any two rational numbers r and s with $r < s$, there is another rational number between r and s . An important consequence is that there are infinitely many rational numbers in between any two distinct rational numbers. See Section 7.4.

Use the properties of even and odd integers that are listed in Example 4.3.3 to do exercises 21–23. Indicate which properties you use to justify your reasoning.

21. True or false? If m is any even integer and n is any odd integer, then $m^2 + 3n$ is odd. Explain.

22. True or false? If a is any odd integer, then $a^2 + a$ is even. Explain.

23. True or false? If k is any even integer and m is any odd integer, then $(k+2)^2 - (m-1)^2$ is even. Explain.

Derive the statements in 24–26 as corollaries of Theorems 4.3.1, 4.3.2, and the results of exercises 12, 13, 14, 15, and 17.

24. For any rational numbers r and s , $2r + 3s$ is rational.
25. If r is any rational number, then $3r^2 - 2r + 4$ is rational.
26. For any rational number s , $5s^3 + 8s^2 - 7$ is rational.

27. It is a fact that if n is any nonnegative integer, then

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = \frac{1 - (1/2^{n+1})}{1 - (1/2)}.$$

(A more general form of this statement is proved in Section 5.2.) Is the right-hand side of this equation rational? If so, express it as a ratio of two integers.

28. Suppose a , b , c , and d are integers and $a \neq c$. Suppose also that x is a real number that satisfies the equation

$$\frac{ax+b}{cx+d} = 1.$$

Must x be rational? If so, express x as a ratio of two integers.

- * 29. Suppose a , b , and c are integers and x , y , and z are nonzero real numbers that satisfy the following equations:

$$\frac{xy}{x+y} = a \quad \text{and} \quad \frac{xz}{x+z} = b \quad \text{and} \quad \frac{yz}{y+z} = c.$$

Is x rational? If so, express it as ratio of two integers.

30. Prove that if one solution for a quadratic equation of the form $x^2 + bx + c = 0$ is rational (where b and c are rational), then the other solution is also rational. (Use the fact that if the solutions of the equation are r and s , then $x^2 + bx + c = (x-r)(x-s)$.)

31. Prove that if a real number c satisfies a polynomial equation of the form

$$r_3x^3 + r_2x^2 + r_1x + r_0 = 0,$$

where r_0, r_1, r_2 , and r_3 are rational numbers, then c satisfies an equation of the form

$$n_3x^3 + n_2x^2 + n_1x + n_0 = 0,$$

where n_0, n_1, n_2 , and n_3 are integers.

Definition: A number c is called a **root** of a polynomial $p(x)$ if, and only if, $p(c) = 0$.

- * 32. Prove that for every real number c , if c is a root of a polynomial with rational coefficients, then c is a root of a polynomial with integer coefficients.

Use the properties of even and odd integers that are listed in Example 4.3.3 to do exercises 33 and 34.

33. When expressions of the form $(x - r)(x - s)$ are multiplied out, a quadratic polynomial is obtained. For instance, $(x - 2)(x - (-7)) = (x - 2)(x + 7) = x^2 + 5x - 14$.

H a. What can be said about the coefficients of the polynomial obtained by multiplying out $(x - r)(x - s)$ when both r and s are odd integers? When both r and s are even integers? When one of r and s is even and the other is odd?

- b.** It follows from part (a) that $x^2 - 1253x + 255$ cannot be written as a product of two polynomials with integer coefficients. Explain why this is so.

- * 34. Observe that

$$\begin{aligned} (x - r)(x - s)(x - t) \\ = x^3 - (r + s + t)x^2 + (rs + rs + st)x - rst. \end{aligned}$$

- a.** Derive a result for cubic polynomials similar to the result in part (a) of exercise 33 for quadratic polynomials.
- b.** Can $x^3 + 7x^2 - 8x - 27$ be written as a product of three polynomials with integer coefficients? Explain.

In 35–39 find the mistakes in the “proofs” that the sum of any two rational numbers is a rational number.

35. “**Proof:** Any two rational numbers produce a rational number when added together. So if r and s are particular but arbitrarily chosen rational numbers, then $r + s$ is rational.”

36. “**Proof:** Let rational numbers $r = \frac{1}{4}$ and $s = \frac{1}{2}$ be given. Then $r + s = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$, which is a rational number. This is what was to be shown.”

37. “**Proof:** Suppose r and s are rational numbers. By definition of rational, $r = a/b$ for some integers a and b with $b \neq 0$, and $s = a/b$ for some integers a and b with $b \neq 0$. Then

$$r + s = \frac{a}{b} + \frac{a}{b} = \frac{2a}{b}.$$

Let $p = 2a$. Then p is an integer since it is a product of integers. Hence $r + s = p/b$, where p and b are integers and $b \neq 0$. Thus $r + s$ is a rational number by definition of rational. This is what was to be shown.”

38. “**Proof:** Suppose r and s are rational numbers. Then $r = a/b$ and $s = c/d$ for some integers a, b, c , and d with $b \neq 0$ and $d \neq 0$ (by definition of rational). Then

$$r + s = \frac{a}{b} + \frac{c}{d}.$$

But this is a sum of two fractions, which is a fraction. So $r + s$ is a rational number since a rational number is a fraction.”

39. “**Proof:** Suppose r and s are rational numbers. If $r + s$ is rational, then by definition of rational $r + s = a/b$ for some integers a and b with $b \neq 0$. Also since r and s are rational, $r = i/j$ and $s = m/n$ for some integers i, j, m , and n with $j \neq 0$ and $n \neq 0$. It follows that

$$r + s = \frac{i}{j} + \frac{m}{n} = \frac{a}{b},$$

which is a quotient of two integers with a nonzero denominator. Hence it is a rational number. This is what was to be shown.”

EXERCISE SET 4.4

Direct Proof and Counterexample IV: Divisibility
50 of 50 exercises in scope

Give a reason for your answer in each of 1–13. Assume that all variables represent integers.

1. Is 52 divisible by 13?
2. Does $7 \mid 56$?

3. Does $5 \mid 0$?
4. Does 3 divide $(3k + 1)(3k + 2)(3k + 3)$?
5. Is $6m(2m + 10)$ divisible by 4?
6. Is 29 a multiple of 3?
7. Is -3 a factor of 66?
8. Is $6a(a + b)$ a multiple of $3a$?
9. Is 4 a factor of $2a \cdot 34b$?
10. Does $7 \mid 34$?
11. Does $13 \mid 73$?
12. If $n = 4k + 1$, does 8 divide $n^2 - 1$?
13. If $n = 4k + 3$, does 8 divide $n^2 - 1$?
14. Fill in the blanks in the following proof that for all integers a and b , if $a \mid b$ then $a \mid (-b)$.

Proof: Suppose a and b are any integers such that (a). By definition of divisibility, there exists an integer r such that (b). By substitution,

$$-b = -(ar) = a(-r).$$

Let $t = \underline{(c)}$. Then t is an integer because $t = (-1) \cdot r$, and both -1 and r are integers. Thus, by substitution, $-b = at$, where t is an integer, and so by definition of divisibility, (d), as was to be shown.

Prove statements 15 and 16 directly from the definition of divisibility.

15. For all integers a , b , and c , if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$.
- H 16. For all integers a , b , and c , if $a \mid b$ then $a \mid c$ then $a \mid (b - c)$.
17. For all integers a , b , c , and d , if $a \mid c$ and $b \mid d$ then $ab \mid cd$.

18. Consider the following statement: The negative of any multiple of 3 is a multiple of 3.
 - a. Write the statement formally using a quantifier and a variable.
 - b. Determine whether the statement is true or false and justify your answer.
19. Show that the following statement is false: For all integers a and b , if $3 \mid (a + b)$ then $3 \mid (a - b)$.

For each statement in 20–32, determine whether the statement is true or false. Prove the statement directly from the definitions if it is true, and give a counterexample if it is false.

- H 20. The sum of any three consecutive integers is divisible by 3.
21. The product of any two even integers is a multiple of 4.
- H 22. A necessary condition for an integer to be divisible by 6 is that it be divisible by 2.
23. A sufficient condition for an integer to be divisible by 8 is that it be divisible by 16.
24. For all integers a , b , and c , if $a \mid b$ and $a \mid c$ then $a \mid (2b - 3c)$.
25. For all integers a , b , and c , if a is a factor of c and b is a factor of c then ab is a factor of c .
- H 26. For all integers a , b , and c , if $ab \mid c$ then $a \mid c$ and $b \mid c$.
- H 27. For all integers a , b , and c , if $a \mid (b + c)$ then $a \mid b$ or $a \mid c$.
28. For all integers a , b , and c , if $a \mid bc$ then $a \mid b$ or $a \mid c$.
29. For all integers a and b , if $a \mid b$ then $a^2 \mid b^2$.
30. For all integers a and n , if $a \mid n^2$ and $a \leq n$ then $a \mid n$.
31. For all integers a and b , if $a \mid 10b$ then $a \mid 10$ or $a \mid b$.

32. A fast-food chain has a contest in which a card with numbers on it is given to each customer who makes a purchase. If some of the numbers on the card add up to 100, then the customer wins \$100. A certain customer receives a card containing the numbers

72, 21, 15, 36, 69, 81, 9, 27, 42, and 63.

Will the customer win \$100? Why or why not?

33. Is it possible to have a combination of nickels, dimes, and quarters that add up to \$4.72? Explain.
34. Consider a string consisting of a 's, b 's, and c 's where the number of b 's is three times the number of a 's and the number of c 's is five times the number of a 's. Prove that the length of the string is divisible by 3.
35. Two athletes run a circular track at a steady pace so that the first completes one round in 8 minutes and the second in 10 minutes. If they both start from the same spot at 4 P.M., when will be the first time they return to the start together?
36. It can be shown (see exercises 44–48) that an integer is divisible by 3 if, and only if, the sum of its digits is divisible by 3; an integer is divisible by 9 if, and only if, the sum of its digits is divisible by 9; an integer is divisible by 5 if, and only if, its right-most digit is a 5 or a 0; and an integer is divisible by 4 if, and only if, the number formed by its right-most two digits is divisible by 4. Check the following integers for divisibility by 3, 4, 5, and 9.
- 637,425,403,705,125
 - 12,858,306,120,312
 - 517,924,440,926,512
 - 14,328,083,360,232

37. Use the unique factorization theorem to write the following integers in standard factored form.
- 1,176
 - 5,733
 - 3,675

38. Let $n = 8,424$.

- Write the prime factorization for n .
- Write the prime factorization for n^5 .

H c. Is n^5 divisible by 20? Explain.

H d. What is the least positive integer m so that $8,424 \cdot m$ is a perfect square?

39. Suppose that in standard factored form $a = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, where k is a positive integer; $p_1, p_2, \dots, p_k$ are prime numbers; and $e_1, e_2, \dots, e_k$ are positive integers.

- What is the standard factored form for a^3 ?
- Find the least positive integer k such that $2^4 \cdot 3^5 \cdot 7 \cdot 11^2 \cdot k$ is a perfect cube (that is, it equals an integer to the third power). Write the resulting product as a perfect cube.

40. a. If a and b are integers and $12a = 25b$, does $12|b$? does $25|a$? Explain.
- b. If x and y are integers and $10x = 9y$, does $10|y$? does $9|x$? Explain.

H 41. How many zeros are at the end of $45^8 \cdot 88^5$? Explain how you can answer this question without actually computing the number. (Hint: $10 = 2 \cdot 5$.)

42. If n is an integer and $n > 1$, then $n!$ is the product of n and every other positive integer that is less than n . For example, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.
- Write $6!$ in standard factored form.
 - Write $20!$ in standard factored form.
 - Without computing the value of $(20!)^2$ determine how many zeros are at the end of this number when it is written in decimal form. Justify your answer.

- * 43. At a certain university $2/3$ of the mathematics students and $3/5$ of the computer science students have taken a discrete mathematics course. The number of mathematics students who have taken the course equals the number of computer science students who have taken the course. If there are at least 100 mathematics students at the university, what are the least possible number of mathematics students and the least possible number of computer science students at the university?

Definition: Given any nonnegative integer n , the **decimal representation** of n is an expression of the form

$$d_k d_{k-1} \cdots d_2 d_1 d_0,$$

where k is a nonnegative integer, $d_0, d_1, d_2, \dots, d_k$ (called the **decimal digits** of n) are integers from 0 to 9 inclusive, $dk \neq 0$ unless $n = 0$ and $k = 0$, and

$$n = d_k \cdot 10^k + d_{k-1} \cdot 10^{k-1} + \cdots + d_2 \cdot 10^2 + d_1 \cdot 10 + d_0.$$

(For example, $2,503 = 2 \cdot 10^3 + 5 \cdot 10^2 + 0 \cdot 10 + 3$.)

44. Prove that if n is any nonnegative integer whose decimal representation ends in 0, then $5 \mid n$. (Hint: If the decimal representation of a nonnegative integer n ends in d_0 , then $n = 10m + d_0$ for some integer m .)
45. Prove that if n is any nonnegative integer whose decimal representation ends in 5, then $5 \mid n$.
46. Prove that if the decimal representation of a nonnegative integer n ends in d_1d_0 and if $4 \mid (10d_1 + d_0)$, then $4 \mid n$. (Hint: If the decimal representation of a nonnegative integer n ends in d_1d_0 , then there is an integer s such that $n = 100s + 10d_1 + d_0$.)

†* 47. Observe that

$$\begin{aligned}
 7,524 &= 7 \cdot 1,000 + 5 \cdot 100 + 2 \cdot 10 + 4 \\
 &= 7(999 + 1) + 5(99 + 1) + 2(9 + 1) + 4 \\
 &= (7 \cdot 999 + 7) + (5 \cdot 99 + 5) + (2 \cdot 9 + 2) + 4 \\
 &= (7 \cdot 999 + 5 \cdot 99 + 2 \cdot 9) + (7 + 5 + 2 + 4) \\
 &= (7 \cdot 111 \cdot 9 + 5 \cdot 11 \cdot 9 + 2 \cdot 9) + (7 + 5 + 2 + 4) \\
 &= (7 \cdot 111 + 5 \cdot 11 + 2) \cdot 9 + (7 + 5 + 2 + 4) \\
 &= (\text{an integer divisible by } 9) \\
 &\quad + (\text{the sum of the digits of } 7,524).
 \end{aligned}$$

Since the sum of the digits of 7,524 is divisible by 9, 7,524 can be written as a sum of two integers each of which is divisible by 9. It follows from exercise 15 that 7,524 is divisible by 9.

Generalize the argument given in this example to any nonnegative integer n . In other words, prove that for any nonnegative integer n , if the sum of the digits of n is divisible by 9, then n is divisible by 9.

- * 48. Prove that for any nonnegative integer n , if the sum of the digits of n is divisible by 3, then n is divisible by 3.

- * 49. Given a positive integer n written in decimal form, the alternating sum of the digits of n is obtained by starting with the right-most digit, subtracting the digit immediately to its left, adding the next digit to the left, subtracting the next digit, and so forth. For example, the alternating sum of the digits of 180,928 is $8 - 2 + 9 - 0 + 8 - 1 = 22$. Justify the fact that for any nonnegative integer n , if the alternating sum of the digits of n is divisible by 11, then n is divisible by 11.
50. The integer 123,123 has the form abc,abc , where a , b , and c are integers from 0 through 9. Consider all six-digit integers of this form. Which prime numbers divide every one of these integers? Prove your answer.

EXERCISE SET 4.5

Division into Cases and the Quotient-Remainder Theorem
50 of 50 exercises in scope

For each of the values of n and d given in 1–6, find integers q and r such that $n = dq + r$ and $0 \leq r < d$.

1. $n = 70, d = 9$
2. $n = 62, d = 7$
3. $n = 36, d = 40$
4. $n = 3, d = 11$
5. $n = -45, d = 11$
6. $n = -27, d = 8$

Evaluate the expressions in 7–10.

- | | |
|----------------------------|------------------------|
| 7. a. $43 \text{ div } 9$ | b. $43 \text{ mod } 9$ |
| 8. a. $50 \text{ div } 7$ | b. $50 \text{ mod } 7$ |
| 9. a. $28 \text{ div } 5$ | b. $28 \text{ mod } 5$ |
| 10. a. $30 \text{ div } 2$ | b. $30 \text{ mod } 2$ |

11. Check the correctness of formula (4.5.1) given in Example 4.5.3 for the following values of $\text{Day}T$ and N .
- a. $\text{Day}T = 6$ (Saturday) and $N = 15$
 - b. $\text{Day}T = 0$ (Sunday) and $N = 7$
 - c. $\text{Day}T = 4$ (Thursday) and $N = 12$

- * 12. Justify formula (4.5.1) for general values of $DayT$ and N .
13. On a Monday a friend says he will meet you again in 30 days. What day of the week will that be?
- H 14. If today is Tuesday, what day of the week will it be 1,000 days from today?
15. January 1, 2000, was a Saturday, and 2000 was a leap year. What day of the week will January 1, 2050, be?
16. Suppose d is a positive and n is any integer. If $d|n$, what is the remainder obtained when the quotient-remainder theorem is applied to n with divisor d ?
17. Prove directly from the definitions that for every integer n , $n^2 - n + 3$ is odd. Use division into two cases: n is even and n is odd.
18. a. Prove that the product of any two consecutive integers is even.
b. The result of part (a) suggests that the second approach in the discussion of Example 4.5.7 might be possible after all. Write a new proof of Theorem 4.5.3 based on this observation.
19. Prove directly from the definitions that for all integers m and n , if m and n have the same parity, then $5m + 7n$ is even. Divide into two cases: m and n are both even and m and n are both odd.
20. Suppose a is any integer. If $a \bmod 7 = 4$, what is $5a \bmod 7$? In other words, if division of a by 7 gives a remainder of 4, what is the remainder when $5a$ is divided by 7? Your solution should show that you obtain the same answer no matter what integer you start with.
21. Suppose b is any integer. If $b \bmod 12 = 5$, what is $8b \bmod 12$? In other words, if division of b by 12 gives a remainder of 5, what is the remainder when $8b$ is divided by 12? Your solution should show that you obtain the same answer no matter what integer you start with.
22. Suppose c is any integer. If $c \bmod 15 = 3$, what is $10c \bmod 15$? In other words, if division of c by 15 gives a remainder of 3, what is the remainder when $10c$ is divided by 15? Your solution should show that you obtain the same answer no matter what integer you start with.
23. Prove that for every integer n , if $n \bmod 5 = 3$ then $n^2 \bmod 5 = 4$.
24. Prove that for all integers m and n , if $m \bmod 5 = 2$ and $n \bmod 5 = 1$ then $mn \bmod 5 = 2$.
25. Prove that for all integers a and b , if $a \bmod 7 = 5$ and $b \bmod 7 = 6$ then $ab \bmod 7 = 2$.
- H 26. Prove that a necessary and sufficient condition for an integer n to be divisible by a positive integer d is that $n \bmod d = 0$.
- H 27. Use the quotient-remainder theorem with divisor equal to 2 to prove that the square of any integer can be written in one of the two forms $4k$ or $4k + 1$ for some integer k .
- H 28. a. Prove: Given any set of three consecutive integers, one of the integers is a multiple of 3.
b. Use the result of part (a) to prove that any product of three consecutive integers is a multiple of 3.
- H 29. a. Use the quotient-remainder theorem with divisor equal to 3 to prove that the square of any integer has the form $3k$ or $3k + 1$ for some integer k .
b. Use the *mod* notation to rewrite the result of part (a).
30. a. Use the quotient-remainder theorem with divisor equal to 3 to prove that the product of any two consecutive integers has the form $3k$ or $3k + 2$ for some integer k .
b. Use the *mod* notation to rewrite the result of part (a).
- In 31–33, you may use the properties listed in Example 4.3.3.

31. a. Prove that for all integers m and n , $m + n$ and $m - n$ are either both odd or both even.
 b. Find all solutions to the equation $m^2 - n^2 = 56$ for which both m and n are positive integers.
 c. Find all solutions to the equation $m^2 - n^2 = 88$ for which both m and n are positive integers.
32. Given any integers a , b , and c , if $a - b$ is even and $b - c$ is even, what can you say about the parity of $2a - (b + c)$? Prove your answer.
33. Given any integers a , b , and c , if $a - b$ is odd and $b - c$ is even, what can you say about the parity of $a - c$? Prove your answer.
- H 34. Given any integer n , if $n > 3$, could n , $n + 2$, and $n + 4$ all be prime? Prove or give a counterexample.

Prove each of the statements in 35–43.

35. The fourth power of any integer has the form $8m$ or $8m + 1$ for some integer m .
- H 36. The product of any four consecutive integers is divisible by 8.
- H 37. For any integer n , $n^2 + 5$ is not divisible by 4.
38. For every integer m , $m^2 = 5k$, or $m^2 = 5k + 1$, or $m^2 = 5k + 4$ for some integer k .
- H 39. Every prime number except 2 and 3 has the form $6q + 1$ or $6q + 5$ for some integer q .
40. If n is any odd integer, then $n^4 \bmod 16 = 1$.
- H 41. For all real numbers x and y , $|x| \cdot |y| = |xy|$.
42. For all real numbers r and c with $c \geq 0$, $-c \leq r \leq c$ if, and only if, $|r| \leq c$. (Hint: Proving A if, and only if, B requires proving both if A then B and if B then A .)
- H 43. For all real numbers a and b , $||a| - |b|| \leq |a - b|$.

44. A matrix $\mathbf{M}$ has 3 rows and 4 columns.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

The 12 entries in the matrix are to be stored in row major form in locations 7,609 to 7,620 in a computer's memory. This means that the entries in the first row (reading left to right) are stored first, then the entries in the second row, and finally the entries in the third row.

- a. Which location will a_{22} be stored in?
 b. Write a formula (in i and j) that gives the integer n so that a_{ij} is stored in location $7,609 + n$.
 c. Find formulas (in n) for r and s so that a_{rs} is stored in location $7,609 + n$.
45. Let $\mathbf{M}$ be a matrix with m rows and n columns, and suppose that the entries of $\mathbf{M}$ are stored in a computer's memory in row major form (see exercise 44) in locations N , $N + 1$, $N + 2$, $\dots$, $N + mn - 1$. Find formulas in k for r and s so that a_{rs} is stored in location $N + k$.
- * 46. If m , n , and d are integers, $d > 0$, and $m \bmod d = n \bmod d$, does it necessarily follow that $m = n$? That $m - n$ is divisible by d ? Prove your answers.
- * 47. If m , n , and d are integers, $d > 0$, and $d \mid (m - n)$, what is the relation between $m \bmod d$ and $n \bmod d$? Prove your answer.
- * 48. If m , n , a , b , and d are integers, $d > 0$, and $m \bmod d = a$ and $n \bmod d = b$, is $(m + n) \bmod d = a + b$? Is $(m + n) \bmod d = (a + b) \bmod d$? Prove your answers.
- * 49. If m , n , a , b , and d are integers, $d > 0$, and $m \bmod d = a$ and $n \bmod d = b$, is $(mn) \bmod d = ab$? Is $(mn) \bmod d = ab \bmod d$? Prove your answers.
50. Prove that if m , d , and k are integers and $d > 0$, then $(m + dk) \bmod d = m \bmod d$.

EXERCISE SET 4.7

Indirect Argument: Contradiction and Contraposition
36 of 36 exercises in scope

1. Fill in the blanks in the following proof by contradiction that there is no least positive real number.

Proof: Suppose not. That is, suppose that there is a least positive real number x . [We must deduce (a)] Consider the number $x/2$. Since x is a positive real number, $x/2$ is also (b). In addition, we can deduce that $x/2 < x$ by multiplying both sides of the inequality $1 < 2$ by (c) and dividing (d). Hence $x/2$ is a positive real number that is less than the least positive real number. This is a (e). [Thus the supposition is false, and so there is no least positive real number.]

2. Is $\frac{1}{0}$ an irrational number? Explain.
3. Use proof by contradiction to show that for every integer n , $3n + 2$ is not divisible by 3.
4. Use proof by contradiction to show that for every integer m , $7m + 4$ is not divisible by 7.

Carefully formulate the negations of each of the statements in 5–7. Then prove each statement by contradiction.

5. There is no greatest even integer.
6. There is no greatest negative real number.
7. There is no least positive rational number.
8. Fill in the blanks for the following proof that the difference of any rational number and any irrational number is irrational.

Proof (by contradiction): Suppose not. That is, suppose that there exist (a) x and (b) y such that $x - y$ is rational. By definition of rational, there exist integers a, b, c , and d with $b \neq 0$ and $d \neq 0$ so that $x = \frac{a}{b}$ (c) and $x - y = \frac{c}{d}$ (d). By substitution,

$$\frac{a}{b} - y = \frac{c}{d}$$

Adding y and subtracting $\frac{c}{d}$ on both sides gives

$$\begin{aligned} y &= \frac{(e)}{\quad} && \text{by substitution} \\ &= \frac{ad}{bd} - \frac{bc}{bd} \\ &= \frac{ad - bc}{bd} && \text{by algebra.} \end{aligned}$$

Now both $ad - bc$ and bd are integers because products and differences of (f) are (g). And $bd \neq 0$ by the (h). Hence y is a ratio of integers with a nonzero denominator, and thus y is (i) by definition of rational. We therefore have both that y is irrational and that y is rational, which is a contradiction. [Thus the supposition is false and the statement to be proved is true.]

9. a. When asked to prove that the difference of any irrational number and any rational number is irrational, a student began, “Suppose not. That is, suppose the difference of any irrational number and any rational number is rational.”
What is wrong with beginning the proof in this way? (Hint: If needed, review the answer to exercise 11 in Section 3.2.)
b. Prove that the difference of any irrational number and any rational number is irrational.

10. Let S be the statement: For all positive real numbers r and s , $\sqrt{r+s} \neq \sqrt{r} + \sqrt{s}$. Statement S is true, but the following “proof” is incorrect. Find the mistake.

“Proof by contradiction: Suppose not. That is, suppose that for all positive real numbers r and s , $\sqrt{r+s} = \sqrt{r} + \sqrt{s}$. This means that the equation will be true no matter what positive real numbers are substituted for r and s . So let $r = 9$ and $s = 16$. Then r and s are positive real numbers and

$$\sqrt{r+s} = \sqrt{9+16} = \sqrt{25} = 5$$

whereas

$$\sqrt{r} + \sqrt{s} = \sqrt{9} + \sqrt{16} = 3 + 4 = 7.$$

Since $5 \neq 7$, we have that $\sqrt{r+s} \neq \sqrt{r} + \sqrt{s}$, which contradicts the supposition that $\sqrt{r+s} = \sqrt{r} + \sqrt{s}$. This contradiction shows that the supposition is false, and hence statement S is true.”

11. Let T be the statement: The sum of any two rational numbers is rational. Then T is true, but the following “proof” is incorrect. Find the mistake.
- “Proof by contradiction:** Suppose not. That is, suppose that the sum of any two rational numbers is not rational. This means that no matter what two rational numbers are chosen their sum is not rational. Now both 1 and 3 are rational because $1 = 1/1$ and $3 = 3/1$, and so both are ratios of integers with a nonzero denominator. Hence, by a supposition, the sum of 1 and 3, which is 4, is not rational. But 4 is rational because $4 = 4/1$, which is a ratio of integers with a nonzero denominator. Hence 4 is both rational and not rational, which is a contradiction. This contradiction shows that the supposition is false, and hence statement T is true.”
12. Let R be the statement: The square root of any irrational number is irrational.
- Write a negation for R .
 - Prove R by contradiction.
13. Let S be the statement: The product of any irrational number and any nonzero rational number is irrational.
- Write a negation for S .
 - Prove S by contradiction.
14. Let T be the statement: For every integer a , if $a \bmod 6 = 3$, then $a \bmod 3 \neq 2$.
- Write a negation for T .
 - Prove T by contradiction.
- H 15. Do there exist integers a , b , and c such that a , b , and c are all odd and $a^2 + b^2 = c^2$? Prove your answer.
- Prove each statement in 16–19 by contradiction.
16. For all odd integers a and b , $b^2 - a^2 \neq 4$.
(Hint: $b^2 - a^2 = (b+a)(b-a)$ and the only way to factor 4 is either $4 = 2 \cdot 2$ or $4 = 4 \cdot 1$.)
- H 17. For all prime numbers a , b , and c , $a^2 + b^2 \neq c^2$.
18. If a and b are rational numbers, $b \neq 0$, and r is an irrational number, then $a + br$ is irrational.
- H 19. For any integer n , $n^2 - 2$ is not divisible by 4.
20. Fill in the blanks in the following proof by contraposition that for every integer n , if $5 \nmid n^2$ then $5 \nmid n$.
- Proof (by contraposition):** [The contrapositive is: For every integer n , if $5 \mid n$ then $5 \mid n^2$.] Suppose n is any integer such that (a) _____. [We must show that (b) _____.] By definition of divisibility, $n = \frac{(c)}{(d)}$ for some integer k . By substitution, $n^2 = \frac{(d)}{(d)} = 5(5k^2)$. But $5k^2$ is an integer because it is a product of integers. Hence $n^2 = 5 \cdot (\text{an integer})$, and so (e) _____ [as was to be shown].
21. Consider the statement “For every integer n , if n^2 is odd then n is odd.”
- Write what you would suppose and what you would need to show to prove this statement by contradiction.
 - Write what you would suppose and what you would need to show to prove this statement by contraposition.

22. Consider the statement “For every real number r , if r^2 is irrational then r is irrational.”
- Write what you would suppose and what you would need to show to prove this statement by contradiction.
 - Write what you would suppose and what you would need to show to prove this statement by contraposition.

Prove each of the statements in 23–25 in two ways:
(a) by contraposition and (b) by contradiction.

23. The negative of any irrational number is irrational.

24. The reciprocal of any irrational number is irrational.
(The **reciprocal** of a nonzero real number x is $1/x$.)

H 25. For every integer n , if n^2 is odd then n is odd.

Use any method to prove the statements in 26–29.

26. For all integers a , b , and c , if $a \nmid bc$ then $a \nmid b$.

27. For all positive real numbers r and s ,
 $\sqrt{r+s} \neq \sqrt{r} + \sqrt{s}$.

H 28. For all integers a , b , and c , if $a \mid b$ and $a \nmid c$, then $a \nmid (b+c)$.

H 29. For all integers m and n , if $m+n$ is even then m and n are both even or m and n are both odd.

30. a. Let $n = 53$. Find an approximate value for $\sqrt{n}$ and write a list of all the prime numbers less than or equal to $\sqrt{n}$. Is the following statement true or false? When $n = 53$, n is not divisible by any prime number less than or equal to $\sqrt{n}$.
- b. Suppose n is a fixed integer. Let S be the statement, “ n is not divisible by any prime number less than or equal to $\sqrt{n}$.” The following statement is equivalent to S :

$\forall$ prime number p , if p is less than or equal to $\sqrt{n}$ then n is not divisible by p .

Which of the following are negations for S ?

- $\exists$ a prime number p such that $p \leq \sqrt{n}$ and n is divisible by p .
- n is divisible by every prime number less than or equal to $\sqrt{n}$.
- $\exists$ a prime number p such that p is a multiple of n and p is less than or equal to $\sqrt{n}$.
- n is divisible by some prime number that is less than or equal to $\sqrt{n}$.
- $\forall$ prime number p , if p is less than or equal to $\sqrt{n}$, then n is divisible by p .

31. a. Prove by contraposition: For all positive integers n , r , and s , if $rs \leq n$, then $r \leq \sqrt{n}$ or $s \leq \sqrt{n}$.

(Hint: Use Theorem T27 in Appendix A.)

- b. Prove: For each integer $n > 1$, if n is not prime then there exists a prime number p such that $p \leq \sqrt{n}$ and n is divisible by p . (Hint: Use the results of part (a), Theorems 4.4.1, 4.4.3, and 4.4.4, and the transitive property of order.)
- c. State the contrapositive of the result of part (b). The results of exercise 31 provide a way to test whether an integer is prime.

Test for Primality

Given an integer $n > 1$, to test whether n is prime check to see if it is divisible by a prime number less than or equal to its square root. If it is not divisible by any of these numbers, then it is prime.

32. Use the test for primality to determine whether the following numbers are prime or not.

a. 667 b. 557 c. 527 d. 613

33. The sieve of Eratosthenes, named after its inventor, the Greek scholar Eratosthenes (276–194 B.C.E.), provides a way to find all prime numbers less than or equal to some fixed number n . To construct it, write out all the integers from 2 to n . Cross out all multiples of 2 except 2 itself, then all multiples of 3 except 3 itself, then all multiples of 5 except 5 itself, and so forth. Continue crossing out the multiples of each successive prime number up to $\sqrt{n}$. The numbers that are not crossed out are all the prime numbers from 2 to n . Here is a sieve of Eratosthenes that includes the numbers from 2 to 27. The multiples of 2 are crossed out with a /, the multiples of 3 with a \, and the multiples of 5 with a —.

2 3 4 5 6 7 8 9 10 11 12 13 14
15 16 17 18 19 20 21 22 23 24 25 26 27

Use the sieve of Eratosthenes to find all prime numbers less than 100.

34. Use the test for primality and the result of exercise 33 to determine whether the following numbers are prime.
a. 9,269 b. 9,103 c. 8,623 d. 7,917

- * 35.** Use proof by contradiction to show that every integer greater than 11 is a sum of two composite numbers.
- * 36.** For all odd integers a , b , and c , if z is a solution of $ax^2 + bx + c = 0$ then z is irrational. (In the proof, use the properties of even and odd integers that are listed in Example 4.3.3.)

EXERCISE SET 4.8

Indirect Argument: Two Famous Theorems
38 of 38 exercises in scope

- A calculator display shows that $\sqrt{2} = 1.414213562$. Because $1.414213562 = \frac{1414213562}{1000000000}$, this suggests that $\sqrt{2}$ is a rational number, which contradicts Theorem 4.8.1. Explain the discrepancy.
 - Example 4.3.1(h) illustrates a technique for showing that any repeating decimal number is rational. A calculator display shows the result of a certain calculation as 40.72727272727. Can you be sure that the result of the calculation is a rational number? Explain.
 - Could there be a rational number whose first trillion digits are the same as the first trillion digits of $\sqrt{2}$? Explain.
 - A calculator display shows that the result of a certain calculation is 0.2. Can you be sure that the result of the calculation is a rational number?
 - Let S be the statement: The cube root of every irrational number is irrational. This statement is true, but the following “proof” is incorrect. Explain the mistake.
“Proof (by contradiction): Suppose not. Suppose the cube root of every irrational number is rational. But $2\sqrt{2}$ is irrational because it is a product of a rational and an irrational number, and the cube root of $2\sqrt{2}$ is $\sqrt{2}$, which is irrational. This is a contradiction, and hence it is not true that the cube root of every irrational number is rational. Thus the statement to be proved is true.”
- Determine which statements in 6–16 are true and which are false. Prove those that are true and disprove those that are false.**
- $6 - 7\sqrt{2}$ is irrational.
 - $3\sqrt{2} - 7$ is irrational.
 - $\sqrt{4}$ is irrational.
 - $\sqrt{2}/6$ is irrational.
 - The sum of any two irrational numbers is irrational.
 - The difference of any two irrational numbers is irrational.
 - The positive square root of a positive irrational number is irrational.
 - If r is any rational number and s is any irrational number, then r/s is irrational.
 - The sum of any two positive irrational numbers is irrational.
 - The product of two irrational numbers is irrational.

- H 16.** If an integer greater than 1 is a perfect square, then its cube root is irrational.
- 17.** Consider the following sentence: If x is rational then $\sqrt{x}$ is irrational. Is this sentence always true, sometimes true and sometimes false, or always false? Justify your answer.
- 18. a.** Prove that for every integer a , if a^3 is even then a is even.
b. Prove that $\sqrt[3]{2}$ is irrational.
- 19. a.** Use proof by contradiction to show that for any integer n , it is impossible for n to equal both $3q_1 + r_1$ and $3q_2 + r_2$, where q_1, q_2, r_1 , and r_2 are integers, $0 \leq r_1 < 3$, $0 \leq r_2 < 3$, and $r_1 \neq r_2$.
b. Use proof by contradiction, the quotient-remainder theorem, division into cases, and the result of part (a) to prove that for every integer n , if n^2 is divisible by 3 then n is divisible by 3.
c. Prove that $\sqrt{3}$ is irrational.
- 20.** Give an example to show that if d is not prime and n^2 is divisible by d , then n need not be divisible by d .

H 21. The quotient-remainder theorem says not only that there exist quotients and remainders but also that the quotient and remainder of a division are unique. Prove the uniqueness. That is, prove that if a and d are integers with $d > 0$ and if q_1, r_1, q_2 , and r_2 are integers such that

$$a = dq_1 + r_1 \quad \text{where} \quad 0 \leq r_1 < d$$

and

$$a = dq_2 + r_2 \quad \text{where} \quad 0 \leq r_2 < d,$$

then

$$q_1 = q_2 \quad \text{and} \quad r_1 = r_2.$$

H 22. Prove that $\sqrt{5}$ is irrational.

H 23. Prove that for any integer a , $9 \nmid (a^2 - 3)$.

24. An alternative proof of the irrationality of $\sqrt{2}$ counts the number of 2's on the two sides of the equation $2n^2 = m^2$ and uses the unique factorization of integers theorem to deduce a contradiction. Write a proof that uses this approach.

25. Use the proof technique illustrated in exercise 24 to prove that if n is any positive integer that is not a perfect square, then $\sqrt{n}$ is irrational.

H 26. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

*** 27.** Prove that $\log_5(2)$ is irrational. (*Hint:* Use the unique factorization of integers theorem.)

H 28. Let $N = 2 \cdot 3 \cdot 5 \cdot 7 + 1$. What remainder is obtained when N is divided by 2? 3? 5? 7? Justify your answer.

H 29. Suppose a is an integer and p is a prime number such that $p \mid a$ and $p \mid (a + 3)$. What can you deduce about p ? Why?

30. Let $p_1, p_2, p_3, \dots$ be a list of all prime numbers in ascending order. Here is a table of the first six:

p_1	p_2	p_3	p_4	p_5	p_6
2	3	5	7	11	13

- H a.** Let $N_1 = p_1$, $N_2 = p_1 \cdot p_2$, $N_3 = p_1 \cdot p_2 \cdot p_3$, $\dots$, $N_6 = p_1 \cdot p_2 \cdot p_3 \cdot p_4 \cdot p_5 \cdot p_6$. Calculate N_1, N_2, N_3, N_4, N_5 , and N_6 .
b. For each $i = 1, 2, 3, 4, 5, 6$, find whether N_i is itself prime or just has a prime factor less than itself. (*Hint:* Use the test for primality from exercise 31 in Section 4.7 to determine your answers.)

For exercises 31 and 32, use the fact that for every integer n ,

$$n! = n(n-1) \cdots 3 \cdot 2 \cdot 1.$$

31. An alternative proof of the infinitude of the prime numbers begins as follows:

Proof: Suppose there are only finitely many prime numbers. Then one is the largest. Call it p . Let $M = p! + 1$. We will show that there is a prime number q such that $q > p$. Complete this proof.

*** 32.** Prove that for every integer n , if $n > 2$ then there is a prime number p such that $n < p < n!$.

*** 33.** Prove that if $p_1, p_2, \dots$, and p_n are distinct prime numbers with $p_1 = 2$ and $n > 1$, then $p_1 p_2 \cdots p_n + 1$ can be written in the form $4k + 3$ for some integer k .

- H 34.** a. Fermat's last theorem says that for every integer $n > 2$, the equation $x^n + y^n = z^n$ has no positive integer solution (solution for which x , y , and z are positive integers). Prove the following: If for every prime number $p > 2$, $x^p + y^p = z^p$ has no positive integer solution, then for any integer $n > 2$ that is not a power of 2, $x^n + y^n = z^n$ has no positive integer solution.
- b. Fermat proved that there are no integers x , y , and z such that $x^4 + y^4 = z^4$. Use this result to remove the restriction in part (a) that n not be a power of 2. That is, prove that if n is a power of 2 and $n > 4$, then $x^n + y^n = z^n$ has no positive integer solution.

For exercises 35–38 note that to show there is a unique object with a certain property, show that (1) there is an object with the property and (2) if objects A and B have the property, then $A = B$.

- 35.** Prove that there exists a unique prime number of the form $n^2 - 1$, where n is an integer that is greater than or equal to 2.
- 36.** Prove that there exists a unique prime number of the form $n^2 + 2n - 3$, where n is a positive integer.
- 37.** Prove that there is at most one real number a with the property that $a + r = r$ for every real number r . (Such a number is called an *additive identity*.)
- 38.** Prove that there is at most one real number b with the property that $br = r$ for every real number r . (Such a number is called a *multiplicative identity*.)

EXERCISE SET 6.1

Set Theory: Definitions and the Element Method of Proof
38 of 38 exercises in scope

- In each of (a)–(f), answer the following questions: Is $A \subseteq B$? Is $B \subseteq A$? Is either A or B a proper subset of the other?
 - $A = \{2, \{2\}, (\sqrt{2})^2\}$, $B = \{2, \{2\}, \{\{2\}\}\}$
 - $A = \{3, \sqrt{5^2 - 4^2}, 24 \bmod 7\}$, $B = \{8 \bmod 5\}$
 - $A = \{\{1, 2\}, \{2, 3\}\}$, $B = \{1, 2, 3\}$
 - $A = \{a, b, c\}$, $B = \{\{a\}, \{b\}, \{c\}\}$
 - $A = \{\sqrt{16}, \{4\}\}$, $B = \{4\}$
 - $A = \{x \in \mathbf{R} \mid \cos x \in \mathbf{Z}\}$,
 $B = \{x \in \mathbf{R} \mid \sin x \in \mathbf{Z}\}$
- Complete the proof from Example 6.1.3: Prove that $B \subseteq A$ where

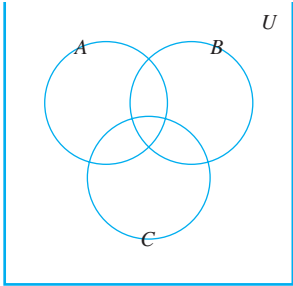
$$A = \{m \in \mathbf{Z} \mid m = 2a \text{ for some integer } a\}$$
 and

$$B = \{n \in \mathbf{Z} \mid n = 2b - 2 \text{ for some integer } b\}$$
- Let sets R , S , and T be defined as follows:

$$R = \{x \in \mathbf{Z} \mid x \text{ is divisible by } 2\}$$

$$S = \{y \in \mathbf{Z} \mid y \text{ is divisible by } 3\}$$

$$T = \{z \in \mathbf{Z} \mid z \text{ is divisible by } 6\}.$$
 Prove or disprove each of the following statements.
 - $R \subseteq T$
 - $T \subseteq R$
 - $T \subseteq S$
- Let $A = \{n \in \mathbf{Z} \mid n = 5r \text{ for some integer } r\}$ and $B = \{m \in \mathbf{Z} \mid m = 20s \text{ for some integer } s\}$. Prove or disprove each of the following statements.
 - $A \subseteq B$
 - $B \subseteq A$
- Let $C = \{n \in \mathbf{Z} \mid n = 6r - 5 \text{ for some integer } r\}$ and $D = \{m \in \mathbf{Z} \mid m = 3s + 1 \text{ for some integer } s\}$. Prove or disprove each of the following statements.
 - $C \subseteq D$
 - $D \subseteq C$
- Let $A = \{x \in \mathbf{Z} \mid x = 5a + 2 \text{ for some integer } a\}$, $B = \{y \in \mathbf{Z} \mid y = 10b - 3 \text{ for some integer } b\}$, and $C = \{z \in \mathbf{Z} \mid z = 10c + 7 \text{ for some integer } c\}$. Prove or disprove each of the following statements.
 - $A \subseteq B$
 - $B \subseteq A$
 - $B \subseteq C$
- Let $A = \{x \in \mathbf{Z} \mid x = 6a + 4 \text{ for some integer } a\}$, $B = \{y \in \mathbf{Z} \mid y = 18b - 2 \text{ for some integer } b\}$, and $C = \{z \in \mathbf{Z} \mid z = 18c + 16 \text{ for some integer } c\}$. Prove or disprove each of the following statements.
 - $A \subseteq B$
 - $B \subseteq A$
 - $B = C$

8. Write in words how to read each of the following out loud. Then write each set using the symbols for union, intersection, set difference, or set complement.
- $\{x \in U \mid x \in A \text{ and } x \in B\}$
 - $\{x \in U \mid x \in A \text{ or } x \in B\}$
 - $\{x \in U \mid x \in A \text{ and } x \notin B\}$
 - $\{x \in U \mid x \notin A\}$
9. Complete the following sentences without using the symbols $\cup$, $\cap$, or $-$.
- $x \notin A \cup B$ if, and only if, _____.
 - $x \notin A \cap B$ if, and only if, _____.
 - $x \notin A - B$ if, and only if, _____.
10. Let $A = \{1, 3, 5, 7, 9\}$, $B = \{3, 6, 9\}$, and $C = \{2, 4, 6, 8\}$. Find each of the following:
- $A \cup B$
 - $A \cap B$
 - $A \cup C$
 - $A \cap C$
 - $A - B$
 - $B - A$
 - $B \cup C$
 - $B \cap C$
11. Let the universal set be $\mathbf{R}$, the set of all real numbers, and let $A = \{x \in \mathbf{R} \mid 0 < x \leq 2\}$, $B = \{x \in \mathbf{R} \mid 1 \leq x < 4\}$, and $C = \{x \in \mathbf{R} \mid 3 \leq x < 9\}$. Find each of the following:
- $A \cup B$
 - $A \cap B$
 - A^c
 - $A \cup C$
 - $A \cap C$
 - B^c
 - $A^c \cap B^c$
 - $A^c \cup B^c$
 - $(A \cap B)^c$
 - $(A \cup B)^c$
12. Let the universal set be $\mathbf{R}$, the set of all real numbers, and let $A = \{x \in \mathbf{R} \mid -3 \leq x \leq 0\}$, $B = \{x \in \mathbf{R} \mid -1 < x < 2\}$, and $C = \{x \in \mathbf{R} \mid 6 < x \leq 8\}$. Find each of the following:
- $A \cup B$
 - $A \cap B$
 - A^c
 - $A \cup C$
 - $A \cap C$
 - B^c
 - $A^c \cap B^c$
 - $A^c \cup B^c$
 - $(A \cap B)^c$
 - $(A \cup B)^c$
13. Let S be the set of all strings of 0's and 1's of length 4, and let A and B be the following subsets of S : $A = \{1110, 1111, 1000, 1001\}$ and $B = \{1100, 0100, 1111, 0111\}$. Find each of the following:
- $A \cap B$
 - $A \cup B$
 - $A - B$
 - $B - A$
14. In each of the following, draw a Venn diagram for sets A , B , and C that satisfy the given conditions.
- $A \subseteq B$, $C \subseteq B$, $A \cap C = \emptyset$
 - $C \subseteq A$, $B \cap C = \emptyset$
15. In each of the following, draw a Venn diagram for sets A , B , and C that satisfy the given conditions.
- $A \cap B = \emptyset$, $A \subseteq C$, $C \cap B \neq \emptyset$
 - $A \subseteq B$, $C \subseteq B$, $A \cap C \neq \emptyset$
 - $A \cap B \neq \emptyset$, $B \cap C \neq \emptyset$, $A \cap C = \emptyset$, $A \not\subseteq B$, $C \not\subseteq B$
16. Let $A = \{a, b, c\}$, $B = \{b, c, d\}$, and $C = \{b, c, e\}$.
- Find $A \cup (B \cap C)$, $(A \cup B) \cap C$, and $(A \cup B) \cap (A \cup C)$. Which of these sets are equal?
 - Find $A \cap (B \cup C)$, $(A \cap B) \cup C$, and $(A \cap B) \cup (A \cap C)$. Which of these sets are equal?
 - Find $(A - B) - C$ and $A - (B - C)$. Are these sets equal?
17. Consider the following Venn diagram. For each of (a)–(f), copy the diagram and shade the region corresponding to the indicated set.
- $A \cap B$
 - $B \cup C$
 - A^c
 - $A - (B \cup C)$
 - $(A \cup B)^c$
 - $A^c \cap B^c$
- 
18.
 - Is the number 0 in $\emptyset$? Why?
 - Is $\emptyset = \{\emptyset\}$? Why?
 - Is $\emptyset \in \{\emptyset\}$? Why?
 - Is $\emptyset \in \emptyset$? Why?
19. Let $A_i = \{i, i^2\}$ for each integer $i = 1, 2, 3, 4$.
- $A_1 \cup A_2 \cup A_3 \cup A_4 = ?$
 - $A_1 \cap A_2 \cap A_3 \cap A_4 = ?$
 - Are A_1, A_2, A_3 , and A_4 mutually disjoint? Explain.
20. Let $B_i = \{x \in \mathbf{R} \mid 0 \leq x \leq i\}$ for each integer $i = 1, 2, 3, 4$.
- $B_1 \cup B_2 \cup B_3 \cup B_4 = ?$
 - $B_1 \cap B_2 \cap B_3 \cap B_4 = ?$
 - Are B_1, B_2, B_3 , and B_4 mutually disjoint? Explain.

21. Let $C_i = \{i, -i\}$ for each nonnegative integer i .

- a. $\bigcup_{i=0}^4 C_i = ?$
- b. $\bigcap_{i=0}^4 C_i = ?$
- c. Are $C_0, C_1, C_2, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n C_i = ?$
- e. $\bigcap_{i=0}^n C_i = ?$
- f. $\bigcup_{i=0}^{\infty} C_i = ?$
- g. $\bigcap_{i=0}^{\infty} C_i = ?$

22. Let $D_i = \{x \in \mathbf{R} \mid -i \leq x \leq i\} = [-i, i]$ for each nonnegative integer i .

- a. $\bigcup_{i=0}^4 D_i = ?$
- b. $\bigcap_{i=0}^4 D_i = ?$
- c. Are $D_0, D_1, D_2, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n D_i = ?$
- e. $\bigcap_{i=0}^n D_i = ?$
- f. $\bigcup_{i=0}^{\infty} D_i = ?$
- g. $\bigcap_{i=0}^{\infty} D_i = ?$

23. Let $V_i = \{x \in \mathbf{R} \mid -\frac{1}{i} \leq x \leq \frac{1}{i}\} = [-\frac{1}{i}, \frac{1}{i}]$ for each positive integer i .

- a. $\bigcup_{i=0}^4 V_i = ?$
- b. $\bigcap_{i=0}^4 V_i = ?$
- c. Are $V_1, V_2, V_3, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n V_i = ?$
- e. $\bigcap_{i=0}^n V_i = ?$
- f. $\bigcup_{i=0}^{\infty} V_i = ?$
- g. $\bigcap_{i=0}^{\infty} V_i = ?$

24. Let $W_i = \{x \in \mathbf{R} \mid x > i\} = (i, \infty)$ for each non-negative integer i .

- a. $\bigcup_{i=0}^4 W_i = ?$
- b. $\bigcap_{i=0}^4 W_i = ?$
- c. Are $W_0, W_1, W_2, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n W_i = ?$
- e. $\bigcap_{i=0}^n W_i = ?$
- f. $\bigcup_{i=0}^{\infty} W_i = ?$
- g. $\bigcap_{i=0}^{\infty} W_i = ?$

25. Let $R_i = \{x \in \mathbf{R} \mid 1 \leq x \leq 1 + \frac{1}{i}\} = [1, 1 + \frac{1}{i}]$ for each positive integer i .

- a. $\bigcup_{i=0}^4 R_i = ?$
- b. $\bigcap_{i=0}^4 R_i = ?$
- c. Are $R_1, R_2, R_3, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n R_i = ?$
- e. $\bigcap_{i=0}^n R_i = ?$
- f. $\bigcup_{i=0}^{\infty} R_i = ?$
- g. $\bigcap_{i=0}^{\infty} R_i = ?$

26. Let $S_i = \{x \in \mathbf{R} \mid 1 < x < 1 + \frac{1}{i}\} = (1, 1 + \frac{1}{i})$ for each positive integer i .

- a. $\bigcup_{i=0}^4 S_i = ?$
- b. $\bigcap_{i=0}^4 S_i = ?$
- c. Are $S_1, S_2, S_3, \dots$ mutually disjoint? Explain.
- d. $\bigcup_{i=0}^n S_i = ?$
- e. $\bigcap_{i=0}^n S_i = ?$
- f. $\bigcup_{i=0}^{\infty} S_i = ?$
- g. $\bigcap_{i=0}^{\infty} S_i = ?$

27. a. Is $\{\{a, d, e\}, \{b, c\}, \{d, f\}\}$ a partition of $\{a, b, c, d, e, f\}$?
- b. Is $\{\{w, x, v\}, \{u, y, q\}, \{p, z\}\}$ a partition of $\{p, q, u, v, w, x, y, z\}$?
- c. Is $\{\{5, 4\}, \{7, 2\}, \{1, 3, 4\}, \{6, 8\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8\}$?
- d. Is $\{\{3, 7, 8\}, \{2, 9\}, \{1, 4, 5\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$?
- e. Is $\{\{1, 5\}, \{4, 7\}, \{2, 8, 6, 3\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8\}$?

28. Let E be the set of all even integers and O the set of all odd integers. Is $\{E, O\}$ a partition of $\mathbf{Z}$, the set of all integers? Explain your answer.

29. Let $\mathbf{R}$ be the set of all real numbers. Is $\{\mathbf{R}^+, \mathbf{R}^-, \{0\}\}$ a partition of $\mathbf{R}$? Explain your answer.

30. Let $\mathbf{Z}$ be the set of all integers and let

$$A_0 = \{n \in \mathbf{Z} \mid n = 4k, \text{ for some integer } k\}$$

$$A_1 = \{n \in \mathbf{Z} \mid n = 4k + 1, \text{ for some integer } k\}$$

$$A_2 = \{n \in \mathbf{Z} \mid n = 4k + 2, \text{ for some integer } k\}$$

and

$$A_3 = \{n \in \mathbf{Z} \mid n = 4k + 3, \text{ for some integer } k\}.$$

Is $\{A_0, A_1, A_2, A_3\}$ a partition of $\mathbf{Z}$? Explain your answer.

31. Suppose $A = \{1, 2\}$ and $B = \{2, 3\}$. Find each of the following:

- $\mathcal{P}(A \cap B)$
- $\mathcal{P}(A)$
- $\mathcal{P}(A \cup B)$
- $\mathcal{P}(A \times B)$

32. a. Suppose $A = \{1\}$ and $B = \{u, v\}$. Find $\mathcal{P}(A \times B)$.

b. Suppose $X = \{a, b\}$ and $Y = \{x, y\}$. Find $\mathcal{P}(X \times Y)$.

33. a. Find $\mathcal{P}(\emptyset)$.

b. Find $\mathcal{P}(\mathcal{P}(\emptyset))$.

c. Find $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$.

34. Let $A_1 = \{1\}$, $A_2 = \{u, v\}$, and $A_3 = \{m, n\}$. Find each of the following sets:

- $A_1 \cup (A_2 \times A_3)$
- $(A_1 \cup A_2) \times A_3$

35. Let $A = \{a, b\}$, $B = \{1, 2\}$, and $C = \{2, 3\}$. Find each of the following sets.

- $A \times (B \cup C)$
- $(A \times B) \cup (A \times C)$
- $A \times (B \cap C)$
- $(A \times B) \cap (A \times C)$

36. Trace the action of Algorithm 6.1.1 on the variables i, j , *found*, and *answer* for $m = 3$, $n = 3$, and sets A and B represented as the arrays $a[1] = u$, $a[2] = v$, $a[3] = w$, $b[1] = w$, $b[2] = u$, and $b[3] = v$.

37. Trace the action of Algorithm 6.1.1 on the variables i, j , *found*, and *answer* for $m = 4$, $n = 4$, and sets A and B represented as the arrays $a[1] = u$, $a[2] = v$, $a[3] = w$, $a[4] = x$, $b[1] = r$, $b[2] = u$, $b[3] = y$, $b[4] = z$.

38. Write an algorithm to determine whether a given element x belongs to a given set that is represented as the array $a[1], a[2], \dots, a[n]$.

EXERCISE SET 6.2

Properties of Sets

44 of 44 exercises in scope

- To say that an element is in $A \cap (B \cup C)$ means that it is in (1) and in (2).
 - To say that an element is in $(A \cap B) \cup C$ means that it is in (1) or in (2).
 - To say that an element is in $A - (B \cap C)$ means that it is in (1) and not in (2).
 - To prove that $(A \cup B) \cap C \subseteq A \cup (B \cap C)$, we suppose that x is any element in (1). Then we must show that (2).
 - If A, B , and C are any sets such that $B \subseteq C$, to prove that $A \cap B \subseteq A \cap C$, we suppose that x is any element in (1). Then we must show that (2).
- The following are two proofs that for all sets A and B , $A - B \subseteq A$. The first is less formal, and the second is more formal. Fill in the blanks.
 - Proof: Suppose A and B are any sets. To show that $A - B \subseteq A$, we must show that every element in (1) is in (2). But any element in $A - B$ is in (3) and not in (4) (by definition of $A - B$). In particular, such an element is in A .
 - Proof: Suppose A and B are any sets and $x \in A - B$. [We must show that (1).] By definition of set difference, $x \in$ (2) and $x \notin$ (3). In particular, $x \in$ (4) [which is what was to be shown].

In 3 and 4, supply explanations of the steps in the given proofs.

3. Theorem: For all sets A, B , and C , if $A \subseteq B$, $B \subseteq C$, then $A \subseteq C$.

Proof:

Statement	Explanation
Suppose A, B , and C are any sets such that $A \subseteq B$ and $B \subseteq C$.	starting point
We must show that $A \subseteq C$.	conclusion to be shown
Let x be any element in A .	start of an element proof

Then x is in B .	<u>(a)</u>
It follows that x is in C .	<u>(b)</u>
Thus every element in A is in C .	since x could be any element of A
Therefore, $A \subseteq C$ [as was to be shown].	<u>(c)</u>

4. **Theorem:** For all sets A and B , if $A \subseteq B$, then $A \cup B \subseteq B$.

Proof:

Statement	Explanation
Suppose A , B , and C are any sets such that $A \subseteq B$.	starting point
We must show that $A \cup B \subseteq B$.	conclusion to be shown
Let x be any element in $A \cup B$.	start of an element proof
Then x is in A or x is in B .	<u>(a)</u>
In case x is in A , then x is in B .	<u>(b)</u>
In case x is in B , then x is in B .	tautology ($p \rightarrow p$)
So in either case x is in B .	proof by division into cases
Thus every element in $A \cup B$ is in B .	since x could be any element of $A \cup B$
Therefore, $A \cup B \subseteq B$ [as was to be shown].	<u>(c)</u>

5. Prove that for all sets A and B , $(B - A) = B \cap A^c$.

- H 6.** Let $\cap$ and $\cup$ stand for the words “intersection” and “union,” respectively. Fill in the blanks in the following proof that for all sets A , B , and C , $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof: Suppose A , B , and C are any sets.

(1) Proof that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$:

Let $x \in A \cap (B \cup C)$. [We must show that $x \in \underline{(a)}$.]

By definition of $\cap$, $x \in \underline{(b)}$ and $x \in B \cup C$.

Thus $x \in A$ and, by definition of $\cup$, $x \in B$ or (c).

Case 1 ($x \in A$ and $x \in B$): In this case, $x \in A \cap B$ by definition of $\cap$.

Case 2 ($x \in A$ and $x \in C$): In this case, $x \in A \cap C$ by definition of $\cap$.

By cases 1 and 2, $x \in A \cap B$ or $x \in A \cap C$, and so, by definition of $\cup$, (d).

[So $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ by definition of subset.]

(2) Proof that $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$:

Let $x \in (A \cap B) \cup (A \cap C)$. [We must show that $x \in A \cap (B \cup C)$.]

By definition of $\cup$, $x \in A \cap B$ (a) $x \in A \cap C$.

Case 1 ($x \in A \cap B$): In this case, by definition of $\cap$, $x \in A$ and $x \in B$.

Since $x \in B$, then $x \in B \cup C$ by definition of $\cup$.

Case 2 ($x \in A \cap C$): In this case, by definition of $\cap$, $x \in A$ (b) $x \in C$.

Since $x \in C$, then $x \in B \cup C$ by definition of $\cup$.

In both cases $x \in A$ and $x \in B \cup C$, and so, by definition of $\cap$, (c).

[So $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ by definition of (d).]

(3) Conclusion: [Since both subset relations have been proved, it follows, by definition of set equality, that (a).]

Use an element argument to prove each statement in 7–22. Assume that all sets are subsets of a universal set U .

- H 7.** For all sets A and B , $(A \cap B)^c = A^c \cup B^c$.

8. For all sets A and B , $(A \cap B) \cup (A \cap B^c) = A$.
(This property is used in Section 9.9.)

- H 9.** For all sets A , B , and C ,

$$(A - B) \cup (C - B) = (A \cup C) - B.$$

10. For all sets A , B , and C ,
 $(A \cup B) \cap C \subseteq A \cup (B \cap C)$.

11. For all sets A , B , and C ,
 $A \cap (B - C) \subseteq (A \cap B) - (A \cap C)$.

12. For all sets A , B , and C ,
 $(A \cup B) - C \subseteq (A - C) \cup (B - C)$.

13. For all sets A , B , and C ,
 $(A - B) \cap (C - B) = (A \cap C) - B$.

- H 14. For all sets A and B , $A \cup (A \cap B) = A$.

15. For every set A , $A \cup \emptyset = A$.

16. For all sets A , B , and C , if
 $A \subseteq B$ then $A \cap C \subseteq B \cap C$.

17. For all sets A , B , and C , if $A \subseteq B$ then
 $A \cup C \subseteq B \cup C$.

18. For all sets A and B , if $A \subseteq B$ then $B^c \subseteq A^c$.

- H 19. For all sets A , B , and C , if $A \subseteq B$ and $A \subseteq C$ then
 $A \subseteq B \cap C$.

20. For all sets A , B , and C , if $A \subseteq C$ and $B \subseteq C$ then
 $A \cup B \subseteq C$.

21. For all sets A , B , and C ,
 $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

22. For all sets A , B , and C ,
 $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

23. Find the mistake in the following “proof” that for all sets A , B , and C , if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

“**Proof:** Suppose A , B , and C are any sets such that $A \subseteq B$ and $B \subseteq C$. Since $A \subseteq B$, there is an element x such that $x \in A$ and $x \in B$, and since $B \subseteq C$, there is an element x such that $x \in B$ and $x \in C$. Hence there is an element x such that $x \in A$ and $x \in C$ and so $A \subseteq C$.”

- H 24. Find the mistake in the following “proof.”

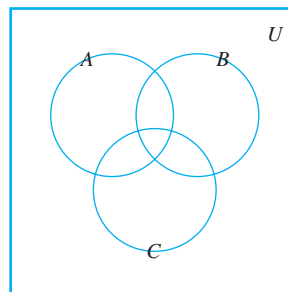
“**Theorem:**” For all sets A and B , $A^c \cup B^c \subseteq (A \cup B)^c$.

“**Proof:** Suppose A and B are any sets, and $x \in A^c \cup B^c$. Then $x \in A^c$ or $x \in B^c$ by definition of union. It follows that $x \notin A$ or $x \notin B$ by definition of complement, and so $x \notin A \cup B$ by definition of union. Thus $x \in (A \cup B)^c$ by definition of complement, and hence $A^c \cup B^c \subseteq (A \cup B)^c$.”

25. Find the mistake in the following “proof” that for all sets A and B , $(A - B) \cup (A \cap B) \subseteq A$.

“**Proof:** Suppose A and B are any sets, and suppose $x \in (A - B) \cup (A \cap B)$. If $x \in A$ then $x \in A - B$, and so, by definition of difference, $x \in A$ and $x \notin B$. In particular, $x \in A$, and, therefore, $(A - B) \cup (A \cap B) \subseteq A$ by definition of subset.”

26. Consider the Venn diagram below.



- Illustrate one of the distributive laws by shading in the region corresponding to $A \cup (B \cap C)$ on one copy of the diagram and $(A \cup B) \cap (A \cup C)$ on another.
- Illustrate the other distributive law by shading in the region corresponding to $A \cap (B \cup C)$ on one copy of the diagram and $(A \cap B) \cup (A \cap C)$ on another.
- Illustrate one of De Morgan's laws by shading in the region corresponding to $(A \cup B)^c$ on one copy of the diagram and $A^c \cap B^c$ on the other. (Leave the set C out of your diagrams.)
- Illustrate the other De Morgan's law by shading in the region corresponding to $(A \cap B)^c$ on one copy of the diagram and $A^c \cup B^c$ on the other. (Leave the set C out of your diagrams.)

27. Fill in the blanks in the following proof that for all sets A and B , $(A - B) \cap (B - A) = \emptyset$.

Proof: Let A and B be any sets and suppose $(A - B) \cap (B - A) \neq \emptyset$. That is, suppose there is an element x in (a). By definition of (b), $x \in A - B$ and $x \in$ (c). Then by definition of set difference, $x \in A$ and $x \notin B$ and $x \in$ (d) and $x \notin$ (e). In particular $x \in A$ and $x \notin$ (f), which is a contradiction. Hence [the supposition that $(A - B) \cap (B - A) \neq \emptyset$ is false, and so] (g).

Use the element method for proving a set equals the empty set to prove each statement in 28–38. Assume that all sets are subsets of a universal set U .

28. For all sets A and B , $(A \cap B) \cap (A \cap B^c) = \emptyset$.
(This property is used in Section 9.9.)

29. For all sets A , B , and C ,

$$(A - C) \cap (B - C) \cap (A - B) = \emptyset.$$

30. For every subset A of a universal set U ,
 $A \cap A^c = \emptyset$.

31. If U denotes a universal set, then $U^c = \emptyset$.

32. For every set A , $A \times \emptyset = \emptyset$.

33. For all sets A and B , if $A \subseteq B$ then $A \cap B^c = \emptyset$.

34. For all sets A and B , if $B \subseteq A^c$ then $A \cap B = \emptyset$.

35. For all sets A , B , and C , if $A \subseteq B$ and $B \cap C = \emptyset$ then $A \cap C = \emptyset$.

36. For all sets A , B , and C , if $C \subseteq B - A$, then
 $A \cap C = \emptyset$.

37. For all sets A , B , and C ,
if $B \cap C \subseteq A$, then $(C - A) \cap (B - A) = \emptyset$.

38. For all sets A , B , C , and D ,
if $A \cap C = \emptyset$ then $(A \times B) \cap (C \times D) = \emptyset$.

Prove each statement in 39–44.

- H 39. For all sets A and B ,

- $(A - B) \cup (B - A) \cup (A \cap B) = A \cup B$
- The sets $(A - B)$, $(B - A)$, and $(A \cap B)$ are mutually disjoint.

40. For every positive integer n , if A and $B_1, B_2, B_3, \dots$ are any sets, then

$$A \cap \left(\bigcup_{i=1}^n B_i \right) = \bigcup_{i=1}^n (A \cap B_i).$$

- H 41. For every positive integer n , if $A_1, A_2, A_3, \dots$ and B are any sets, then

$$\bigcup_{i=1}^n (A_i - B) = \left(\bigcup_{i=1}^n A_i \right) - B.$$

42. For every positive integer n , if $A_1, A_2, A_3, \dots$ and B are any sets, then

$$\bigcap_{i=1}^n (A_i - B) = \left(\bigcap_{i=1}^n A_i \right) - B.$$

43. For every positive integer n , if A and $B_1, B_2, B_3, \dots$ are any sets, then

$$\bigcup_{i=1}^n (A \times B_i) = A \times \left(\bigcup_{i=1}^n B_i \right).$$

44. For every positive integer n , if A and $B_1, B_2, B_3, \dots$ are any sets, then

$$\bigcap_{i=1}^n (A \times B_i) = A \times \left(\bigcap_{i=1}^n B_i \right).$$

EXERCISE SET 6.3

Disproofs and Algebraic Proofs

53 of 54 exercises in scope · omitted: mathematical induction (Epp Ch. 5)

For each of 1–4 find a counterexample to show that the statement is false. Assume all sets are subsets of a universal set U .

1. For all sets A , B , and C ,

$$(A \cup B) \cap C = A \cup (B \cap C).$$

2. For all sets A and B , $(A \cup B)^c = A^c \cup B^c$.

3. For all sets A , B , and C , if $A \not\subseteq B$ and $B \not\subseteq C$ then $A \not\subseteq C$.

4. For all sets A , B , and C , if $B \cup C \subseteq A$ then

$$(A - B) \cap (A - C) = \emptyset.$$

For each of 5–21 prove each statement that is true and find a counterexample for each statement that is false. Assume all sets are subsets of a universal set U .

5. For all sets A , B , and C ,

$$A - (B - C) = (A - B) - C.$$

6. For all sets A and B , $A \cap (A \cup B) = A$.

7. For all sets A , B , and C ,

$$(A - B) \cap (C - B) = A - (B \cup C).$$

8. For all sets A and B , if $A^c \subseteq B$ then $A \cup B = U$.

9. For all sets A , B , and C , if $A \subseteq C$ and $B \subseteq C$ then $A \cup B \subseteq C$.

10. For all sets A and B , if $A \subseteq B$ then $A \cap B^c = \emptyset$.

H 11. For all sets A , B , and C , if $A \subseteq B$ then $A \cap (B \cap C)^c = \emptyset$.

H 12. For all sets A , B , and C ,

$$A \cap (B - C) = (A \cap B) - (A \cap C).$$

13. For all sets A , B , and C ,

$$A \cup (B - C) = (A \cup B) - (A \cap C).$$

H 14. For all sets A , B , and C , if $A \cap C = B \cap C$ and $A \cup C = B \cup C$, then $A = B$.

H 15. For all sets A , B , and C , $(A - B) \cup C \subseteq A \cup (C - B)$.

16. For all sets A and B , if $A \cap B = \emptyset$ then $A \times B = \emptyset$.

17. For all sets A and B , if $A \subseteq B$ then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

18. For all sets A and B , $\mathcal{P}(A \cup B) \subseteq \mathcal{P}(A) \cup \mathcal{P}(B)$.

H 19. For all sets A and B , $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

20. For all sets A and B , $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

21. For all sets A and B , $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$.

22. Write a negation for each of the following statements. Indicate which is true, the statement or its negation. Justify your answers.

a. $\forall$ sets S , $\exists$ a set T such that $S \cap T = \emptyset$.

b. $\exists$ a set S such that $\forall$ sets T , $S \cup T = \emptyset$.

H*23. Let $S = \{a, b, c\}$, and for each integer $i = 0, 1, 2, 3$, let S_i be the set of all subsets of S that have i elements. List the elements in S_0 , S_1 , S_2 , and S_3 . Is $\{S_0, S_1, S_2, S_3\}$ a partition of $\mathcal{P}(S)$?

24. Let $A = \{t, u, v, w\}$, and let S_1 be the set of all subsets of A that do not contain w and S_2 the set of all subsets of A that contain w .

a. Find S_1 .

b. Find S_2 .

c. Are S_1 and S_2 disjoint?

d. Compare the sizes of S_1 and S_2 .

e. How many elements are in $S_1 \cup S_2$?

f. What is the relation between $S_1 \cup S_2$ and $\mathcal{P}(A)$?

H*26. The following problem, devised by Ginger Bolton, appeared in the January 1989 issue of the *College Mathematics Journal* (Vol. 20, No. 1, p. 68): Given a positive integer $n \geq 2$, let S be the set of all nonempty subsets of $\{2, 3, \dots, n\}$. For each $S_i \in S$, let P_i be the product of the elements of S_i . Prove or disprove that

$$\sum_{i=1}^{2^{n-1}-1} P_i = \frac{(n+1)!}{2} - 1.$$

In 27 and 28 supply a reason for each step in the derivation.

27. For all sets A , B , and C ,

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C).$$

Proof: Suppose A , B , and C are any sets. Then

$$\begin{aligned} (A \cup B) \cap C &= C \cap (A \cup B) && \text{by (a)} \\ &= (C \cap A) \cup (C \cap B) && \text{by (b)} \\ &= (A \cap C) \cup (B \cap C) && \text{by (c).} \end{aligned}$$

H 28. For all sets A , B , and C ,

$$(A \cup B) - (C - A) = A \cup (B - C).$$

Proof: Suppose A , B , and C are any sets. Then

$$\begin{aligned} (A \cup B) - (C - A) &= (A \cup B) \cap (C - A)^c && \text{by (a)} \\ &= (A \cup B) \cap (C \cap A^c)^c && \text{by (b)} \\ &= (A \cup B) \cap (A^c \cap C)^c && \text{by (c)} \\ &= (A \cup B) \cap ((A^c)^c \cup C^c) && \text{by (d)} \\ &= (A \cup B) \cap (A \cup C^c) && \text{by (e)} \\ &= A \cup (B \cap C^c) && \text{by (f)} \\ &= A \cup (B - C) && \text{by (g)}. \end{aligned}$$

H 29. Some steps are missing from the following proof that for all sets A and B , $(A \cup B^c) - B = (A - B) \cup B^c$. Indicate what they are, and then write the proof correctly.

Proof: Let any sets A and B be given. Then

$$\begin{aligned} (A \cup B^c) - B &= (A \cup B^c) \cap B^c && \text{by the set difference law} \\ &= (B^c \cap A) \cup (B^c \cap B^c) && \text{by the distributive law} \\ &= (B^c \cap A) \cup B^c && \text{by the idempotent law for } \cup \\ &= (A - B) \cup B^c && \text{by the set difference law.} \end{aligned}$$

In 30–40, construct an algebraic proof for the given statement. Cite a property from Theorem 6.2.2 for every step.

30. For all sets A , B , and C ,

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C).$$

31. For all sets A and B , $A \cup (B - A) = A \cup B$.

32. For all sets A and B , $(A - B) \cup (A \cap B) = A$.

33. For all sets A and B , $(A - B) \cap (A \cap B) = \emptyset$.

34. For all sets A , B , and C ,

$$(A - B) - C = A - (B \cup C).$$

35. For all sets A and B , $A - (A - B) = A \cap B$.

36. For all sets A and B , $((A^c \cup B^c) - A)^c = A$.

37. For all sets A and B , $(B^c \cup (B^c - A))^c = B$.

38. For all sets A and B , $(A \cap B)^c \cap A = A - B$.

39. For all sets A and B ,

$$(A - B) \cup (B - A) = (A \cup B) - (A \cap B).$$

40. For all sets A , B , and C ,

$$(A - B) - (B - C) = A - B.$$

In 41–43 simplify the given expression. Cite a property from Theorem 6.2.2 for every step.

H 41. $A \cap ((B \cup A^c) \cap B^c)$

42. $(A - (A \cap B)) \cap (B - (A \cap B))$

43. $((A \cap (B \cup C)) \cap (A - B)) \cap (B \cup C^c)$

44. Consider the following set property: For all sets A and B , $A - B$ and B are disjoint.

- Use an element argument to derive the property.
- Use an algebraic argument to derive the property (by applying properties from Theorem 6.2.2).
- Comment on which method you found easier.

45. Consider the following set property: For all sets A , B , and C ,

$$(A - B) \cup (B - C) = (A \cup B) - (B \cap C).$$

- Use an element argument to derive the property.
- Use an algebraic argument to derive the property (by applying properties from Theorem 6.2.2).
- Comment on which method you found easier.

Definition: Given sets A and B , the **symmetric difference of A and B** , denoted $A \Delta B$, is

$$A \Delta B = (A - B) \cup (B - A).$$

46. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{5, 6, 7, 8\}$. Find each of the following sets:

- $A \Delta B$
- $B \Delta C$
- $A \Delta C$
- $(A \Delta B) \Delta C$

Refer to the definition of symmetric difference given above. Prove each of 47–52, assuming that A , B , and C are all subsets of a universal set U .

47. $A \Delta B = B \Delta A$

48. $A \Delta \emptyset = A$

49. $A \Delta A^c = U$

50. $A \Delta A = \emptyset$

H* 51. If $A \Delta C = B \Delta C$, then $A = B$.

H* 52. $(A \Delta B) \Delta C = A \Delta (B \Delta C)$

H 53. Derive the set identity $A \cup (A \cap B) = A$ from the properties listed in Theorem 6.2.2(1)–(9). Start by showing that for every subset B of a universal set U , $U \cup B = U$. Then intersect both sides with A and deduce the identity.

54. Derive the set identity $A \cap (A \cup B) = A$ from the properties listed in Theorem 6.2.2(1)–(9). Start by showing that for every subset B of a universal set U , $\emptyset = \emptyset \cap B$. Then take the union of both sides with A and deduce the identity.

EXERCISE SET 6.4

Boolean Algebras, Russell's Paradox, Halting Problem

5 of 29 exercises in scope · omitted: Boolean-algebra axioms (the lecture only names Boolean algebra, via Table 6.4.1), liar sentences, and the halting problem

24. Can there exist a computer program that has as output a list of all the computer programs that do not list themselves in their output? Explain your answer.
25. Can there exist a book that refers to all those books and only those books that do not refer to themselves? Explain your answer.
26. Some English adjectives are descriptive of themselves (for instance, the word *polysyllabic* is polysyllabic) whereas others are not (for instance, the word *monosyllabic* is not monosyllabic). The word *heterological* refers to an adjective that does not describe itself. Is *heterological* heterological? Explain your answer.
27. As strange as it may seem, it is possible to give a precise-looking verbal definition of an integer that, in fact, is not a definition at all. The following was devised by an English librarian, G. G. Berry, and reported by Bertrand Russell. Explain how it leads to a contradiction. Let n be “the smallest integer not describable in fewer than 12 English words.” (Note that the total number of strings consisting of 11 or fewer English words is finite.)

29. Use a technique similar to that used to derive Russell's paradox to prove that for any set A , $\mathcal{P}(A) \not\subseteq A$.

EXERCISE SET 8.1

Relations on Sets

23 of 24 exercises in scope · omitted: one-to-one / onto functions (Epp Ch. 7)

1. As in Example 8.1.2, the **congruence modulo 2** relation E is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For every ordered pair $(m, n) \in \mathbf{Z} \times \mathbf{Z}$,

$$m E n \iff m - n \text{ is even.}$$

- a. Is $0 E 0$? Is $5 E 2$? Is $(6, 6) \in E$?
Is $(-1, 7) \in E$?
- b. Prove that for any even integer n , $n E 0$.

H 2. Prove that for all integers m and n , $m - n$ is even if, and only if, both m and n are even or both m and n are odd.

3. The **congruence modulo 3** relation, T , is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For all integers m and n ,

$$m T n \iff 3 \mid (m - n).$$

- a. Is $10 T 1$? Is $1 T 10$? Is $(2, 2) \in T$?
Is $(8, 1) \in T$?
- b. List five integers n such that $n T 0$.
- c. List five integers n such that $n T 1$.
- d. List five integers n such that $n T 2$.

H e. Make and prove a conjecture about which integers are related by T to 0, which integers are related by T to 1, and which integers are related by T to 2.

4. Define a relation P on $\mathbf{Z}$ as follows: For every ordered pair $(m, n) \in \mathbf{Z} \times \mathbf{Z}$,

$$m P n \iff m \text{ and } n \text{ have a common prime factor.}$$

- a. Is $15 P 25$? b. Is $22 P 27$?
- c. Is $0 P 5$? d. Is $8 P 8$?

5. Let $X = \{a, b, c\}$. Recall that $\mathcal{P}(X)$ is the power set of X . Define a relation $\mathbf{S}$ on $\mathcal{P}(X)$ as follows: For all sets A and B in $\mathcal{P}(X)$,

$$A \mathbf{S} B \iff A \text{ has the same number of elements as } B.$$

- a. Is $\{a, b\} \mathbf{S} \{b, c\}$? b. Is $\{a\} \mathbf{S} \{a, b\}$?
- c. Is $\{c\} \mathbf{S} \{b\}$?

6. Let $X = \{a, b, c\}$. Define a relation $\mathbf{J}$ on $\mathcal{P}(X)$ as follows: For all sets A and B in $\mathcal{P}(X)$,

$$A \mathbf{J} B \iff A \cap B \neq \emptyset.$$

- a. Is $\{a\} \mathbf{J} \{c\}$? b. Is $\{a, b\} \mathbf{J} \{b, c\}$?
c. Is $\{a, b\} \mathbf{J} \{a, b, c\}$?

7. Define a relation R on $\mathbf{Z}$ as follows: For all integers m and n ,

$$m R n \iff 5 \mid (m^2 - n^2).$$

- a. Is $1 R (-9)$? b. Is $2 R 13$?
c. Is $2 R (-8)$? d. Is $(-8) R 2$?

8. Let A be the set of all strings of a 's and b 's of length 4. Define a relation R on A as follows: For every $s, t \in A$,

$$s R t \iff s \text{ has the same first two characters as } t.$$

- a. Is $abaa R abba$? b. Is $aabb R bbaa$?
c. Is $aaaa R aaab$? d. Is $baaa R abaa$?

9. Let A be the set of all strings of 0's, 1's, and 2's of length 4. Define a relation R on A as follows: For every $s, t \in A$,

$$s R t \iff \text{the sum of the characters in } s \text{ equals the sum of the characters in } t.$$

- a. Is $0121 R 2200$? b. Is $1011 R 2101$?
c. Is $2212 R 2121$? d. Is $1220 R 2111$?

10. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let R be the "less than" relation. That is, for every ordered pair $(x, y) \in A \times B$,

$$x R y \iff x < y.$$

State explicitly which ordered pairs are in R and R^{-1} .

11. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let S be the "divides" relation. That is, for every ordered pair $(x, y) \in A \times B$,

$$x S y \iff x \mid y.$$

State explicitly which ordered pairs are in S and S^{-1} .

13. Define a relation R on $A = \{0, 1, 2, 3\}$ by $R = \{(0, 0), (1, 2), (2, 2)\}$.

14. Define a relation S on $B = \{a, b, c, d\}$ by $S = \{(a, b), (a, c), (b, c), (d, d)\}$.

15. Let $A = \{2, 3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows: For every $x, y \in A$,

$$x R y \iff x \mid y.$$

- H 16. Let $A = \{5, 6, 7, 8, 9, 10\}$ and define a relation S on A as follows: For every $x, y \in A$,

$$x S y \iff 2 \mid (x - y).$$

17. Let $A = \{2, 3, 4, 5, 6, 7, 8\}$ and define a relation T on A as follows: For every $x, y \in A$,

$$x T y \iff 3 \mid (x - y).$$

18. Let $A = \{0, 1, 3, 4, 5, 6\}$ and define a relation V on A as follows: For every $x, y \in A$,

$$x V y \iff 5 \mid (x^2 - y^2).$$

Exercises 19–20 refer to unions and intersections of relations. Since relations are subsets of Cartesian products, their unions and intersections can be calculated as for any subsets. Given two relations R and S from A to B ,

$$R \cup S = \{(x, y) \in A \times B \mid (x, y) \in R \text{ or } (x, y) \in S\}$$

$$R \cap S = \{(x, y) \in A \times B \mid (x, y) \in R \text{ and } (x, y) \in S\}.$$

19. Let $A = \{2, 4\}$ and $B = \{6, 8, 10\}$ and define relations R and S from A to B as follows: For every $(x, y) \in A \times B$,

$$x R y \iff x \mid y \text{ and}$$

$$x S y \iff y - 4 = x.$$

State explicitly which ordered pairs are in $A \times B$, R , S , $R \cup S$, and $R \cap S$.

Draw the directed graphs of the relations defined in 13–18.

20. Let $A = \{-1, 1, 2, 4\}$ and $B = \{1, 2\}$ and define relations R and S from A to B as follows: For every $(x, y) \in A \times B$,

$$x R y \iff |x| = |y| \text{ and}$$

$$x S y \iff x - y \text{ is even.}$$

State explicitly which ordered pairs are in $A \times B$, R , S , $R \cup S$, and $R \cap S$.

21. Define relations R and S on $\mathbf{R}$ as follows:

$$R = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid x < y\} \text{ and}$$

$$S = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid x = y\}.$$

That is, R is the “less than” relation and S is the “equals” relation on $\mathbf{R}$. Graph R , S , $R \cup S$, and $R \cap S$ in the Cartesian plane.

22. Define relations R and S on $\mathbf{R}$ as follows:

$$R = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid x^2 + y^2 = 4\} \text{ and}$$

$$S = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid x = y\}.$$

Graph R , S , $R \cup S$, and $R \cap S$ in the Cartesian plane.

23. Define relations R and S on $\mathbf{R}$ as follows:

$$R = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid y = |x|\} \text{ and}$$

$$S = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid y = 1\}.$$

Graph R , S , $R \cup S$, and $R \cap S$ in the Cartesian plane.

24. In Example 8.1.7 consider the query `SELECT Patient_ID#, Name FROM S WHERE Primary_Diagnosis = X`. The response to the query is the projection onto the first two coordinates of the intersection of the database with the set $A_1 \times A_2 \times A_3 \times \{X\}$.
- Find the result of the query `SELECT Patient_ID#, Name FROM S WHERE Primary_Diagnosis = pneumonia`.
 - Find the result of the query `SELECT Patient_ID#, Name FROM S WHERE Primary_Diagnosis = appendicitis`.

EXERCISE SET 8.2

Reflexivity, Symmetry, and Transitivity
56 of 56 exercises in scope

In 1–8, a number of relations are defined on the set $A = \{0, 1, 2, 3\}$. For each relation:

- Draw the directed graph.
- Determine whether the relation is reflexive.
- Determine whether the relation is symmetric.
- Determine whether the relation is transitive.

Give a counterexample in each case in which the relation does not satisfy one of the properties.

1. $R_1 = \{(0, 0), (0, 1), (0, 3), (1, 1), (1, 0), (2, 3), (3, 3)\}$

2. $R_2 = \{(0, 0), (0, 1), (1, 1), (1, 2), (2, 2), (2, 3)\}$

3. $R_3 = \{(2, 3), (3, 2)\}$

4. $R_4 = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$

5. $R_5 = \{(0, 0), (0, 1), (0, 2), (1, 2)\}$

6. $R_6 = \{(0, 1), (0, 2)\}$

7. $R_7 = \{(0, 3), (2, 3)\}$

8. $R_8 = \{(0, 0), (1, 1)\}$

In 9–33, determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answers.

9. R is the “greater than or equal to” relation on the set of real numbers: For every $x, y \in \mathbf{R}$, $x R y \iff x \geq y$.

10. C is the circle relation on the set of real numbers: For every $x, y \in \mathbf{R}$, $x C y \iff x^2 + y^2 = 1$.

11. D is the relation defined on $\mathbf{R}$ as follows: For every $x, y \in \mathbf{R}$, $x D y \iff xy \geq 0$.

12. E is the congruence modulo 4 relation on $\mathbf{Z}$: For every $m, n \in \mathbf{Z}$, $m E n \iff 4 \mid (m - n)$.

13. F is the congruence modulo 5 relation on $\mathbf{Z}$: For every $m, n \in \mathbf{Z}$, $m F n \iff 5 \mid (m - n)$.

14. O is the relation defined on $\mathbf{Z}$ as follows: For every $m, n \in \mathbf{Z}$, $m O n \iff m - n$ is odd.

15. D is the “divides” relation on $\mathbf{Z}^+$: For all positive integers m and n , $m D n \Leftrightarrow m \mid n$.
16. A is the “absolute value” relation on $\mathbf{R}$: For all real numbers x and y , $x A y \Leftrightarrow |x| = |y|$.
17. Recall that a prime number is an integer that is greater than 1 and has no positive integer divisors other than 1 and itself. (In particular, 1 is not prime.) A relation P is defined on $\mathbf{Z}$ as follows: For every $m, n \in \mathbf{Z}$, $m P n \Leftrightarrow \exists$ a prime number p such that $p \mid m$ and $p \mid n$.
- H 18. Define a relation Q on $\mathbf{R}$ as follows: For all real numbers x and y , $x Q y \Leftrightarrow x - y$ is rational.
19. Define a relation I on $\mathbf{R}$ as follows: For all real numbers x and y , $x I y \Leftrightarrow x - y$ is irrational.
20. Let $X = \{a, b, c\}$ and $\mathcal{P}(X)$ be the power set of X (the set of all subsets of X). A relation E is defined on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A E B \Leftrightarrow$ the number of elements in A equals the number of elements in B .
21. Let $X = \{a, b, c\}$ and $\mathcal{P}(X)$ be the power set of X . A relation L is defined on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A L B \Leftrightarrow$ the number of elements in A is less than the number of elements in B .
22. Let $X = \{a, b, c\}$ and $\mathcal{P}(X)$ be the power set of X . A relation N is defined on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A N B \Leftrightarrow$ the number of elements in A is not equal to the number of elements in B .
23. Let X be a nonempty set and $\mathcal{P}(X)$ the power set of X . Define the “subset” relation S on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A S B \Leftrightarrow A \subseteq B$.
24. Let X be a nonempty set and $\mathcal{P}(X)$ the power set of X . Define the “not equal to” relation U on $\mathcal{P}(X)$ as follows: For every $A, B \in \mathcal{P}(X)$, $A U B \Leftrightarrow A \neq B$.
25. Let A be the set of all strings of a ’s and b ’s of length 4. Define a relation R on A as follows: For every $s, t \in A$, $s R t \Leftrightarrow s$ has the same first two characters as t .
26. Let A be the set of all strings of 0’s, 1’s, and 2’s that have length 4 and for which the sum of the characters in the string is less than or equal to 2. Define a relation R on A as follows: For every $s, t \in A$, $s R t \Leftrightarrow$ the sum of the characters of s equals the sum of the characters of t .
27. Let A be the set of all English statements. A relation I is defined on A as follows: For every $p, q \in A$,

$$p I q \Leftrightarrow p \rightarrow q \text{ is true.}$$
28. Let $A = \mathbf{R} \times \mathbf{R}$. A relation F is defined on A as follows: For every (x_1, y_1) and (x_2, y_2) in A ,

$$(x_1, y_1) F (x_2, y_2) \Leftrightarrow x_1 = x_2.$$
29. Let $A = \mathbf{R} \times \mathbf{R}$. A relation S is defined on A as follows: For every (x_1, y_1) and (x_2, y_2) in A ,

$$(x_1, y_1) S (x_2, y_2) \Leftrightarrow y_1 = y_2.$$
30. Let A be the “punctured plane”; that is, A is the set of all points in the Cartesian plane except the origin $(0, 0)$. A relation R is defined on A as follows: For every p_1 and p_2 in A , $p_1 R p_2 \Leftrightarrow p_1$ and p_2 lie on the same half line emanating from the origin.
31. Let A be the set of people living in the world today. A relation R is defined on A as follows: For all people p and q in A ,

$$p R q \Leftrightarrow p \text{ lives within 100 miles of } q.$$
32. Let A be the set of all lines in the plane. A relation R is defined on A as follows: For every l_1 and l_2 in A , $l_1 R l_2 \Leftrightarrow l_1$ is parallel to l_2 . (Assume that a line is parallel to itself.)
33. Let A be the set of all lines in the plane. A relation R is defined on A as follows: For every l_1 and l_2 in A ,

$$l_1 R l_2 \Leftrightarrow l_1 \text{ is perpendicular to } l_2.$$

In 34–36, assume that R is a relation on a set A . Prove or disprove each statement.

34. If R is reflexive, then R^{-1} is reflexive.
35. If R is symmetric, then R^{-1} is symmetric.
36. If R is transitive, then R^{-1} is transitive.

In 37–42, assume that R and S are relations on a set A . Prove or disprove each statement.

37. If R and S are reflexive, is $R \cap S$ reflexive? Why?
- H 38. If R and S are symmetric, is $R \cap S$ symmetric? Why?
39. If R and S are transitive, is $R \cap S$ transitive? Why?
40. If R and S are reflexive, is $R \cup S$ reflexive? Why?
41. If R and S are symmetric, is $R \cup S$ symmetric? Why?
42. If R and S are transitive, is $R \cup S$ transitive? Why?

In 43–50, the following definitions are used: A relation on a set A is defined to be

irreflexive if, and only if, for every $x \in A$, $x \not R x$;
 asymmetric if, and only if, for every $x, y \in A$ if $x R y$ then $y \not R x$;
 intransitive if, and only if, for every $x, y, z \in A$, if $x R y$ and $y R z$ then $x \not R z$.

For each of the relations in the referenced exercise, determine whether the relation is irreflexive, asymmetric, intransitive, or none of these.

43. Exercise 1
44. Exercise 2
45. Exercise 3
46. Exercise 4
47. Exercise 5
48. Exercise 6
49. Exercise 7
50. Exercise 8

In 51–53, R , S , and T are relations defined on $A = \{0, 1, 2, 3\}$.

51. Let $R = \{(0, 1), (0, 2), (1, 1), (1, 3), (2, 2), (3, 0)\}$. Find R^t , the transitive closure of R .
52. Let $S = \{(0, 0), (0, 3), (1, 0), (1, 2), (2, 0), (3, 2)\}$. Find S^t , the transitive closure of S .

53. Let $T = \{(0, 2), (1, 0), (2, 3), (3, 1)\}$. Find T^t , the transitive closure of T .
54. Write a computer algorithm to test whether a relation R defined on a finite set A is reflexive, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$
55. Write a computer algorithm to test whether a relation R defined on a finite set A is symmetric, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$
56. Write a computer algorithm to test whether a relation R defined on a finite set A is transitive, where
- $$A = \{a[1], a[2], \dots, a[n]\}.$$

EXERCISE SET 8.3

Equivalence Relations

46 of 47 exercises in scope · omitted: digital logic circuits (Epp 2.4)

1. Suppose that $S = \{a, b, c, d, e\}$ and R is a relation on S such that $a R b$, $b R c$, and $d R e$. List all of the following that must be true if R is (a) reflexive (but not symmetric or transitive), (b) symmetric (but not reflexive or transitive), (c) transitive (but not reflexive or symmetric), and (d) an equivalence relation.
- $c R b$ $c R c$ $a R c$ $b R a$
 $a R d$ $e R a$ $e R d$ $c R a$
2. Each of the following partitions of $\{0, 1, 2, 3, 4\}$ induces a relation R on $\{0, 1, 2, 3, 4\}$. In each case, find the ordered pairs in R .
- a. $\{0, 2\}, \{1\}, \{3, 4\}$ b. $\{0\}, \{1, 3, 4\}, \{2\}$
 c. $\{0\}, \{1, 2, 3, 4\}$

In each of 3–6, the relation R is an equivalence relation on A . As in Example 8.3.5, first find the specified equivalence classes. Then state the number of distinct equivalence classes for R and list them.

3. $A = \{0, 1, 2, 3, 4\}$
 $R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)\}$
 equivalence classes: $[0], [1], [2], [3]$
4. $A = \{a, b, c, d\}$
 $R = \{(a, a), (b, b), (b, d), (c, c), (d, b), (d, d)\}$
 equivalence classes: $[a], [b], [c], [d]$

5. $A = \{1, 2, 3, 4, \dots, 20\}$. R is defined on A as follows:

$$\text{For all } x, y \in A, \quad x R y \iff 4 \mid (x - y).$$

equivalence classes: $[1], [2], [3], [4], [5]$

6. $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$. R is defined on A as follows:

$$\text{For all } x, y \in A, \quad x R y \iff 3 \mid (x - y).$$

equivalence classes: $[0], [1], [2], [3]$

In each of 7–14, the relation R is an equivalence relation on the set A . Find the distinct equivalence classes of R .

7. $A = \{(1, 3), (2, 4), (-4, -8), (3, 9), (1, 5), (3, 6)\}$. R is defined on A as follows: For every $(a, b), (c, d) \in A$,

$$(a, b) R (c, d) \iff ad = bc.$$

8. $X = \{a, b, c\}$ and $A = \mathcal{P}(X)$. R is defined on A as follows: For all sets u and v in $\mathcal{P}(X)$,

$$u R v \iff N(u) = N(v).$$

(That is, the number of elements in u equals the number of elements in v .)

9. $X = \{-1, 0, 1\}$ and $A = \mathcal{P}(X)$. R is defined on $\mathcal{P}(X)$ as follows: For all sets s and t in $\mathcal{P}(X)$,

$$s R t \iff \text{the sum of the elements in } s \text{ equals the sum of the elements in } t.$$

10. $A = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$. R is defined on A as follows: For all $m, n \in \mathbb{Z}$,

$$m R n \iff 3 \mid (m^2 - n^2).$$

11. $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. R is defined on A as follows: For every $(m, n) \in A$,

$$m R n \iff 4 \mid (m^2 - n^2).$$

12. $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. R is defined on A as follows: For all $(m, n) \in A$,

$$m R n \iff 5 \mid (m^2 - n^2).$$

13. A is the set of all strings of length 4 in a 's and b 's. R is defined on A as follows: For all strings s and t in A ,

$$s R t \iff s \text{ has the same first two characters as } t.$$

14. A is the set of all strings of 0's, 1's, and 2's that have length 4 and for which the sum of the characters in the string is less than or equal to 2. R is defined on A as follows: For every $s, t \in A$,

$$s R t \iff \text{the sum of the characters of } s \text{ equals the sum of the characters of } t.$$

15. Determine which of the following congruence relations are true and which are false.

- a. $17 \equiv 2 \pmod{5}$
- b. $4 \equiv -5 \pmod{7}$
- c. $-2 \equiv -8 \pmod{3}$
- d. $-6 \equiv 22 \pmod{2}$

16. a. Let R be the relation of congruence modulo 3. Which of the following equivalence classes are equal?

$$[7], [-4], [-6], [17], [4], [27], [19]$$

- b. Let R be the relation of congruence modulo 7. Which of the following equivalence classes are equal?

$$[35], [3], [-7], [12], [0], [-2], [17]$$

17. a. Prove that for all integers m and n , $m \equiv n \pmod{3}$ if, and only if, $m \bmod 3 = n \bmod 3$.
b. Prove that for all integers m and n and any positive integer d , $m \equiv n \pmod{d}$ if, and only if, $m \bmod d = n \bmod d$.
18. a. Give an example of two sets that are distinct but not disjoint.
b. Find sets A_1 and A_2 and elements x, y , and z such that x and y are in A_1 and y and z are in A_2 but x and z are not both in either of the sets A_1 or A_2 .

In 19–31, (1) prove that the relation is an equivalence relation, and (2) describe the distinct equivalence classes of each relation.

19. A is the set of all students at your college.

- a. R is the relation defined on A as follows: For every x and y in A ,

$$x R y \iff x \text{ has the same major (or double major) as } y.$$

(Assume “undeclared” is a major.)

- b. S is the relation defined on A as follows: For every $x, y \in A$,

$$x S y \iff x \text{ is the same age as } y.$$

20. E is the relation defined on $\mathbf{Z}$ as follows:

For every $m, n \in \mathbf{Z}$, $m E n \iff 4 \mid (m - n)$.

H 21. R is the relation defined on $\mathbf{Z}$ as follows:

For every $m, n \in \mathbf{Z}$, $m R n \iff 7m - 5n$ is even.

22. Let A be the set of all statement forms in three variables p, q , and r . $\mathbf{R}$ is the relation defined on A as follows: For all P and Q in A ,

$P \mathbf{R} Q \iff P$ and Q have the same truth table.

23. Let P be a set of parts shipped to a company from various suppliers. S is the relation defined on P as follows: For every $x, y \in P$,

$x S y \iff x$ has the same part number and is shipped from the same supplier as y .

24. Let A be the set of identifiers in a computer program. It is common for identifiers to be used for only a short part of the execution time of a program and not to be used again to execute other parts of the program. In such cases, arranging for identifiers to share memory locations makes efficient use of a computer's memory capacity. Define a relation R on A as follows: For all identifiers x and y ,

$x R y \iff$ the values of x and y are stored in the same memory location during execution of the program.

25. A is the "absolute value" relation defined on $\mathbf{R}$ as follows:

For every $x, y \in \mathbf{R}$, $x A y \iff |x| = |y|$.

H 26. D is the relation defined on $\mathbf{Z}$ as follows: For every $m, n \in \mathbf{Z}$,

$m D n \iff 3 \mid (m^2 - n^2)$.

27. R is the relation defined on $\mathbf{Z}$ as follows: For every $(m, n) \in \mathbf{Z}$,

$m R n \iff 4 \mid (m^2 - n^2)$.

28. I is the relation defined on $\mathbf{R}$ as follows:

For every $x, y \in \mathbf{R}$, $m I n \iff x - y$ is an integer.

29. Define P on the set $\mathbf{R} \times \mathbf{R}$ of ordered pairs of real numbers as follows: For every $(w, x), (y, z) \in \mathbf{R} \times \mathbf{R}$,

$(w, x) P (y, z) \iff w = y$.

30. Define Q on the set $\mathbf{R} \times \mathbf{R}$ as follows: For every $(w, x), (y, z) \in \mathbf{R} \times \mathbf{R}$,

$(w, x) Q (y, z) \iff x = z$.

31. Let P be the set of all points in the Cartesian plane except the origin. R is the relation defined on P as follows: For every p_1 and p_2 in P ,
 $p_1 R p_2 \iff$ the line segment $\overline{p_1 p_2}$ emanates from the origin.

32. Let A be the set of all straight lines in the Cartesian plane. Define a relation $\parallel$ on A as follows:

For every l_1 and l_2 in A ,

$l_1 \parallel l_2 \iff l_1$ is parallel to l_2 .

Then $\parallel$ is an equivalence relation on A . Describe the equivalence classes of this relation.

33. Let A be the set of points in the rectangle with x and y coordinates between 0 and 1. That is,

$A = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1\}$.

Define a relation R on A as follows: For all (x_1, y_1) and (x_2, y_2) in A ,

$(x_1, y_1) R (x_2, y_2) \iff$

$(x_1, y_1) = (x_2, y_2)$; or

$x_1 = 0$ and $x_2 = 1$ and $y_1 = y_2$; or

$x_1 = 1$ and $x_2 = 0$ and $y_1 = y_2$; or

$y_1 = 0$ and $y_2 = 1$ and $x_1 = x_2$; or

$y_1 = 1$ and $y_2 = 0$ and $x_1 = x_2$.

In other words, all points along the top edge of the rectangle are related to the points along the bottom edge directly beneath them, and all points directly opposite each other along the left and right edges are related to each other. The points in the interior of the rectangle are not related to anything other than themselves. Then R is an equivalence relation on A . Imagine gluing together all the points that are in the same equivalence class. Describe the resulting figure.

- 34.** The documentation for the computer language Java recommends that when an “equals method” is defined for an object, it be an equivalence relation. That is, if R is defined as follows:

$x R y \iff x.equals(y)$ for all objects in the class, then R should be an equivalence relation. Suppose that in trying to optimize some of the mathematics of a graphics application, a programmer creates an object called a point, consisting of two coordinates in the plane. The programmer defines an equals method as follows: If p and q are any points, then

$$p.equals(q) \iff \text{the distance from } p \text{ to } q \text{ is less than or equal to } c$$

where c is a small positive number that depends on the resolution of the computer display. Is the programmer’s equals method an equivalence relation? Justify your answer.

Let R be an equivalence relation on a set A . Prove each of the statements in 36–41 directly from the definitions of equivalence relation and equivalence class without using the results of Lemma 8.3.2, Lemma 8.3.3, or Theorem 8.3.4.

- 36.** For every a in A , $a \in [a]$.
- 37.** For every a and b in A , if $b \in [a]$ then $a R b$.
- 38.** For every a , b , and c in A , if $b R c$ and $c \in [a]$ then $b \in [a]$.
- 39.** For every a and b in A , if $[a] = [b]$ then $a R b$.
- 40.** For every a , b , and x in A , if $a R b$ and $x \in [a]$ then $x \in [b]$.
- H 41.** For every a and b in A , if $a \in [b]$ then $[a] = [b]$.
- 42.** Let R be the relation defined in Example 8.3.12.
- Prove that R is reflexive.
 - Prove that R is symmetric.
 - List four distinct elements in $[(1, 3)]$.
 - List four distinct elements in $[(2, 5)]$.

- * 43.** In Example 8.3.12, define operations of addition (+) and multiplication ($\cdot$) as follows: For every $(a, b), (c, d) \in A$,

$$[(a, b)] + [(c, d)] = [(ad + bc, bd)]$$

$$[(a, b)] \cdot [(c, d)] = [(ac, bd)].$$

- Prove that this addition is well defined. That is, show that if $[(a, b)] = [(a', b')]$ and $[(c, d)] = [(c', d')]$, then $[(ad + bc, bd)] = [(a'd' + b'c', b'd')]$.
- Prove that this multiplication is well defined. That is, show that if $[(a, b)] = [(a', b')]$ and $[(c, d)] = [(c', d')]$, then $[(ac, bd)] = [(a'c', b'd')]$.
- Show that $[(0, 1)]$ is an identity element for addition. That is, show that for any $(a, b) \in A$,

$$[(a, b)] + [(0, 1)] = [(0, 1)] + [(a, b)] = [(a, b)].$$
- Find an identity element for multiplication. That is, find (i, j) in A so that for every (a, b) in A , $[(a, b)] \cdot [(i, j)] = [(i, j)] \cdot [(a, b)] = [(a, b)]$.
- For any $(a, b) \in A$, show that $[(-a, b)]$ is an inverse for $[(a, b)]$ for addition. That is, show that $[(-a, b)] + [(a, b)] = [(a, b)] + [(-a, b)] = [(0, 1)]$.
- Given any $(a, b) \in A$ with $a \neq 0$, find an inverse for $[(a, b)]$ for multiplication. That is, find (c, d) in A so that $[(a, b)] \cdot [(c, d)] = [(c, d)] \cdot [(a, b)] = [(i, j)]$, where $[(i, j)]$ is the identity element you found in part (d).

- 44.** Let $A = \mathbf{Z}^+ \times \mathbf{Z}^+$. Define a relation R on A as follows: For every (a, b) and (c, d) in A ,

$$(a, b) R (c, d) \iff a + d = c + b.$$

- Prove that R is reflexive.
- Prove that R is symmetric.
- H c.** Prove that R is transitive.
- List five elements in $[(1, 1)]$.
- List five elements in $[(3, 1)]$.
- List five elements in $[(1, 2)]$.
- Describe the distinct equivalence classes of R .

45. The following argument claims to prove that the requirement that an equivalence relation be reflexive is redundant. In other words, it claims to show that if a relation is symmetric and transitive, then it is reflexive. Find the mistake in the argument.

“**Proof:** Let R be a relation on a set A and suppose R is symmetric and transitive. For any two elements x and y in A , if $x R y$ then $y R x$ since R is symmetric. Thus it follows by transitivity that $x R x$, and hence R is reflexive.”

46. Let R be a relation on a set A and suppose R is symmetric and transitive. Prove the following: If for every x in A there is a y in A such that $x R y$, then R is an equivalence relation.
47. Refer to the quote at the beginning of this section to answer the following questions.
- What is the name of the Knight’s song called?
 - What is the name of the Knight’s song?
 - What is the Knight’s song called?
 - What is the Knight’s song?
 - What is your (full, legal) name?
 - What are you called?
 - What *are* you? (Do not answer this on paper; just think about it.)

EXERCISE SET 8.5

Partial Order Relations

48 of 51 exercises in scope · omitted: lexicographic order, mathematical induction, and PERT/CPM timing

- Each of the following is a relation on $\{0, 1, 2, 3\}$. Draw directed graphs for each relation, and indicate which relations are antisymmetric.
 - $R_1 = \{(0, 0), (0, 2), (1, 0), (1, 3), (2, 2), (3, 0), (3, 1)\}$
 - $R_2 = \{(0, 1), (0, 2), (1, 1), (1, 2), (1, 3), (2, 2), (3, 2)\}$
 - $R_3 = \{(0, 0), (0, 3), (1, 0), (1, 3), (2, 2), (3, 3), (3, 2)\}$
 - $R_4 = \{(0, 0), (1, 0), (1, 2), (1, 3), (2, 0), (2, 1), (3, 2), (3, 0)\}$
- Let P be the set of all people in the world and define a relation R on P as follows: For all people x and y ,

$$x R y \iff x \text{ is no older than } y.$$
 Is R antisymmetric? Prove or give a counterexample.
- Let S be the set of all strings of a ’s and b ’s. Define a relation R on S as follows: For every $s, t \in S$,

$$s R t \iff L(s) \leq L(t),$$
 where $L(x)$ denotes the length of a string x . Is R antisymmetric? Prove or give a counterexample.
- Let R be the “less than” relation on $\mathbf{R}$, the set of all real numbers: For every $x, y \in \mathbf{R}$,

$$x R y \iff x < y.$$
 Is R antisymmetric? Prove or give a counterexample.
- Let $\mathbf{R}$ be the set of all real numbers and define a relation R on $\mathbf{R} \times \mathbf{R}$ as follows: For every (a, b) and (c, d) in $\mathbf{R} \times \mathbf{R}$,

$$(a, b) R (c, d) \iff \text{either } a < c \text{ or both } a = c \text{ and } b \leq d.$$
 Is R a partial order relation? Prove or give a counterexample.
- Let P be the set of all people who have ever lived and define a relation R on P as follows: For every $r, s \in P$,

$$r R s \iff r \text{ is an ancestor of } s \text{ or } r = s.$$
 Is R a partial order relation? Prove or give a counterexample.
- Define a relation R on $\mathbf{Z}$, the set of all integers as follows: For every $m, n \in \mathbf{Z}$,

$$m R n \iff \begin{array}{l} \text{every prime factor of } m \\ \text{is a prime factor of } n. \end{array}$$
 Is R a partial order relation? Prove or give a counterexample.
- Define a relation R on $\mathbf{Z}$, the set of all integers as follows: For every $m, n \in \mathbf{Z}$,

$$m R n \iff m + n \text{ is even.}$$
 Is R a partial order relation? Prove or give a counterexample.
- Define a relation R on $\mathbf{R}$, the set of all real numbers as follows: For every $x, y \in \mathbf{R}$,

$$x R y \iff x^2 \leq y^2.$$
 Is R a partial order relation? Prove or give a counterexample.

10. Suppose R and S are antisymmetric relations on a set A . Must $R \cup S$ also be antisymmetric? Explain.

12. Prove Theorem 8.5.1.

13. Let $A = \{a, b\}$. Describe all partial order relations on A .

14. Let $A = \{a, b, c\}$.

- Describe all partial order relations on A for which a is a maximal element.
- Describe all partial order relations on A for which a is a minimal element.

H 15. Suppose a relation R on a set A is reflexive, symmetric, transitive, and antisymmetric. What can you conclude about R ? Prove your answer.

16. Consider the “divides” relation on each of the following sets A . Draw the Hasse diagram for each relation.

- $A = \{1, 2, 4, 5, 10, 15, 20\}$
- $A = \{2, 3, 4, 6, 8, 9, 12, 18\}$

17. Consider the “subset” relation on $\mathcal{P}(S)$ for each of the following sets S . Draw the Hasse diagram for each relation.

- $S = \{0, 1\}$
- $S = \{0, 1, 2\}$

18. Let $S = \{0, 1\}$ and consider the partial order relation R defined on $S \times S$ as follows: For all ordered pairs (a, b) and (c, d) in $S \times S$,

$$(a, b) R (c, d) \iff \text{either } a < c \text{ or both } a = c \text{ and } b \leq d,$$

where $<$ denotes the usual “less than” and $\leq$ denotes the usual “less than or equal to” relation for real numbers. Draw the Hasse diagram for R .

19. Let $S = \{0, 1\}$ and consider the partial order relation R defined on $S \times S$ as follows: For all ordered pairs (a, b) and (c, d) in $S \times S$,

$$(a, b) R (c, d) \iff a \leq c \text{ and } b \leq d,$$

where $\leq$ denotes the usual “less than or equal to” relation for real numbers. Draw the Hasse diagram for R .

20. Let $S = \{0, 1\}$ and consider the partial order relation R defined on $S \times S \times S$ as follows: For all ordered triples (a, b, c) and (d, e, f) in $S \times S \times S$,

$$(a, b, c) R (d, e, f) \iff a \leq d, b \leq e, \text{ and } c \leq f,$$

where $\leq$ denotes the usual “less than or equal to” relation for real numbers. Draw the Hasse diagram for R .

21. Consider the “divides” relation defined on the set $A = \{1, 2, 2^2, 2^3, \dots, 2^n\}$, where n is a nonnegative integer.

- Prove that this relation is a total order relation on A .
- Draw the Hasse diagram for this relation for $n = 4$.

In 22–29, find all greatest, least, maximal, and minimal elements for the relations in each of the referenced exercises.

22. Exercise 16(a)

23. Exercise 16(b)

24. Exercise 17(a)

25. Exercise 17(b)

26. Exercise 18

27. Exercise 19

28. Exercise 20

29. Exercise 21

30. Each of the following sets is partially ordered with respect to the “less than or equal to” relation, $\leq$, for real numbers. In each case, determine whether the set has a greatest or least element.

- $\mathbf{R}$
- $\{x \in \mathbf{R} \mid 0 \leq x \leq 1\}$
- $\{x \in \mathbf{R} \mid 0 < x < 1\}$
- $\{x \in \mathbf{Z} \mid 0 < x < 10\}$

31. Let $A = \{a, b, c, d\}$, and let R be the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (c, a), (a, d), (c, d), (b, c), (b, d), (b, a)\}.$$

Is R a total order on A ? Justify your answer.

32. Let $A = \{a, b, c, d\}$, and let R be the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (c, b), (a, d), (b, a), (b, d), (c, d), (c, a)\}.$$

Is R a total order on A ? Justify your answer.

33. Consider the set $A = \{12, 24, 48, 3, 9\}$ ordered by the “divides” relation. Is A totally ordered with respect to the relation? Justify your answer.

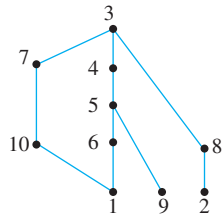
H 34. Suppose that R is a partial order relation on a set A and that B is a subset of A . The **restriction of R to B** is defined as follows:

The restriction of R to B

$$= \{(x, y) \mid x \in B, y \in B, \text{ and } (x, y) \in R\}.$$

In other words, two elements of B are related by the restriction of R to B if, and only if, they are related by R . Prove that the restriction of R to B is a partial order relation on B . (In less formal language, this says that a subset of a partially ordered set is partially ordered.)

35. The set $\mathcal{P}(\{w, x, y, z\})$ is partially ordered with respect to the “subset” relation $\subseteq$. Find a chain of length 4 in $\mathcal{P}(\{w, x, y, z\})$.
36. The set $A = \{2, 4, 3, 6, 12, 18, 24\}$ is partially ordered with respect to the “divides” relation. Find a chain of length 3 in A .
37. Find a chain of length 2 for the relation defined in exercise 19.
38. Prove that a partially ordered set is totally ordered if, and only if, it is a chain.
40. Prove that a nonempty, finite, partially ordered set has
- at least one minimal element,
 - at least one maximal element.
41. Prove that a finite, partially ordered set has
- at most one greatest element,
 - at most one least element.
42. Draw a Hasse diagram for a partially ordered set that has two maximal elements and two minimal elements and is such that each element is comparable to exactly two other elements.
43. Draw a Hasse diagram for a partially ordered set that has three maximal elements and three minimal elements and is such that each element is either greater than or less than exactly two other elements.
44. Use the algorithm given in the text to find a topological sorting for the relation of exercise 16(a) that is different from the “less than or equal to” relation $\leq$.
45. Use the algorithm given in the text to find a topological sorting for the relation of exercise 16(b) that is different from the “less than or equal to” relation $\leq$.
46. Use the algorithm given in the text to find a topological sorting for the relation of exercise 19.
47. Use the algorithm given in the text to find a topological sorting for the relation of exercise 20.
48. Use the algorithm given in the text to find a topological sorting for the “subset” relation on $\mathcal{P}(\{a, b, c, d\})$.
49. Refer to the prerequisite structure shown in Figure 8.5.1.
- Find a list of six noncomparable courses that is different from the list given in the text.
 - Find two topological sortings that are different from the one given in the text.
50. A set S of jobs can be ordered by writing $x \preceq y$ to mean that either $x = y$ or x must be done before y , for all x and y in S . The following is a Hasse diagram for this relation for a particular set S of jobs.



- a. If one person is to perform all the jobs, one after another, find an order in which the jobs can be done.
- b. Suppose enough people are available to perform any number of jobs simultaneously.
 - (i) If each job requires one day to perform, what is the least number of days needed to perform all ten jobs?
 - (ii) What is the maximum number of jobs that can be performed at the same time?